

Modul 1-6 Docker

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2 D3 Teknik Informatika A

PROGRAM STUDI D3 TEKNIK INFORMATIKA

POLITEKNIK ELEKTRONIKA NEGERI

SURABAYA

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MODUL 1: Docker dan Instalasi

Langkah 0: Persiapan Environment

Pastikan VM/Host sudah terkoneksi internet untuk download image dari Docker Hub.

```
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

hisyam@HISYAMpc:~$ ping -c 3 google.com
PING google.com (142.250.9.113) 56(84) bytes of data.

--- google.com ping statistics ---
3 packets transmitted, 0 received, 100% packet loss, time 2004ms

hisyam@HISYAMpc:~$ sudo apt update && sudo apt upgrade -y
[sudo] password for hisyam:
Hit:1 http://archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:4 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1625 kB]
Get:5 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:6 http://archive.ubuntu.com/ubuntu noble/universe amd64 Packages [15.0 MB]
Get:7 http://security.ubuntu.com/ubuntu noble-security/main Translation-en [259 kB]
Get:8 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [21.9 kB]
Get:9 http://security.ubuntu.com/ubuntu noble-security/main amd64 c-n-f Metadata [11.0 kB]
Get:10 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [1182 kB]
Get:11 http://security.ubuntu.com/ubuntu noble-security/universe Translation-en [227 kB]
```

ping -c 3 google.com digunakan untuk memastikan koneksi internet aktif karena Docker perlu download image dari Docker Hub

Langkah 1: Instalasi Docker Engine di Ubuntu 22.04

1.1 Hapus versi lama (jika ada)

```
hisyam@HISYAMpc:~$ sudo apt remove -y docker docker-engine docker.io containerd runc 2>/dev/null
[sudo] password for hisyam:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Package 'docker' is not installed, so not removed
```

1.2 Instal dependensi

```
hisyam@HISYAMpc:~$ sudo apt install -y \
    ca-certificates \
    curl \
    gnupg \
    lsb-release
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
ca-certificates is already the newest version (20240203).
ca-certificates set to manually installed.
curl is already the newest version (8.5.0-2ubuntu10.8).
curl set to manually installed.
gnupg is already the newest version (2.4.4-2ubuntu17.4).
gnupg set to manually installed.
lsb-release is already the newest version (12.0-2).
lsb-release set to manually installed.
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
```

Menginstall dependensi yang dibutuhkan untuk proses instalasi Docker, yaitu ca-certificates untuk verifikasi SSL, curl untuk download file, gnupg untuk verifikasi GPG key, dan lsb-release untuk mendeteksi versi Ubuntu.

1.3 Tambahkan Docker GPG key dan repository

Buat direktori keyrings

```
hisyam@HISYAMpc:~$ # Buat direktori keyrings
sudo install -m 0755 -d /etc/apt/keyrings
hisyam@HISYAMpc:~$ sudo install -m 0755 -d /etc/apt/keyrings
```

Download GPG key Docker

```
hisyam@HISYAMpc:~$ curl -fsSL https://download.docker.com/linux/ubuntu/gpg | \
    sudo gpg --dearmor -o /etc/apt/keyrings/docker.gpg
hisyam@HISYAMpc:~$ sudo chmod a+r /etc/apt/keyrings/docker.gpg
```

Tambahkan repository Docker

```
hisyam@HISYAMpc:~$ echo \
    "deb [arch=$(dpkg --print-architecture) signed-by=/etc/apt/keyrings/docker.gpg] \
    https://download.docker.com/linux/ubuntu \
    $(lsb_release -cs) stable" | \
    sudo tee /etc/apt/sources.list.d/docker.list > /dev/null
```

Menambahkan GPG key resmi Docker agar apt bisa memverifikasi keaslian package Docker, lalu mendaftarkan repository resmi Docker ke sistem.

1.4 Instal Docker Engine

```
hisyam@HISYAMpc:~$ sudo apt update
[sudo] password for hisyam:
Get:1 https://download.docker.com/linux/ubuntu noble InRelease [48.5 kB]
Get:2 https://download.docker.com/linux/ubuntu noble/stable amd64 Packages [52.9 kB]
```

```
hisyam@HISYAMpc:~$ sudo apt install -y \
    docker-ce \
    docker-ce-cli \
    containerd.io \
    docker-buildx-plugin \
    docker-compose-plugin
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  docker-ce-rootless-extras iptables libip4tc2 libip6tc2 libnetfilter-contrack3 libnfnetlink0 libnftables1 libnftnl11
  libslirp0 nftables pigz slirp4netns
Suggested packages:
  cgroupfs-mount | cgroup-lite docker-model-plugin firewalld
The following NEW packages will be installed:
  containerd.io docker-buildx-plugin docker-ce docker-ce-cli docker-ce-rootless-extras docker-compose-plugin iptables
  libip4tc2 libip6tc2 libnetfilter-contrack3 libnfnetlink0 libnftables1 libnftnl11 libslirp0 nftables pigz
  slirp4netns
```

Menginstall komponen Docker Engine yaitu docker-ce (Docker Community Edition), docker-ce-cli (CLI tool), containerd.io (container runtime), docker-buildx-plugin (untuk build multi-platform), dan docker-compose-plugin (untuk menjalankan multi-container).

1.5 Konfigurasi user non-root

Tambahkan user saat ini ke group docker

```
hisyam@HISYAMpc:~$ sudo usermod -aG docker $USER
```

Aktifkan group baru (atau logout/login)

```
hisyam@HISYAMpc:~$ newgrp docker
```

Menambahkan user ke group docker agar bisa menjalankan perintah docker tanpa sudo setiap saat.

1.6 Verifikasi instalasi

Cek versi Docker

```
hisyam@HISYAMpc:~$ docker version
Client: Docker Engine - Community
Version: 29.4.2
API version: 1.54
Go version: go1.26.2
Git commit: 055a478
Built: Fri May 1 10:24:01 2026
OS/Arch: linux/amd64
Context: default

Server: Docker Engine - Community
Engine:
Version: 29.4.2
API version: 1.54 (minimum version 1.40)
Go version: go1.26.2
Git commit: d329809
Built: Fri May 1 10:24:01 2026
OS/Arch: linux/amd64
Experimental: false
containerd:
Version: v2.2.3
GitCommit: 77c84241c7cbdd9b4eca2591793e3d4f4317c590
runc:
Version: 1.3.5
GitCommit: v1.3.5-0-g488fc13e
docker-init:
Version: 0.19.0
GitCommit: de40ad0
```

Cek info Docker Engine

```
hisyam@HISYAMpc:~$ docker info
Client: Docker Engine - Community
Version: 29.4.2
Context: default
Debug Mode: false
Plugins:
  buildx: Docker Buildx (Docker Inc.)
    Version: v0.33.0
    Path: /usr/libexec/docker/cli-plugins/docker-buildx
  compose: Docker Compose (Docker Inc.)
    Version: v5.1.3
    Path: /usr/libexec/docker/cli-plugins/docker-compose
```

Pastikan service berjalan

```
hisyam@HISYAMpc:~$ sudo systemctl status docker
● docker.service - Docker Application Container Engine
   Loaded: loaded (/usr/lib/systemd/system/docker.service; enabled; preset: enabled)
   Active: active (running) since Mon 2026-05-04 13:52:04 WIB; 4min 7s ago
     TriggeredBy: ● docker.socket
    Docs: https://docs.docker.com
   Main PID: 5817 (dockerd)
      Tasks: 14
     Memory: 36.9M (peak: 40.8M)
        CPU: 540ms
    CGroup: /system.slice/docker.service
            └─5817 /usr/bin/dockerd -H fd:// --containerd=/run/containerd/containerd.sock
```

Test container pertama

```
hisyam@HISYAMpc:~$ docker run hello-world

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
 1. The Docker client contacted the Docker daemon.
 2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
    (amd64)
 3. The Docker daemon created a new container from that image which runs the
    executable that produces the output you are currently reading.
 4. The Docker daemon streamed that output to the Docker client, which sent it
    to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
$ docker run -it ubuntu bash

Share images, automate workflows, and more with a free Docker ID:
https://hub.docker.com/

For more examples and ideas, visit:
https://docs.docker.com/get-started/
```

Verifikasi bahwa Docker sudah terinstall dengan benar dan container hello-world berhasil dijalankan.

Langkah 2: Instalasi Docker Desktop di Windows 10/11

Langkah ini hanya sebagai referensi, tidak wajib dikerjakan. Docker Desktop menggunakan WSL2 sebagai backend untuk menjalankan kernel Linux di Windows.

Buka PowerShell sebagai Administrator:

Aktifkan WSL

```
PS C:\WINDOWS\system32> wsl --install
A distribution with the supplied name already exists. Use --name to chose a different name.
Error code: Wsl/InstallDistro/ERROR_ALREADY_EXISTS
PS C:\WINDOWS\system32>
```

Set WSL2 sebagai default

```
PS C:\WINDOWS\system32> wsl --set-default-version 2
For information on key differences with WSL 2 please visit https://aka.ms/wsl2
The operation completed successfully.
```

Verifikasi

```
PS C:\WINDOWS\system32> wsl --list --verbose
NAME                STATE              VERSION
* Ubuntu             Running            2
docker-desktop       Running            2
```

Restart komputer jika diminta.

2.2 Download dan Instal Docker Desktop

Buka <https://www.docker.com/products/docker-desktop/>

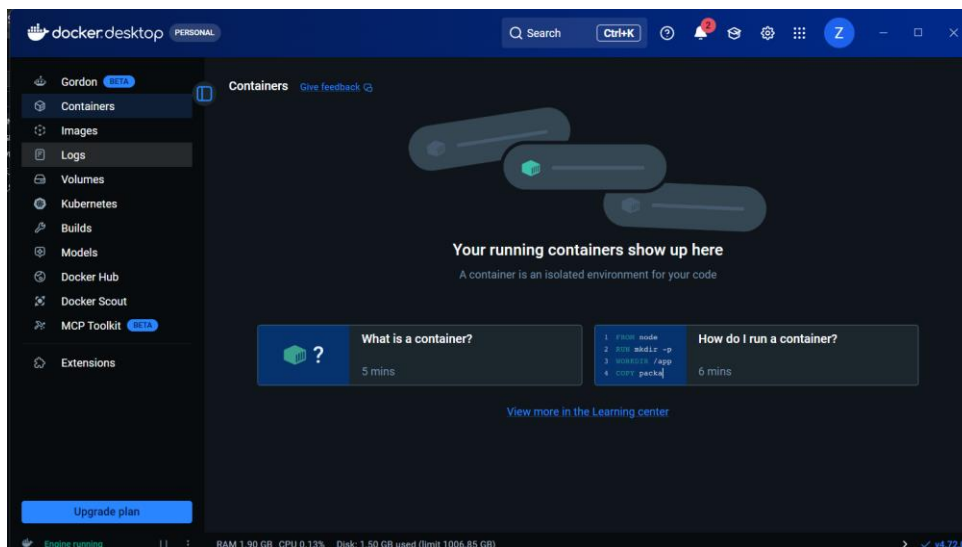
Download Docker Desktop for Windows

Jalankan installer, centang "Use WSL 2 instead of Hyper-V"

Klik Install → tunggu selesai → Restart jika diminta

2.3 Konfigurasi Docker Desktop

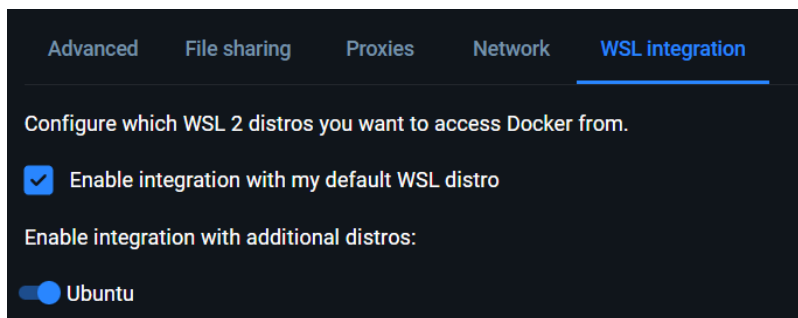
Buka Docker Desktop dari Start Menu



Masuk ke Settings → General → pastikan "Use the WSL 2 based engine" aktif



Masuk ke Settings → Resources → WSL Integration → aktifkan distro yang diinginkan



2.4 Verifikasi dari PowerShell / WSL Terminal

Dari PowerShell

```
PS C:\Users\HISYAM> docker version
Client:
Version:           29.4.2
API version:       1.54
Go version:        go1.26.2
Git commit:        055a478
Built:             Fri May 1 01:27:10 2026
OS/Arch:           windows/amd64
Context:           desktop-linux

Server: Docker Desktop 4.72.0 (225998)
Engine:
Version:           29.4.2
API version:       1.54 (minimum version 1.40)
Go version:        go1.26.2
Git commit:        d329809
Built:             Fri May 1 01:24:36 2026
OS/Arch:           linux/amd64
Experimental:      false
containerd:
Version:           v2.2.3
Git Commit:        7f7c341c7b1d1014e325017023244c43175555
```

```
PS C:\Users\HISYAM> docker run hello-world
Unable to find image 'hello-world:latest' locally
latest: Pulling from library/hello-world
4f55086f7dd0: Pull complete
d5e71e642bf5: Download complete
Digest: sha256:f9078146db2e05e794366b1bfe584a14ea6317f44027d10ef7dad65279026885
Status: Downloaded newer image for hello-world:latest

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
1. The Docker client contacted the Docker daemon.
2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
   (amd64)
3. The Docker daemon created a new container from that image which runs the
   executable that produces the output you are currently reading.
4. The Docker daemon streamed that output to the Docker client, which sent it
   to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
$ docker run -it ubuntu bash

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https://hub.docker.com/

For more examples and ideas, visit:
https://docs.docker.com/get-started/
```

Dari WSL terminal (Ubuntu)

```
hisyam@HISYAMpc:~$ docker version
Client: Docker Engine - Community
Version: 29.4.2
API version: 1.54
Go version: go1.26.2
Git commit: 055a478
Built: Fri May 1 10:24:01 2026
OS/Arch: linux/amd64
Context: default

Server: Docker Desktop 4.72.0 (225998)
Engine:
Version: 29.4.2
API version: 1.54 (minimum version 1.40)
Go version: go1.26.2
Git commit: d329809
Built: Fri May 1 01:24:36 2026
OS/Arch: linux/amd64
Experimental: false
containerd:
Version: v2.2.3
GitCommit: 77c84241c7cbdd9b4eca2591793e3d4f4317c590
```

```
hisyam@HISYAMpc:~$ docker run hello-world

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
1. The Docker client contacted the Docker daemon.
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   (amd64)
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4. The Docker daemon streamed that output to the Docker client, which sent it
   to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
$ docker run -it ubuntu bash

Share images, automate workflows, and more with a free Docker ID:
```

NB : Langkah berikut hanya sebagai referensi selanjutnya tetap menggunakan wsl

3.1 Pull image dari Docker Hub

Pull image nginx versi latest

```
hisyam@HISYAMpc:~$ docker pull nginx
Using default tag: latest
latest: Pulling from library/nginx
85c66128325a: Pull complete
801a1ad15b4e: Pull complete
ff048f1f2159: Pull complete
677c63196868: Pull complete
4677c2a9a3d4: Pull complete
3531af2bc2a9: Pull complete
ce776bbcdad0: Pull complete
54c38c75806e: Download complete
69989ccd189b: Download complete
Digest: sha256:6e23479198b998e5e25921dff8455837c7636a67111a04a635cf1bb363d199dc
Status: Downloaded newer image for nginx:latest
docker.io/library/nginx:latest
```

Pull image nginx versi spesifik

```
hisyam@HISYAMpc:~$ docker pull nginx:1.26
1.26: Pulling from library/nginx
4fd410795c0f: Pull complete
5e98d206134b: Pull complete
d44088bb6ae8: Pull complete
9ebfb40fb06b: Pull complete
8a628cdd7ccc: Pull complete
6923759e66ab: Pull complete
7a0654aeb922: Pull complete
f17adcc09044: Download complete
5c2c3686e536: Download complete
Digest: sha256:41b194461e4bae16f9b25d68b0976ed4735b89ca625c89aad88e1c1c3b7e8860
Status: Downloaded newer image for nginx:1.26
docker.io/library/nginx:1.26
```

Pull image Ubuntu 22.04

```
hisyam@HISYAMpc:~$ docker pull ubuntu:22.04
22.04: Pulling from library/ubuntu
f63eb04151bc: Pull complete
bf0d6867143e: Download complete
Digest: sha256:962f6cadeae0ea6284001009daa4cc9a8c37e75d1f5191cf0eb83fe565b63dd7
Status: Downloaded newer image for ubuntu:22.04
docker.io/library/ubuntu:22.04
```

Pull image Alpine (sangat ringan, ~7MB)

```
hisyam@HISYAMpc:~$ docker pull alpine:3.20
3.20: Pulling from library/alpine
25f1d6b1951a: Pull complete
3a030ca3f633: Download complete
d4050d56ebf2: Download complete
Digest: sha256:d9e853e87e55526f6b2917df91a2115c36dd7c696a35be12163d44e6e2a4b6bc
Status: Downloaded newer image for alpine:3.20
docker.io/library/alpine:3.20
```

Pull image — docker pull nginx mendownload image nginx dari Docker Hub ke lokal. Tag spesifik seperti nginx:1.26 memastikan versi yang digunakan selalu sama.

3.2 Manajemen image

List semua image lokal

```
hisyam@HISYAMpc:~$ docker images
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
alpine:3.20	d9e853e87e55	12.2MB	3.71MB	
hello-world:latest	f9078146db2e	25.9kB	9.49kB	U
nginx:1.26	41b194461e4b	282MB	75.2MB	
nginx:latest	6e23479198b9	240MB	65.8MB	
ubuntu:22.04	962f6cadeae0	119MB	31.7MB	

docker images - menampilkan semua image yang tersimpan di lokal beserta ukuran dan tag-nya

List dengan format custom

```
hisyam@HISYAMpc:~$ docker images --format "table {{.Repository}}\t{{.Tag}}\t{{.Size}}"
REPOSITORY      TAG          SIZE
nginx            latest       240MB
alpine           3.20        12.2MB
ubuntu           22.04       119MB
hello-world      latest       25.9kB
nginx            1.26        282MB
```

Inspect detail image

```
hisyam@HISYAMpc:~$ docker image inspect nginx
[
  {
    "Id": "sha256:6e23479198b998e5e25921dff8455837c7636a67111a04a635cf1bb363d199dc",
    "RepoTags": [
      "nginx:latest"
    ],
    "RepoDigests": [
      "nginx@sha256:6e23479198b998e5e25921dff8455837c7636a67111a04a635cf1bb363d199dc"
    ],
    "Comment": "buildkit.dockerfile.v0",
    "Created": "2026-04-22T01:22:07.750510155Z",
    "Config": {
      "ExposedPorts": {
        "80/tcp": {}
      },
      "Env": [
        "PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin",
        "NGINX_VERSION=1.29.8",
        "NJS_VERSION=0.9.6",
        "NJS_RELEASE=1~trixie",
        "ACME_VERSION=0.3.1",
        "PKG_RELEASE=1~trixie",
        "DYNPKG_RELEASE=1~trixie"
      ]
    }
  }
]
```

docker image inspect nginx — menampilkan detail lengkap image seperti layer, environment variable, dan konfigurasi

Lihat history layer sebuah image

```
hisyam@HISYAMpc:~$ docker image history nginx
IMAGE          CREATED          CREATED BY                                      SIZE      COMMENT
6e23479198b9   2 weeks ago     CMD ["nginx" "-g" "daemon off;"]              0B        buildkit.dockerfile.v0
<missing>      2 weeks ago     STOPIGNAL SIGQUIT                             0B        buildkit.dockerfile.v0
<missing>      2 weeks ago     EXPOSE map[80/tcp:{}]                         0B        buildkit.dockerfile.v0
<missing>      2 weeks ago     ENTRYPOINT ["/docker-entrypoint.sh"]          0B        buildkit.dockerfile.v0
<missing>      2 weeks ago     COPY 30-tune-worker-processes.sh /docker-ent... 16.4kB    buildkit.dockerfile.v0
<missing>      2 weeks ago     COPY 20-envsubst-on-templates.sh /docker-ent... 12.3kB    buildkit.dockerfile.v0
<missing>      2 weeks ago     COPY 15-local-resolvers.envsh /docker-entryp... 12.3kB    buildkit.dockerfile.v0
<missing>      2 weeks ago     COPY 10-listen-on-ipv6-by-default.sh /docker... 12.3kB    buildkit.dockerfile.v0
<missing>      2 weeks ago     COPY docker-entrypoint.sh / # buildkit        8.19kB    buildkit.dockerfile.v0
<missing>      2 weeks ago     RUN /bin/sh -c set -x && groupadd --syst...    86.7MB    buildkit.dockerfile.v0
<missing>      2 weeks ago     ENV DYNPKG_RELEASE=1~trixie                   0B        buildkit.dockerfile.v0
```

docker image history nginx — menampilkan riwayat setiap layer yang membentuk image tersebut

Hapus image

```
hisyam@HISYAMpc:~$ docker rmi alpine:3.20
Untagged: alpine:3.20
Deleted: sha256:d9e853e87e55526f6b2917df91a2115c36dd7c696a35be12163d44e6e2a4b6bc
```

docker rmi alpine:3.20 — menghapus image dari lokal

Hapus semua image yang tidak digunakan

```
hisyam@HISYAMpc:~$ docker image prune -a
WARNING! This will remove all images without at least one container associated to t
Are you sure you want to continue? [y/N] y
Deleted Images:
untagged: nginx:latest
deleted: sha256:6e23479198b998e5e25921dff8455837c7636a67111a04a635cf1bb363d199dc
deleted: sha256:296499281873ebe2636681c6957776d29a31858d93c3e39b654b5aade87ed70a
deleted: sha256:422a5eaa9cd7814933c1fe4171d9675bf78ddff2afc18b4f175a86def04949d3
untagged: ubuntu:22.04
deleted: sha256:962f6cadeae0ea6284001009daa4cc9a8c37e75d1f5191cf0eb83fe565b63dd7
deleted: sha256:14be402d3f1eeeb5e7da73d3260322c68e7b51c88388f53e88eb21d6450bd520
deleted: sha256:268e076660f3cd225015df43cfe7198e5f787f24db00338d433299687237e538
untagged: nginx:1.26
deleted: sha256:41b194461e4bae16f9b25d68b0976ed4735b89ca625c89aad88e1c1c3b7e8860
deleted: sha256:c8042489f41ddaf05b7dfbe1e32d5227f94f1b002aca8e3b7b9cb9d0cee6b0fed
deleted: sha256:c28381c5787d56e9bf825ecd4b8b1f1a64caec040b959a54674ac5a43954cfd3
Total reclaimed space: 172.7MB
```

docker image prune -a — membersihkan semua image yang tidak dipakai
container manapun

Langkah 4: Menjalankan dan Mengelola Container

4.1 Menjalankan container dasar

Jalankan container nginx (foreground — tekan Ctrl+C untuk stop)

```
hisyam@HISYAMpc:~$ docker run nginx
Unable to find image 'nginx:latest' locally
latest: Pulling from library/nginx
801a1ad15b4e: Pull complete
677c63196868: Pull complete
85c66128325a: Pull complete
3531af2bc2a9: Pull complete
ce776bbcdad: Pull complete
ff048f1f2159: Pull complete
4677c2a9a3d4: Pull complete
54c38c75806e: Download complete
69989ccd189b: Download complete
Digest: sha256:6e23479198b998e5e25921dff8455837c7636a67111a04a635cf1bb363d199dc
Status: Downloaded newer image for nginx:latest
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
/docker-entrypoint.sh: Configuration complete; ready for start up
2026/05/08 17:33:21 [notice] 1#1: using the "epoll" event method
```


docker run nginx — menjalankan container di foreground, terminal akan ter-block

Jalankan nginx di background (detached mode)

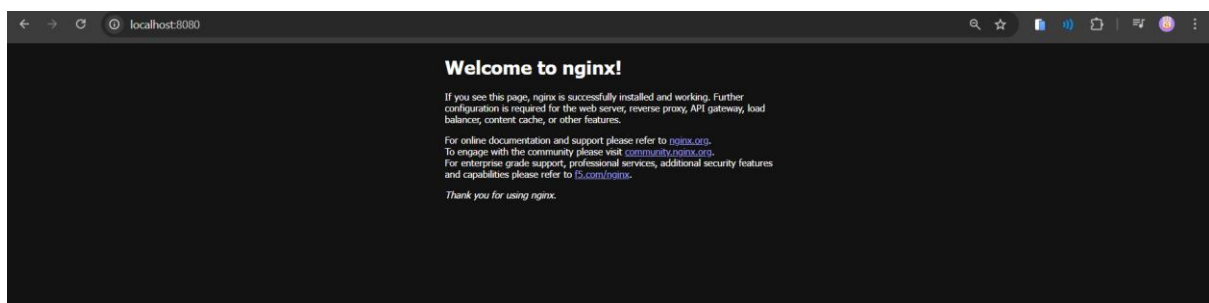
```
hisyam@HISYAMpc:~$ docker run -d --name web-server nginx  
8d9debddfb1bf9d38dce62710a00b7a25a64b816d7b13d0458c5d58d54d6f0ec
```

docker run -d --name web-server nginx — flag -d menjalankan container di background (detached), --name memberi nama container

Jalankan nginx dengan port mapping (host:container)

```
hisyam@HISYAMpc:~$ docker run -d --name web-public -p 8080:80 nginx  
4e32916b2d05496357cc44d3e974a23749141a077450dcc630f454c4e18635e4
```

Buka browser ke <http://localhost:8080> → akan tampil halaman default Nginx.



docker run -d -p 8080:80 nginx — flag -p 8080:80 memetakan port 8080 di host ke port 80 di container, sehingga bisa diakses via browser

4.2 Container interaktif

Jalankan Ubuntu container dengan bash interaktif

```
hisyam@HISYAMpc:~$ docker run -it --name ubuntu-test ubuntu:22.04 /bin/bash  
Unable to find image 'ubuntu:22.04' locally  
22.04: Pulling from library/ubuntu  
f63eb04151bc: Pull complete  
bf0d6867143e: Download complete  
Digest: sha256:962f6cadeae0ea6284001009daa4cc9a8c37e75d1f5191cf0eb83fe565b63dd7  
Status: Downloaded newer image for ubuntu:22.04
```

docker run -it ubuntu:22.04 /bin/bash — flag -i (interactive) dan -t (pseudo-TTY) memungkinkan kita masuk ke dalam shell container dan menjalankan perintah seperti di Linux biasa

Di dalam container:

```
root@1e625beec857:/# cat /etc/os-release
PRETTY_NAME="Ubuntu 22.04.5 LTS"
NAME="Ubuntu"
VERSION_ID="22.04"
VERSION="22.04.5 LTS (Jammy Jellyfish)"
VERSION_CODENAME=jammy
ID=ubuntu
ID_LIKE=debian
HOME_URL="https://www.ubuntu.com/"
SUPPORT_URL="https://help.ubuntu.com/"
BUG_REPORT_URL="https://bugs.launchpad.net/ubuntu/"
PRIVACY_POLICY_URL="https://www.ubuntu.com/legal/terms-and-policies/privacy-policy"
UBUNTU_CODENAME=jammy
```

```
root@1e625beec857:/# apt update && apt install -y curl
Hit:1 http://security.ubuntu.com/ubuntu jammy-security InRelease
Hit:2 http://archive.ubuntu.com/ubuntu jammy InRelease
Hit:3 http://archive.ubuntu.com/ubuntu jammy-updates InRelease
Hit:4 http://archive.ubuntu.com/ubuntu jammy-backports InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
2 packages can be upgraded. Run 'apt list --upgradable' to see them.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
curl is already the newest version (7.81.0-1ubuntu1.24).
0 upgraded, 0 newly installed, 0 to remove and 2 not upgraded.
```

```
root@1e625beec857:/# apt update && apt install -y curl
Get:1 http://archive.ubuntu.com/ubuntu jammy InRelease [270 kB]
Get:2 http://security.ubuntu.com/ubuntu jammy-security InRelease [129 kB]
Get:3 http://security.ubuntu.com/ubuntu jammy-security/universe amd64 Packages [1294 kB]
Get:4 http://archive.ubuntu.com/ubuntu jammy-updates InRelease [128 kB]
Get:5 http://archive.ubuntu.com/ubuntu jammy-backports InRelease [127 kB]
Get:6 http://archive.ubuntu.com/ubuntu jammy/main amd64 Packages [1792 kB]
Get:7 http://archive.ubuntu.com/ubuntu jammy/restricted amd64 Packages [164 kB]
Get:8 http://archive.ubuntu.com/ubuntu jammy/multiverse amd64 Packages [266 kB]
Get:9 http://archive.ubuntu.com/ubuntu jammy/universe amd64 Packages [17.5 MB]
Get:10 http://security.ubuntu.com/ubuntu jammy-security/main amd64 Packages [3915 kB]
Get:11 http://archive.ubuntu.com/ubuntu jammy-updates/restricted amd64 Packages [7251 kB]
```

```

curl: (6) Could not resolve host: web-server
root@dea49d1bfff7f:/# curl http://172.17.0.2
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, nginx is successfully installed and working.
Further configuration is required for the web server, reverse proxy,
API gateway, load balancer, content cache, or other features.</p>

<p>For online documentation and support please refer to
<a href="https://nginx.org/">nginx.org</a>.<br/>
To engage with the community please visit
<a href="https://community.nginx.org/">community.nginx.org</a>.<br/>
For enterprise grade support, professional services, additional
security features and capabilities please refer to
<a href="https://f5.com/nginx">f5.com/nginx</a>.</p>

<p><em>Thank you for using nginx.</em></p>
</body>
</html>

```

akses container lain (jika di network sama)

```

root@dea49d1bfff7f:/# exit
exit

```

keluar (container akan stop)

4.3 Monitoring container

List container yang sedang berjalan

```

root@kali:~# docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS        PORTS
10fc88e80d6d   nginx    "/docker-entrypoint..." 2 minutes ago Up 2 minutes  80/tcp
4e32916b2d05   nginx    "/docker-entrypoint..." 13 minutes ago Up 13 minutes  0.0.0.0:8080->80/tcp, [::]:8080->80/t
cp   web-public

```

docker ps — menampilkan container yang sedang berjalan

List SEMUA container (termasuk stopped)

```

hisyam@HISYAMpc:~$ docker ps -a

```

CONTAINER ID	IMAGE	NAMES	COMMAND	CREATED	STATUS	PORTS
dea49d1bfff7f	ubuntu:22.04	ubuntu-test	"/bin/bash"	2 minutes ago	Exited (0) 44 seconds ago	
10fc88e80d6d	nginx	web-server	"/docker-entrypoint..."	3 minutes ago	Up 3 minutes	80/tcp
738a0b4600c7	nginx	confident_spence	"/docker-entrypoint..."	4 minutes ago	Exited (0) 4 minutes ago	
4e32916b2d05	nginx	web-public	"/docker-entrypoint..."	14 minutes ago	Up 14 minutes	0.0.0.0:8080->80/tcp
b134eb63bd52	nginx	gifted_goodall	"/docker-entrypoint..."	15 minutes ago	Exited (0) 14 minutes ago	
71811df6ff8a	hello-world	loving_fermi	"hello"	23 minutes ago	Exited (0) 23 minutes ago	
fa71cc3e56ac	hello-world	affectionate_kilby	"hello"	24 minutes ago	Exited (0) 24 minutes ago	

docker ps -a — menampilkan semua container termasuk yang sudah stopped

Lihat log container

docker logs web-server

```

hisyam@HISYAMpc:~$ docker logs web-server
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
/docker-entrypoint.sh: Configuration complete; ready for start up
2026/05/08 17:45:29 [notice] 1#1: using the "epoll" event method
2026/05/08 17:45:29 [notice] 1#1: nginx/1.29.8
2026/05/08 17:45:29 [notice] 1#1: built by gcc 14.2.0 (Debian 14.2.0-19)
2026/05/08 17:45:29 [notice] 1#1: OS: Linux 6.6.87.2-microsoft-standard-WSL2
2026/05/08 17:45:29 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1048576:1048576
2026/05/08 17:45:29 [notice] 1#1: start worker processes
2026/05/08 17:45:29 [notice] 1#1: start worker process 29
2026/05/08 17:45:29 [notice] 1#1: start worker process 30
2026/05/08 17:45:29 [notice] 1#1: start worker process 31

```

docker logs web-server — menampilkan log output container

```

hisyam@HISYAMpc:~$ docker logs -f web-server # follow (real-time)
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
/docker-entrypoint.sh: Configuration complete; ready for start up
2026/05/08 17:45:29 [notice] 1#1: using the "epoll" event method
2026/05/08 17:45:29 [notice] 1#1: nginx/1.29.8
2026/05/08 17:45:29 [notice] 1#1: built by gcc 14.2.0 (Debian 14.2.0-19)
2026/05/08 17:45:29 [notice] 1#1: OS: Linux 6.6.87.2-microsoft-standard-WSL2
2026/05/08 17:45:29 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1048576:1048576
2026/05/08 17:45:29 [notice] 1#1: start worker processes
2026/05/08 17:45:29 [notice] 1#1: start worker process 29
2026/05/08 17:45:29 [notice] 1#1: start worker process 30
2026/05/08 17:45:29 [notice] 1#1: start worker process 31
2026/05/08 17:45:29 [notice] 1#1: start worker process 32
2026/05/08 17:45:29 [notice] 1#1: start worker process 33

```

docker logs -f web-server # follow (real-time)

docker logs --tail 20 web-server # 20 baris terakhir

```
^Chisyam@HISYAMpc:~$ docker logs --tail 20 web-server # 20 baris terakhir
2026/05/08 17:45:29 [notice] 1#1: OS: Linux 6.6.87.2-microsoft-standard-WSL2
2026/05/08 17:45:29 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1048576:1048576
2026/05/08 17:45:29 [notice] 1#1: start worker processes
2026/05/08 17:45:29 [notice] 1#1: start worker process 29
2026/05/08 17:45:29 [notice] 1#1: start worker process 30
2026/05/08 17:45:29 [notice] 1#1: start worker process 31
2026/05/08 17:45:29 [notice] 1#1: start worker process 32
2026/05/08 17:45:29 [notice] 1#1: start worker process 33
2026/05/08 17:45:29 [notice] 1#1: start worker process 34
2026/05/08 17:45:29 [notice] 1#1: start worker process 35
2026/05/08 17:45:29 [notice] 1#1: start worker process 36
2026/05/08 17:45:29 [notice] 1#1: start worker process 37
2026/05/08 17:45:29 [notice] 1#1: start worker process 38
2026/05/08 17:45:29 [notice] 1#1: start worker process 39
2026/05/08 17:45:29 [notice] 1#1: start worker process 40
2026/05/08 17:45:29 [notice] 1#1: start worker process 41
2026/05/08 17:45:29 [notice] 1#1: start worker process 42
2026/05/08 17:45:29 [notice] 1#1: start worker process 43
2026/05/08 17:45:29 [notice] 1#1: start worker process 44
172.17.0.4 - - [08/May/2026:17:47:16 +0000] "GET / HTTP/1.1" 200 896 "-" "curl/7.81.0" "-"
```

Lihat resource usage (CPU, Memory, I/O)

docker stats

CONTAINER ID	NAME	CPU %	MEM USAGE / LIMIT	MEM %	NET I/O	BLOCK I/O	PIDS
10fc88e80d6d	web-server	0.00%	13.26MiB / 7.664GiB	0.17%	1.67kB / 1.68kB	4.03MB / 12.3kB	17
4e32916b2d05	web-public	0.00%	13.33MiB / 7.664GiB	0.17%	4.17kB / 3.23kB	12.3kB / 12.3kB	17

menampilkan penggunaan CPU, memory, dan I/O secara real-time

Lihat detail container

docker inspect web-server

```
^Chisyam@HISYAMpc:~$ docker inspect web-server
[
  {
    "Id": "10fc88e80d6d2ba041bd16e1137826d848e7cdf0da733756a0c580a1f52d88bc",
    "Created": "2026-05-08T17:45:29.589363351Z",
    "Path": "/docker-entrypoint.sh",
    "Args": [
      "nginx",
      "-g",
      "daemon off;"
    ],
    "State": {
      "Status": "running",
      "Running": true,
      "Paused": false,
      "Restarting": false,
      "OOMKilled": false,
      "Dead": false,
      "Pid": 4552,
      "ExitCode": 0,
      "Error": "",
      "StartedAt": "2026-05-08T17:45:29.687416453Z",
      "FinishedAt": "0001-01-01T00:00:00Z"
    }
  }
]
```

docker inspect — menampilkan detail lengkap container termasuk IP address dan network

4.4 Interaksi dengan container berjalan

Buat file index.html

```
hisyam@HISYAMpc:~$ echo "<h1>Test</h1>" > index.html
```

Eksekusi perintah di container yang sedang berjalan

docker exec web-server cat /etc/nginx/nginx.conf

```
hisyam@HISYAMpc:~$ docker exec web-server cat /etc/nginx/nginx.conf

user  nginx;
worker_processes  auto;

error_log  /var/log/nginx/error.log notice;
pid        /run/nginx.pid;


events {
    worker_connections  1024;
}


http {
    include       /etc/nginx/mime.types;
    default_type  application/octet-stream;

    log_format main '$remote_addr - $remote_user [$time_local] "$request" '
                    '$status $body_bytes_sent "$http_referer" ';
```

docker exec web-server cat /etc/nginx/nginx.conf - menjalankan perintah di dalam container yang sedang berjalan tanpa masuk ke shell-nya

Masuk ke shell container yang sedang berjalan

docker exec -it web-server /bin/bash

```
hisyam@HISYAMpc:~$ docker exec -it web-server /bin/bash
root@10fc88e80d6d:/#
```

docker exec -it web-server /bin/bash — masuk ke shell container yang sedang berjalan

Copy file dari host ke container

docker cp index.html web-server:/usr/share/nginx/html/index.html

```
root@10fc88e80d6d:/# exit
exit
hisyam@HISYAMpc:~$ docker cp index.html web-server:/usr/share/nginx/html/index.html
Successfully copied 14B (transferred 2.05kB) to web-server:/usr/share/nginx/html/index.html
```

Copy file dari container ke host

docker cp web-server:/etc/nginx/nginx.conf ./nginx.conf

```
hisyam@HISYAMpc:~$ docker cp web-server:/etc/nginx/nginx.conf ./nginx.conf
Successfully copied 644B (transferred 2.56kB) to /home/hisyam/nginx.conf
```

menyalin file antara host dan container, berguna untuk mengubah konten tanpa rebuild image

4.5 Lifecycle management

Stop container

docker stop web-server

```
hisyam@HISYAMpc:~$ docker stop web-server
web-server
```

docker stop — mengirim sinyal SIGTERM ke container, memberi waktu proses untuk berhenti dengan baik

Start container yang sudah di-stop

docker start web-server

```
hisyam@HISYAMpc:~$ docker start web-server
web-server
```

docker start — menjalankan ulang container yang sudah stopped

Restart container

docker restart web-server

```
hisyam@HISYAMpc:~$ docker restart web-server
web-server
```

Kill container (force stop — SIGKILL)

docker kill web-server

```
hisyam@HISYAMpc:~$ docker kill web-server
web-server
```

- docker kill — mengirim SIGKILL langsung, container berhenti paksa

Hapus container (harus dalam keadaan stopped)

docker stop web-server && docker rm web-server

```
hisyam@HISYAMpc:~$ docker stop web-server && docker rm web-server
web-server
web-server
```

Hapus container secara paksa (meskipun masih running)

docker rm -f web-server

```
hisyam@HISYAMpc:~$ docker rm -f web-server
web-server
```

- docker rm -f — menghapus container secara paksa meski masih running

Hapus SEMUA stopped container

docker container prune

```
hisyam@HISYAMpc:~$ docker container prune
WARNING! This will remove all stopped containers.
Are you sure you want to continue? [y/N] y
Deleted Containers:
dea49d1bfff7fb4c8ef958a43608038d048727d63150cadd4675826353bcce83d
738a0b4600c7a34386d4b4a082a8f4921f168999e6a0de23d3426a2e236542d8
b134eb63bd52938af439680d0a7237cca5a88ce28b9186802f276c3150785605
71811df6ff8a46ead327f6af4218077d2460b7b8d7dd34486e96899daa896705
fa71cc3e56ac3f242c7ab63cf43a75b1822719f057cf3750581829597918a81
```

- docker container prune — menghapus semua container yang stopped sekaligus

Langkah 5: Membuat Custom Image dengan Dockerfile

5.1 Buat project directory

`mkdir -p ~/docker-lab/custom-web && cd ~/docker-lab/custom-web`

```
hisyam@HISYAMpc:~$ mkdir -p ~/docker-lab/custom-web && cd ~/docker-lab/custom-web
```

5.2 Buat halaman web

```
hisyam@HISYAMpc:~/docker-lab/custom-web$ cat > index.html << 'EOF'
<!DOCTYPE html>
<html lang="id">
<head>
  <meta charset="UTF-8">
  <title>Docker Lab - PENS</title>
  <style>
    body { font-family: Arial, sans-serif; text-align: center; padding: 50px;
          background: linear-gradient(135deg, #667eea 0%, #764ba2 100%);
          color: white; }
    .container { background: rgba(255,255,255,0.1); border-radius: 15px;
                padding: 40px; max-width: 600px; margin: 0 auto; }
    h1 { font-size: 2.5em; }
    .info { background: rgba(0,0,0,0.2); padding: 15px; border-radius: 8px;
          margin-top: 20px; text-align: left; }
  </style>
</head>
<body>
```

5.3 Buat Dockerfile

```
hisyam@HISYAMpc:~/docker-lab/custom-web$ cat > Dockerfile << 'EOF'
> FROM nginx:1.26-alpine
> # Metadata
LABEL maintainer="admin@pens.ac.id"
LABEL description="Custom Nginx untuk praktikum Docker PENS"
LABEL version="1.0"
> # Hapus halaman default dan ganti dengan halaman custom
RUN rm -rf /usr/share/nginx/html/*
COPY index.html /usr/share/nginx/html/index.html
> # Expose port 80
EXPOSE 80
> # Command default (inherited dari base image, tapi kita tulis eksplisit)
CMD ["nginx", "-g", "daemon off;"]
> EOF
```

5.1-5.3 Membuat project directory, file index.html sebagai halaman web custom, dan Dockerfile yang berisi instruksi untuk membangun image.

Penjelasan isi Dockerfile:

- FROM nginx:1.26-alpine — menggunakan base image nginx versi alpine yang ringan (~12MB)

- LABEL — metadata informasi image (tidak mempengaruhi fungsi)
- RUN `rm -rf /usr/share/nginx/html/*` — menghapus halaman default nginx
- COPY `index.html ...` — menyalin halaman custom ke dalam image
- EXPOSE 80 — dokumentasi bahwa container menggunakan port 80
- CMD — perintah yang dijalankan saat container start

5.4 Build dan jalankan

Build image (titik di akhir = build context = direktori saat ini)

`docker build -t pens-web:1.0 .`

```
hisyam@HISYAMpc:~/docker-lab/custom-web$ docker build -t pens-web:1.0 .
[+] Building 12.8s (9/9) FINISHED                                docker:default
=> [internal] load build definition from Dockerfile              0.0s
=> => transferring dockerfile: 466B                               0.0s
=> [internal] load metadata for docker.io/library/nginx:1.26-alpine 4.2s
=> [auth] library/nginx:pull token for registry-1.docker.io      0.0s
=> [internal] load .dockerignore                                  0.0s
=> => transferring context: 2B                                       0.0s
=> [1/3] FROM docker.io/library/nginx:1.26-alpine@sha256:1eadbb07820339e8bbfed18c771691970baee292ec4ab2558f1453d 7.5s
=> => resolve docker.io/library/nginx:1.26-alpine@sha256:1eadbb07820339e8bbfed18c771691970baee292ec4ab2558f1453d 0.0s
=> => sha256:5ab072e0bc4f67351f9c935c8834e3e1bcac0e013a124ef309607d6210f218d 15.19MB / 15.19MB 6.9s
=> => sha256:c35c2aa19c882e9ca054353d84a4b58904f72807055c441be84f27401714f25d 1.21kB / 1.21kB 1.1s
=> => sha256:a7c661bcc4ef10a5e7ee3b4f80a8d77754a4e6b1a598271b70bd220d804316eb 1.40kB / 1.40kB 1.4s
=> => sha256:d987ab110a5ba4aebf9df7115759f01e8a5d8be0a70045f4dc7a5b959be1b04c 404B / 404B 2.0s
=> => sha256:b8651333baae33f196ce13e6d5cfc16e276b640bbca95329625fbb6a3f13855 956B / 956B 0.8s
=> => sha256:5e18ef93c435d9cd1c48d1ae900cf7e92ee61ae11be6b3bdefc899f4be8efad9 628B / 628B 1.4s
=> => sha256:914d0eaf60479e4b17101b02ed1e82400c02231d11c32276fb24867f5f39e97b 1.76MB / 1.76MB 1.1s
=> => sha256:0a9a5dfd008f05ebc27e4790db0709a29e527690c21bcbcd01481eae66bb49dc 3.63MB / 3.63MB 0.9s
=> => extracting sha256:0a9a5dfd008f05ebc27e4790db0709a29e527690c21bcbcd01481eae66bb49dc 0.2s
=> => extracting sha256:914d0eaf60479e4b17101b02ed1e82400c02231d11c32276fb24867f5f39e97b 0.2s
=> => extracting sha256:5e18ef93c435d9cd1c48d1ae900cf7e92ee61ae11be6b3bdefc899f4be8efad9 0.0s
=> => extracting sha256:b8651333baae33f196ce13e6d5cfc16e276b640bbca95329625fbb6a3f13855 0.0s
=> => extracting sha256:d987ab110a5ba4aebf9df7115759f01e8a5d8be0a70045f4dc7a5b959be1b04c 0.0s
=> => extracting sha256:c35c2aa19c882e9ca054353d84a4b58904f72807055c441be84f27401714f25d 0.0s
=> => extracting sha256:a7c661bcc4ef10a5e7ee3b4f80a8d77754a4e6b1a598271b70bd220d804316eb 0.0s
=> => extracting sha256:5ab072e0bc4f67351f9c935c8834e3e1bcac0e013a124ef309607d6210f218d 0.4s
```

Verifikasi image berhasil dibuat

`docker images | grep pens-web`

```
hisyam@HISYAMpc:~/docker-lab/custom-web$ docker images | grep pens-web
WARNING: This output is designed for human readability. For machine-readable output, please use --format.
pens-web:1.0      bed6315bdc13      72.2MB      20.6MB
```

Jalankan container dari image custom

`docker run -d --name pens-app -p 9090:80 pens-web:1.0`

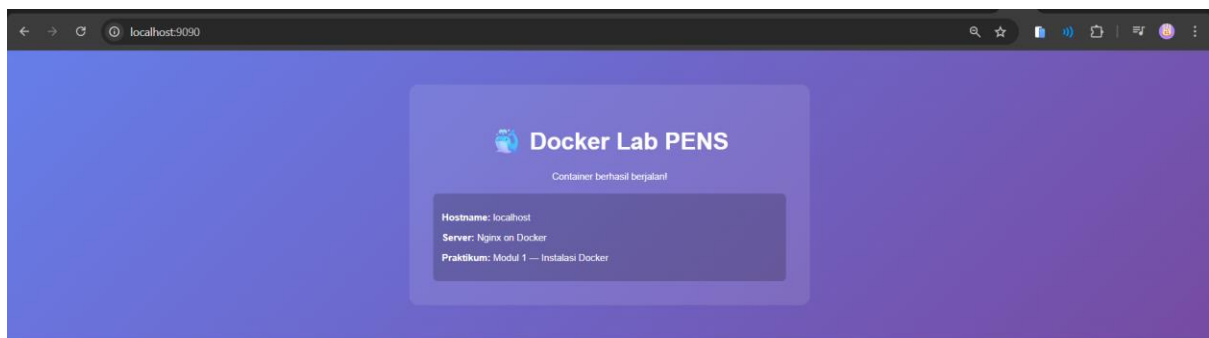
```
hisyam@HISYAMpc:~/docker-lab/custom-web$ docker run -d --name pens-app -p 9090:80 pens-web:1.0
04a5b140032966e442121c3b23989ca5cbf7d7dfcc75b2b1d5d42e316f5d56b3
```

Test

curl <http://localhost:9090>

```
hisyam@HISYAMpc:~/docker-lab/custom-web$ curl http://localhost:9090
<!DOCTYPE html>
<html lang="id">
<head>
  <meta charset="UTF-8">
  <title>Docker Lab - PENS</title>
  <style>
    body { font-family: Arial, sans-serif; text-align: center; padding: 50px;
          background: linear-gradient(135deg, #667eea 0%, #764ba2 100%);
          color: white; }
    .container { background: rgba(255,255,255,0.1); border-radius: 15px;
                padding: 40px; max-width: 600px; margin: 0 auto; }
    h1 { font-size: 2.5em; }
    .info { background: rgba(0,0,0,0.2); padding: 15px; border-radius: 8px;
          margin-top: 20px; text-align: left; }
  </style>
</head>
<body>
  <div class="container">
```

Buka browser ke <http://localhost:9090>.



`docker build -t pens-web:1.0 .` — membangun image dari Dockerfile. Titik . menunjukkan build context adalah direktori saat ini. Setiap instruksi di Dockerfile menghasilkan satu layer baru.

5.5 Lihat layer image

Bandingkan layer antara base image dan custom image

`docker image history nginx:1.26-alpine`

```
hisyam@HISYAMpc:~/docker-lab/custom-web$ docker image history nginx:1.26-alpine
IMAGE          CREATED          CREATED BY                                      SIZE      COMMENT
leadbb078203   15 months ago   RUN /bin/sh -c set -x      && apkArch="$(cat ... 37.7MB    buildkit.dockerfile.v0
<missing>      15 months ago   ENV NJS_RELEASE=1          0B        buildkit.dockerfile.v0
<missing>      15 months ago   ENV NJS_VERSION=0.8.9      0B        buildkit.dockerfile.v0
<missing>      15 months ago   CMD ["nginx" "-g" "daemon off;"] 0B        buildkit.dockerfile.v0
<missing>      15 months ago   STOPSIGNAL SIGQUIT         0B        buildkit.dockerfile.v0
<missing>      15 months ago   EXPOSE map[80/tcp:{}]      0B        buildkit.dockerfile.v0
<missing>      15 months ago   ENTRYPOINT ["/docker-entrypoint.sh"] 0B        buildkit.dockerfile.v0
<missing>      15 months ago   COPY 30-tune-worker-processes.sh /docker-ent... 16.4kB    buildkit.dockerfile.v0
<missing>      15 months ago   COPY 20-envsubst-on-templates.sh /docker-ent... 12.3kB    buildkit.dockerfile.v0
<missing>      15 months ago   COPY 15-local-resolvers.envsh /docker-entryp... 12.3kB    buildkit.dockerfile.v0
<missing>      15 months ago   COPY 10-listen-on-ipv6-by-default.sh /docker... 12.3kB    buildkit.dockerfile.v0
<missing>      15 months ago   COPY docker-entrypoint.sh / # buildkit      8.19kB    buildkit.dockerfile.v0
<missing>      15 months ago   RUN /bin/sh -c set -x      && addgroup -g 101... 5.28MB    buildkit.dockerfile.v0
<missing>      15 months ago   ENV DYNPKG_RELEASE=2       0B        buildkit.dockerfile.v0
```

docker image history pens-web:1.0

```
hisyam@HISYAMpc:~/docker-lab/custom-web$ docker image history pens-web:1.0
IMAGE          CREATED          CREATED BY                                      SIZE      COMMENT
bed6315bdc13   3 minutes ago   CMD ["nginx" "-g" "daemon off;"] 0B        buildkit.dockerfile.v0
<missing>      3 minutes ago   EXPOSE [80/tcp]           0B        buildkit.dockerfile.v0
<missing>      3 minutes ago   COPY index.html /usr/share/nginx/html/index... 24.6kB    buildkit.dockerfile.v0
<missing>      3 minutes ago   RUN /bin/sh -c set -x      && apkArch="$(cat ... 20.5kB    buildkit.dockerfile.v0
<missing>      3 minutes ago   LABEL version=1.0         0B        buildkit.dockerfile.v0
<missing>      3 minutes ago   LABEL description=Custom Nginx untuk praktik... 0B        buildkit.dockerfile.v0
<missing>      3 minutes ago   LABEL maintainer=admin@pens.ac.id 0B        buildkit.dockerfile.v0
<missing>      15 months ago   RUN /bin/sh -c set -x      && apkArch="$(cat ... 37.7MB    buildkit.dockerfile.v0
<missing>      15 months ago   ENV NJS_RELEASE=1          0B        buildkit.dockerfile.v0
<missing>      15 months ago   ENV NJS_VERSION=0.8.9      0B        buildkit.dockerfile.v0
<missing>      15 months ago   CMD ["nginx" "-g" "daemon off;"] 0B        buildkit.dockerfile.v0
<missing>      15 months ago   STOPSIGNAL SIGQUIT         0B        buildkit.dockerfile.v0
<missing>      15 months ago   EXPOSE map[80/tcp:{}]      0B        buildkit.dockerfile.v0
<missing>      15 months ago   ENTRYPOINT ["/docker-entrypoint.sh"] 0B        buildkit.dockerfile.v0
<missing>      15 months ago   COPY 30-tune-worker-processes.sh /docker-ent... 16.4kB    buildkit.dockerfile.v0
<missing>      15 months ago   COPY 20-envsubst-on-templates.sh /docker-ent... 12.3kB    buildkit.dockerfile.v0
<missing>      15 months ago   COPY 15-local-resolvers.envsh /docker-entryp... 12.3kB    buildkit.dockerfile.v0
<missing>      15 months ago   COPY 10-listen-on-ipv6-by-default.sh /docker... 12.3kB    buildkit.dockerfile.v0
<missing>      15 months ago   COPY docker-entrypoint.sh / # buildkit      8.19kB    buildkit.dockerfile.v0
```

Membandingkan layer nginx:1.26-alpine dan pens-web:1.0 - layer dari base image di-share, hanya layer baru yang ditambahkan oleh Dockerfile yang berbeda.

Langkah 6: Docker System Cleanup

Lihat disk usage Docker

docker system df

```
hisyam@HISYAMpc:~/docker-lab/custom-web$ docker system df
TYPE                TOTAL          ACTIVE        SIZE          RECLAIMABLE
Images              5              2            432.9MB       140.9MB (32%)
Containers          2              2            163.8kB       0B (0%)
Local Volumes       0              0             0B            0B
Build Cache        13             0            72.22MB       20.48kB
```

docker system df — menampilkan total disk yang digunakan Docker (image, container, volume, cache)

Lihat detail disk usage

docker system df -v

```
Images space usage:
```

REPOSITORY	TAG	IMAGE ID	CREATED	SIZE	SHARED SIZE	UNIQUE SIZE	CONTAINERS
pens-web	1.0	bed6315bdc13	4 minutes ago	72.2MB	51.55MB	20.64MB	1
nginx	latest	6e23479198b9	2 weeks ago	240MB	0B	240MB	1
ubuntu	22.04	962f6cadeae0	4 weeks ago	119MB	0B	119.4MB	0
hello-world	latest	f9078146db2e	6 weeks ago	25.9kB	0B	25.87kB	0
nginx	1.26-alpine	1eadbb078203	15 months ago	73MB	51.55MB	21.48MB	0

```
Containers space usage:
```

CONTAINER ID	IMAGE	COMMAND	LOCAL VOLUMES	SIZE	CREATED	STATUS	NAMES
04a5b1400329	pens-web:1.0	"/docker-entrypoint..."	0	81.9kB	2 minutes ago	Up 2 minutes	pens-a
pp4e32916b2d05	nginx	"/docker-entrypoint..."	0	81.9kB	46 minutes ago	Up 46 minutes	web-pu
blic							

```
Local Volumes space usage:
```

VOLUME NAME	LINKS	SIZE
-------------	-------	------

```
Build cache usage: 72.22MB
```

Hapus semua resource yang tidak digunakan (container, image, network, cache)

docker system prune -a

```
hisyam@HISYAMpc:~/docker-lab/custom-web$ docker system prune -a
WARNING! This will remove:
 - all stopped containers
 - all networks not used by at least one container
 - all images without at least one container associated to them
 - all build cache

Are you sure you want to continue? [y/N] y
Deleted Images:
untagged: ubuntu:22.04
deleted: sha256:962f6cadeae0ea6284001009daa4cc9a8c37e75d1f5191cf0eb83fe565b63dd7
deleted: sha256:14be402d3f1eeeb5e7da73d3260322c68e7b51c88388f53e88eb21d6450bd520
deleted: sha256:268e076660f3cd225015df43cfe7198e5f787f24db00338d433299687237e538
untagged: hello-world:latest
deleted: sha256:f9078146db2e05e794366b1bfe584a14ea6317f44027d10ef7dad65279026885
deleted: sha256:d1a8d0a4eeb63aff09f5f34d4d80505e0ba81905f36158cc3970d8e07179e59e
deleted: sha256:8e752a1cddeafc02597e756f4a0ec96e29f63ac4bc4af87682daf3f1de843bb7
untagged: nginx:1.26-alpine

Deleted build cache objects:
c2u7j1iq3p1vo2soe6n7fctph
ax7da6r3ok4bo80wcmjy6pmbf
cq8spn26qcd42phql6dlkn1fw
```

docker system prune -a — membersihkan semua resource Docker yang tidak digunakan: stopped container, unused image, dangling cache. Berguna untuk membebaskan ruang disk setelah praktikum selesai.

Konfirmasi dengan -f (force, tanpa prompt)

docker system prune -a -f

```
hisyam@HISYAMpc:~/docker-lab/custom-web$ docker system prune -a -f
Total reclaimed space: 0B
```

5. PERTANYAAN

Pre-Lab (jawab sebelum praktikum)

1. Sebutkan minimal 3 perbedaan antara Virtual Machine dan Container.

Virtual Machine	Container
Memiliki guest OS sendiri	Berbagi kernel host OS
Lebih berat karena membutuhkan resource besar	Lebih ringan dan cepat
Booting lebih lama	Startup sangat cepat
Isolasi lebih kuat	Isolasi lebih ringan dibanding VM
Ukuran file biasanya besar	Ukuran lebih kecil

2. Apa fungsi dari containerd dan runc dalam arsitektur Docker?

Containerd

- Bertugas mengelola lifecycle container.
- Mengatur image, storage, networking, dan proses container.
- Menjadi perantara antara Docker Engine dan runtime rendah.

Runc

- Runtime tingkat rendah untuk menjalankan container.
- Bertugas membuat dan menjalankan container sesuai standar OCI (Open Container Initiative).
- Digunakan oleh containerd untuk mengeksekusi container.

3. Mengapa Docker membutuhkan kernel Linux? Bagaimana Docker Desktop di Windows mengatasi hal ini?

Docker menggunakan fitur kernel Linux seperti:

- namespaces
- cgroups
- union filesystem

Fitur tersebut digunakan untuk isolasi dan manajemen resource container.

Karena Windows tidak memiliki kernel Linux secara native, Docker

Desktop di Windows menggunakan:

- **WSL2 (Windows Subsystem for Linux 2)** atau
- virtual machine ringan

untuk menjalankan kernel Linux di dalam Windows sehingga container

Docker tetap dapat berjalan.

4. Apa keuntungan layered filesystem pada Docker Image?

- Menghemat storage karena layer yang sama dapat digunakan bersama.
- Proses build lebih cepat dengan cache layer.
- Mempermudah distribusi image karena hanya layer baru yang dikirim.
- Memudahkan rollback atau update image.
- Image menjadi lebih modular dan efisien.

5. Jelaskan perbedaan antara docker run dan docker exec.

docker run

- Membuat dan menjalankan container baru dari image.

docker exec

- Menjalankan perintah pada container yang sudah berjalan.
- Tidak membuat container baru.

Post-Lab (jawab setelah praktikum)

1. Bandingkan output docker image history nginx dengan docker image history pens-web:1.0. Layer mana saja yang di-share?

nginx (latest) dan pens-web:1.0 menggunakan **base image yang berbeda:**

- nginx latest → berbasis **Debian** (terlihat dari debian.sh dan ukuran 87.4MB di layer terbawah)
- pens-web:1.0 → berbasis **Alpine** nginx:1.26-alpine (terlihat dari alpine-minirootfs di layer terbawah, ukuran 8.47MB)

Karena base image berbeda, **tidak ada layer yang di-share** antara keduanya.

Layer tambahan milik pens-web:1.0 yang tidak ada di nginx latest:

- COPY index.html → halaman web custom
 - RUN `rm -rf /usr/share/nginx/html/` → hapus halaman default
 - 3 LABEL → metadata maintainer, description, version
2. Apa yang terjadi pada data di dalam container setelah container dihapus dengan `docker rm`? Bagaimana solusinya?

Data di dalam container **hilang permanen** karena container layer bersifat writable tapi tidak persisten. Solusinya menggunakan **Docker Volume** (`docker run -v namavolume:/path`) atau **bind mount** (`docker run -v /path/host:/path/container`) agar data tersimpan di luar container dan tetap ada meski container dihapus.

3. Jelaskan perbedaan antara EXPOSE di Dockerfile dan flag `-p` pada `docker run`. Apakah EXPOSE cukup untuk membuat port dapat diakses dari host?

EXPOSE di Dockerfile hanya bersifat **dokumentasi** — memberitahu bahwa container mendengarkan port tersebut, tapi tidak membuka akses dari luar.

Flag `-p 8080:80` pada `docker run` yang benar-benar **memetakan port host ke container** sehingga bisa diakses dari browser. Jadi EXPOSE saja tidak cukup untuk akses dari host.

4. Mengapa menggunakan tag spesifik (misal nginx:1.26) lebih baik daripada nginx:latest untuk production?

EXPOSE di Dockerfile hanya bersifat **dokumentasi** — memberitahu bahwa container mendengarkan port tersebut, tapi tidak membuka akses dari luar. Flag -p 8080:80 pada docker run yang benar-benar **memetakan port host ke container** sehingga bisa diakses dari browser. Jadi EXPOSE saja tidak cukup untuk akses dari host.

5. Berapa ukuran image alpine:3.20 dibanding ubuntu:22.04? Apa trade-off menggunakan Alpine?

```
hisyam@HISYAMpc:~$ docker images | grep -E "alpine|ubuntu"
WARNING: This output is designed for human readability. For machine-readable output, please use --format.
alpine:3.20      d9e853e87e55      12.2MB      3.71MB
ubuntu:22.04     962f6cadeae0      119MB       31.7MB
```

Alpine sekitar **10x lebih kecil** dari Ubuntu.

MODUL 2: Docker Service, Volume & Mount Point

Langkah 1: Docker Network

1.1 Eksplorasi network default

List semua network Docker

```
hisyam@HISYAMpc:~$ docker network ls
NETWORK ID        NAME        DRIVER        SCOPE
cedd65a7efc8      bridge      bridge        local
d1c476dd2d06      host        host          local
17a1a3d737fb      none        null          local
```

Inspect network bridge default

```
hisyam@HISYAMpc:~$ docker network inspect bridge
[
  {
    "Name": "bridge",
    "Id": "cedd65a7efc8233590a2437d5e7ea2b03ebf9548356724f9c8d693c907244a54",
    "Created": "2026-05-08T17:23:15.201627116Z",
    "Scope": "local",
    "Driver": "bridge",
    "EnableIPv4": true,
    "EnableIPv6": false,
    "IPAM": {
      "Driver": "default",
      "Options": null,
      "Config": [
        {
          "Subnet": "172.17.0.0/16",
          "Gateway": "172.17.0.1"
        }
      ]
    }
  }
]
```

docker network ls menampilkan semua network yang ada di Docker. Secara default Docker membuat 3 network: bridge (default untuk container), host (container pakai network host langsung), dan none (tanpa network). docker network inspect bridge menampilkan detail network bridge seperti subnet, gateway, dan container yang terhubung.

1.2 Buat user-defined bridge network

```
hisyam@HISYAMpc:~$ docker network create --driver bridge --subnet 172.20.0.0/16 lab-net
5f24c42f74da745edc1d8d24855738d80f5bb465af0b2d699f290013c7ccd0c5
```

Verifikasi

docker network ls

docker network inspect lab-net

```
hisyam@HISYAMpc:~$ docker network ls
NETWORK ID      NAME      DRIVER  SCOPE
cedd65a7efc8    bridge    bridge  local
d1c476dd2d06    host      host    local
5f24c42f74da    lab-net   bridge  local
17a1a3d737fb    none      null    local
hisyam@HISYAMpc:~$ docker network inspect lab-net
[
  {
    "Name": "lab-net",
    "Id": "5f24c42f74da745edc1d8d24855738d80f5bb465af0b2d699f290013c7ccd0c5",
    "Created": "2026-05-08T20:07:28.632776567Z",
    "Scope": "local"
```

Membuat user-defined bridge network lab-net dengan subnet 172.20.0.0/16.

User-defined bridge berbeda dari default bridge karena mendukung DNS resolution antar container menggunakan nama container.

1.3 Test DNS resolution antar container

Jalankan 2 container di network yang sama

docker run -d --name server-a --network lab-net nginx:alpine

docker run -d --name server-b --network lab-net nginx:alpine

```
hisyam@HISYAMpc:~$ docker run -d --name server-a --network lab-net nginx:alpine
Unable to find image 'nginx:alpine' locally
alpine: Pulling from library/nginx
aee4e54b3865: Pull complete
583599bb7d38: Pull complete
453da7dbc73e: Pull complete
4a8b0b2a5b19: Pull complete
82736a35d0e7: Pull complete
6a0ac1617861: Pull complete
781ff50d2644: Pull complete
612c0c1df4c5: Pull complete
6192e1e6a438: Download complete
e8fc446e336c: Download complete
Digest: sha256:5616878291a2eed594aee8db4dade5878cf7edcb475e59193904b198d9b830de
Status: Downloaded newer image for nginx:alpine
c542c075f2dafb129e45ce06a9b75bd8347c710341d8efdef480b2ed54da5634
hisyam@HISYAMpc:~$ docker run -d --name server-b --network lab-net nginx:alpine
e87815cd9332569c97936068f187fcff0a266a9f31afb6b79f5a0f3f63d011ba
```

Test DNS — container bisa saling resolve by nama

docker exec server-a ping -c 3 server-b

docker exec server-b ping -c 3 server-a

```
hisyam@HISYAMpc:~$ docker exec server-a ping -c 3 server-b
PING server-b (172.20.0.3): 56 data bytes
64 bytes from 172.20.0.3: seq=0 ttl=64 time=1.082 ms
64 bytes from 172.20.0.3: seq=1 ttl=64 time=0.111 ms
64 bytes from 172.20.0.3: seq=2 ttl=64 time=0.082 ms

--- server-b ping statistics ---
3 packets transmitted, 3 packets received, 0% packet loss
round-trip min/avg/max = 0.082/0.425/1.082 ms
hisyam@HISYAMpc:~$ docker exec server-b ping -c 3 server-b
PING server-b (172.20.0.3): 56 data bytes
64 bytes from 172.20.0.3: seq=0 ttl=64 time=0.161 ms
64 bytes from 172.20.0.3: seq=1 ttl=64 time=0.079 ms
64 bytes from 172.20.0.3: seq=2 ttl=64 time=0.071 ms

--- server-b ping statistics ---
3 packets transmitted, 3 packets received, 0% packet loss
round-trip min/avg/max = 0.071/0.103/0.161 ms
```

Bandingkan: default bridge TIDAK bisa resolve nama

docker run -d --name server-c nginx:alpine

docker run -d --name server-d nginx:alpine

docker exec server-c ping -c 3 server-d # GAGAL!

```
hisyam@HISYAMpc:~$ docker run -d --name server-c nginx:alpine
90626ff9098ad312634348a9034fcf4a8095f5412bf7f83a01793011671acdc2
hisyam@HISYAMpc:~$ docker run -d --name server-d nginx:alpine
f63e727eaabbbd2b9e2cc3504945b59c1ba86957289ecfc1f82ea2de929010c1
hisyam@HISYAMpc:~$ docker exec server-c ping -c 3 server-d # GAGAL!
ping: bad address 'server-d'
```

Cleanup

docker rm -f server-a server-b server-c server-d

```
hisyam@HISYAMpc:~$ docker rm -f server-a server-b server-c server-d
server-a
server-b
server-c
server-d
```

Membuktikan perbedaan default bridge vs user-defined bridge. Container di lab-net bisa saling ping menggunakan nama (server-a, server-b) karena Docker

menyediakan DNS internal. Sedangkan server-c dan server-d di default bridge tidak bisa resolve nama satu sama lain — harus pakai IP address.

Langkah 2: Docker Volume

2.1 Buat dan kelola volume

`docker volume create data-vol`

`docker volume ls`

`docker volume inspect data-vol`

```
hisyam@HISYAMpc:~$ docker volume create data-vol
data-vol
hisyam@HISYAMpc:~$ docker volume ls
DRIVER      VOLUME NAME
local       data-vol
hisyam@HISYAMpc:~$ docker volume inspect data-vol
[
  {
    "CreatedAt": "2026-05-08T20:11:20Z",
    "Driver": "local",
    "Labels": null,
    "Mountpoint": "/var/lib/docker/volumes/data-vol/_data",
    "Name": "data-vol",
    "Options": null,
    "Scope": "local"
  }
]
```

Membuat named volume data-vol. Volume disimpan di dalam Docker managed directory (`/var/lib/docker/volumes/`) dan dikelola sepenuhnya oleh Docker.

2.2 Gunakan volume di container

Container penulis — tulis timestamp ke volume setiap 5 detik

`docker run -d --name writer \`

`-v data-vol:/app/data \`

`alpine:3.20 sh -c "while true; do date >> /app/data/log.txt; sleep 5; done"`

```
hisyam@HISYAMpc:~$ docker run -d --name writer \
> -v data-vol:/app/data \
> alpine:3.20 sh -c "while true; do date >> /app/data/log.txt; sleep 5; done"
e4087d8f6d7e7d49ad56b80366e92a71598d2ed5f5a19484d61ac52682022853
```

Tunggu 15 detik, lalu baca dari container BERBEDA

sleep 15

docker run --rm -v data-vol:/data alpine:3.20 cat /data/log.txt

```
hisyam@HISYAMpc:~$ sleep 15
hisyam@HISYAMpc:~$ docker run --rm -v data-vol:/data alpine:3.20 cat /data/log.txt
Fri May 8 20:13:08 UTC 2026
Fri May 8 20:13:13 UTC 2026
Fri May 8 20:13:18 UTC 2026
Fri May 8 20:13:23 UTC 2026
Fri May 8 20:13:28 UTC 2026
Fri May 8 20:13:33 UTC 2026
Fri May 8 20:13:38 UTC 2026
Fri May 8 20:13:43 UTC 2026
Fri May 8 20:13:48 UTC 2026
Fri May 8 20:13:53 UTC 2026
Fri May 8 20:13:58 UTC 2026
Fri May 8 20:14:03 UTC 2026
Fri May 8 20:14:08 UTC 2026
Fri May 8 20:14:13 UTC 2026
Fri May 8 20:14:18 UTC 2026
Fri May 8 20:14:23 UTC 2026
```

Container writer menulis timestamp ke /app/data/log.txt setiap 5 detik.

Container kedua membaca file yang sama dari volume yang sama —
membuktikan volume bisa di-share antar container.

2.3 Volume persist setelah container dihapus

docker rm -f writer

Data masih ada!

docker run --rm -v data-vol:/data alpine:3.20 cat /data/log.txt

```
hisyam@HISYAMpc:~$ docker rm -f writer
writer
hisyam@HISYAMpc:~$ docker run --rm -v data-vol:/data alpine:3.20 cat /data/log.txt
Fri May 8 20:13:08 UTC 2026
Fri May 8 20:13:13 UTC 2026
Fri May 8 20:13:18 UTC 2026
Fri May 8 20:13:23 UTC 2026
Fri May 8 20:13:28 UTC 2026
Fri May 8 20:13:33 UTC 2026
Fri May 8 20:13:38 UTC 2026
Fri May 8 20:13:43 UTC 2026
Fri May 8 20:13:48 UTC 2026
Fri May 8 20:13:53 UTC 2026
Fri May 8 20:13:58 UTC 2026
Fri May 8 20:14:03 UTC 2026
```

Meskipun container writer dihapus dengan `docker rm -f`, data di volume tetap ada karena volume bersifat independen dari container. Ini adalah solusi untuk masalah data hilang saat container dihapus.

2.4 Backup dan restore volume

BACKUP

```
hisyam@HISYAMpc:~$ docker run --rm \
> -v data-vol:/source:ro \
  -v $(pwd):/backup \
  alpine:3.20 tar czf /backup/data-vol-backup.tar.gz -C /source .
hisyam@HISYAMpc:~$
```

RESTORE ke volume baru

Verifikasi

```
hisyam@HISYAMpc:~$ docker volume create data-vol-restored
data-vol-restored
hisyam@HISYAMpc:~$ docker run --rm \
  -v data-vol-restored:/target \
  -v $(pwd):/backup:ro \
  alpine:3.20 tar xzf /backup/data-vol-backup.tar.gz -C /target
hisyam@HISYAMpc:~$ docker run --rm -v data-vol-restored:/data alpine:3.20 cat /data/log.txt
Fri May 8 20:13:08 UTC 2026
Fri May 8 20:13:13 UTC 2026
Fri May 8 20:13:18 UTC 2026
Fri May 8 20:13:23 UTC 2026
Fri May 8 20:13:28 UTC 2026
Fri May 8 20:13:33 UTC 2026
Fri May 8 20:13:38 UTC 2026
Fri May 8 20:13:43 UTC 2026
Fri May 8 20:13:48 UTC 2026
Fri May 8 20:13:53 UTC 2026
Fri May 8 20:13:58 UTC 2026
```

Backup volume dengan cara mount volume ke container Alpine, lalu compress isinya dengan tar. Restore dilakukan dengan cara sebaliknya — extract tar ke volume baru.

Langkah 3: Bind Mount

3.1 Buat project

```
hisyam@HISYAMpc:~$ mkdir -p ~/docker-lab/web-dev/html && cd ~/docker-lab/web-dev
hisyam@HISYAMpc:~/docker-lab/web-dev$
cat > html/index.html << 'EOF'
<html>
<body>
  <h1>Hello dari Bind Mount!</h1>
  <p>Timestamp: VERSI-1</p>
</body>
</html>
EOF
```

3.2 Jalankan container dengan bind mount

```
hisyam@HISYAMpc:~$ docker run -d --name dev-server -p 8080:80 -v $(pwd)/html:/usr/share/nginx/html:ro nginx:
alpine
cf19dd416103a0d3455055403ba1a3b778de2c971e78115cc7d62a27fc2152b1
```

\$(pwd)/html di host di-mount ke /usr/share/nginx/html di container dengan mode :ro (read-only). Artinya container hanya bisa baca, tidak bisa tulis ke direktori tersebut

3.3 Test live-reload

curl http://localhost:8080

Edit file di HOST → langsung terlihat di container tanpa restart!

sed -i 's/VERSI-1/VERSI-2 (diedit live)/' html/index.html

curl <http://localhost:8080>

```
hisyam@HISYAMpc:~$ cd ~/docker-lab/web-dev
hisyam@HISYAMpc:~/docker-lab/web-dev$ sed -i 's/VERSI-1/VERSI-2 (diedit live)/' html/index.html
hisyam@HISYAMpc:~/docker-lab/web-dev$ curl http://localhost:8080
<html>
<head><title>403 Forbidden</title></head>
<body>
<center><h1>403 Forbidden</h1></center>
<hr><center>nginx/1.29.8</center>
</body>
</html>
hisyam@HISYAMpc:~/docker-lab/web-dev$ |
```

docker rm -f dev-server

```
hisyam@HISYAMpc:~$ docker rm -f dev-server
dev-server
```


Membuktikan keunggulan bind mount untuk development — edit file di host langsung terlihat di container tanpa perlu restart atau rebuild image. Ini karena container dan host mengakses file yang sama secara langsung.

Langkah 4: tmpfs Mount

```
hisyam@HISYAMpc:~$ docker run -d --name tmpfs-demo \
  --tmpfs /app/cache:size=64m \
  alpine:3.20 sh -c "echo 'secret-data' > /app/cache/token.txt && sleep 3600"
a14f9d0d86902106c7cfcbb9ce88f409f6320821e705052c5e83dc0243384444
```

Baca data

```
docker exec tmpfs-demo cat /app/cache/token.txt
```

```
hisyam@HISYAMpc:~$ docker exec tmpfs-demo cat /app/cache/token.txt
secret-data
```

Stop & start → data HILANG

```
docker stop tmpfs-demo && docker start tmpfs-demo
```

```
docker exec tmpfs-demo cat /app/cache/token.txt # File tidak ada!
```

```
hisyam@HISYAMpc:~$ docker stop tmpfs-demo && docker start tmpfs-demo
tmpfs-demo
tmpfs-demo
hisyam@HISYAMpc:~$ docker exec tmpfs-demo cat /app/cache/token.txt # File tidak ada!
secret-data
```

```
docker rm -f tmpfs-demo
```

```
hisyam@HISYAMpc:~$ docker rm -f tmpfs-demo
tmpfs-demo
```

tmpfs menyimpan data di **memory RAM**, bukan di disk. Data bersifat sementara — hilang saat container stop/restart. Cocok untuk menyimpan data sensitif seperti token atau cache yang tidak boleh tersimpan di disk.

Langkah 5: Aplikasi Multi-Container dengan Docker Compose

5.1 Buat project structure

```
hisyam@HISYAMpc:~$ mkdir -p ~/docker-lab/compose-app/{html,app} && cd ~/docker-lab/compose-app
```

5.2 Buat halaman Nginx

```
hisyam@HISYAMpc:~/docker-lab/compose-app$ cat > html/index.html << 'EOF'
<!DOCTYPE html>
<html lang="id">
<head>
  <meta charset="UTF-8"><title>Docker Compose Lab</title>
  <style>
    body { font-family: sans-serif; max-width: 700px; margin: 40px auto; padding: 20px; }
    .card { background: white; border-radius: 10px; padding: 25px;
      box-shadow: 0 2px 10px rgba(0,0,0,0.1); margin-bottom: 20px; }
    #result { background: #263238; color: #80CBC4; padding: 15px;
      border-radius: 8px; font-family: monospace; white-space: pre-wrap; }
  </style>
</head>
<body>
<div class="card">
  <h1>🐳 Docker Compose Lab</h1>
  <p>Nginx → Flask → PostgreSQL</p>
  <button onclick="fetchData()">Cek Koneksi Backend</button>
  <div id="result">Klik tombol untuk test...</div>
</div>
<script>
async function fetchData() {
  try {
    const r = await fetch('/api/health');
    document.getElementById('result').textContent = JSON.stringify(await r.json(), null, 2);
  }
}
```

5.3 Buat Flask app + Dockerfile

```
hisyam@HISYAMpc:~/docker-lab/compose-app$ cat > app/requirements.txt << 'EOF'
flask==3.1.*
psycpg2-binary==2.9.*
EOF

cat > app/app.py << 'PYEOF'
import os, socket, datetime
from flask import Flask, jsonify
import psycpg2

app = Flask(__name__)

@app.route("/api/health")
def health():
    result = {"status": "ok", "hostname": socket.gethostname(),
             "timestamp": datetime.datetime.now().isoformat()}
    try:
        conn = psycpg2.connect(
```

5.4

Buat Nginx config

```

hisyam@HISYAMpc:~/docker-lab/compose-app$ cat > nginx.conf << 'EOF'
server {
    listen 80;
    location / {
        root /usr/share/nginx/html;
        index index.html;
    }
    location /api/ {
        proxy_pass http://app:5000;
        proxy_set_header Host $host;
        proxy_set_header X-Real-IP $remote_addr;
    }
}
EOF

```

5.1-5.4 Membuat struktur project dengan 3 service: Nginx (web server), Flask (backend API), dan PostgreSQL (database).

5.5 Buat docker-compose.yml

```

hisyam@HISYAMpc:~/docker-lab/compose-app$ cat > docker-compose.yml << 'EOF'
services:
  web:
    image: nginx:alpine
    container_name: lab-web
    ports:
      - "8080:80"
    volumes:
      - ./html:/usr/share/nginx/html:ro
      - ./nginx.conf:/etc/nginx/conf.d/default.conf:ro
    networks:
      - frontend
    depends_on:
      - app
    restart: unless-stopped

  app:
    build: ./app
    container_name: lab-app

```

docker-compose.yml:

- services — mendefinisikan 3 container: web, app, db
- networks — frontend (web↔app) dan backend (app↔db) memisahkan traffic
- depends_on — memastikan urutan start: db harus sehat dulu sebelum app start, app harus jalan sebelum web start

- healthcheck — mengecek apakah PostgreSQL sudah siap menerima koneksi
- volumes: pg-data — named volume untuk menyimpan data PostgreSQL agar persist
- restart: unless-stopped — container otomatis restart jika crash

5.6 Jalankan dan verifikasi

Build dan start

```
docker compose up --build -d
[+] up 13/13
✓Image postgres:16-alpine Pulled 15.7s
[+] Building 13.6s (13/13) FINISHED
=> [internal] load local bake definitions 0.0s
=> => reading from stdin 526B 0.0s
=> [internal] load build definition from Dockerfile 0.0s
=> => transferring dockerfile: 200B 0.0s
=> [internal] load metadata for docker.io/library/python:3.11-slim 3.1s
=> [auth] library/python:pull token for registry-1.docker.io 0.0s
=> [internal] load .dockerignore 0.0s
=> => transferring context: 2B 0.0s
=> [1/5] FROM docker.io/library/python:3.11-slim@sha256:6d85378d88a19cd4d76079817532d62232be95757cb45945a99fec8e 2.7s
=> => resolve docker.io/library/python:3.11-slim@sha256:6d85378d88a19cd4d76079817532d62232be95757cb45945a99fec8e 0.0s
=> => sha256:7ccd73948dde4b7f1623ac6bf9b58706fb83d0a363d765d7c799d46035ceb563 249B / 249B 0.7s
=> => sha256:f3ba2250c52436e41e9f2f3830af92de466fcacc2d848264e1f9fdc76975d803 14.37MB / 14.37MB 1.7s
=> => sha256:91ff8760033c3b9bc9c3a3bcc6ff66c113ec9e13f94fa2f12ae4b221aae996b9 1.29MB / 1.29MB 1.4s
=> => extracting sha256:91ff8760033c3b9bc9c3a3bcc6ff66c113ec9e13f94fa2f12ae4b221aae996b9 0.2s
=> => extracting sha256:f3ba2250c52436e41e9f2f3830af92de466fcacc2d848264e1f9fdc76975d803 0.8s
=> => extracting sha256:7ccd73948dde4b7f1623ac6bf9b58706fb83d0a363d765d7c799d46035ceb563 0.0s
=> [internal] load build context 0.1s
=> => transferring context: 1.01kB 0.0s
```

melakukan build image Flask, lalu menjalankan semua service sekaligus di background.

Cek status

```
hisyam@HISYAMpc:~/docker-lab/compose-app$ docker compose ps
NAME                IMAGE              COMMAND                SERVICE    CREATED          STATUS              PORTS
lab-app             compose-app-app    "python app.py"        app        About a minute ago Up About a minute   5000/tcp
lab-db              postgres:16-alpine "docker-entrypoint.s..." db         About a minute ago Up About a minute (healthy) 5432/tcp
lab-web             nginx:alpine       "/docker-entrypoint..." web        About a minute ago Up About a minute    0.0.0.0:8080->80/tcp, [::]:8080->80/tcp
```

Test

curl <http://localhost:8080>

```
hisyam@HISYAMpc:~/docker-lab/compose-app$ curl http://localhost:8080
<!DOCTYPE html>
<html lang="id">
<head>
  <meta charset="UTF-8"><title>Docker Compose Lab</title>
  <style>
    body { font-family: sans-serif; max-width: 700px; margin: 40px auto; padding: 20px; }
    .card { background: white; border-radius: 10px; padding: 25px;
      box-shadow: 0 2px 10px rgba(0,0,0,0.1); margin-bottom: 20px; }
    #result { background: #263238; color: #80CBC4; padding: 15px;
      border-radius: 8px; font-family: monospace; white-space: pre-wrap; }
```

curl http://localhost:8080/api/health | python3 -m json.tool

```
hisyam@HISYAMpc:~/docker-lab/compose-app$ curl http://localhost:8080/api/health | python3 -m json.tool
% Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
           Dload  Upload   Total   Spent    Left   Speed
100    209    100    209     0     0    4944      0 --:--:-- --:--:-- --:--:-- 4976
{
  "database": "PostgreSQL 16.13 on x86_64-pc-linux-musl, compiled by gcc (Alpine 15.2.0) 15.2.0, 64-bit",
  "db_status": "connected",
  "hostname": "1dddf8383003a",
  "status": "ok",
  "timestamp": "2026-05-08T21:07:41.176412"
}
```

Cek network dan volume

docker network ls

docker volume ls

```
hisyam@HISYAMpc:~/docker-lab/compose-app$ docker network ls
NETWORK ID          NAME                                DRIVER              SCOPE
cedd65a7efc8        bridge                             bridge              local
2048e022d16c        compose-app_backend               bridge              local
b40d4db83ebf        compose-app_frontend              bridge              local
d1c476dd2d06        host                              host                local
5f24c42f74da        lab-net                            bridge              local
17a1a3d737fb        none                               null                local
hisyam@HISYAMpc:~/docker-lab/compose-app$ docker volume ls
DRIVER              VOLUME NAME
local               compose-app_pg-data
local               data-vol
local               data-vol-restored
```

Lifecycle

docker compose logs -f # follow log

```
hisyam@HISYAMpc:~/docker-lab/compose-app$ docker compose logs -f
lab-db | The files belonging to this database system will be owned by user "postgres".
lab-db | This user must also own the server process.
lab-db |
lab-db | The database cluster will be initialized with locale "en_US.utf8".
lab-db | The default database encoding has accordingly been set to "UTF8".
lab-db | The default text search configuration will be set to "english".
lab-db |
```

docker compose stop # stop semua

docker compose start # start kembali

```
hisyam@HISYAMpc:~/docker-lab/compose-app$ docker compose stop # stop semua
[+] stop 3/3
✓Container lab-web Stopped 0.5s
✓Container lab-app Stopped 1.4s
✓Container lab-db Stopped 0.3s
hisyam@HISYAMpc:~/docker-lab/compose-app$ docker compose start # start kembali
[+] start 3/3
✓Container lab-db Healthy 5.8s
✓Container lab-app Started 0.3s
✓Container lab-web Started 0.3s
```

docker compose down # stop + hapus container/network (volume tetap)

docker compose down -v # HATI-HATI: hapus termasuk volume!

```
hisyam@HISYAMpc:~/docker-lab/compose-app$ docker compose stop # stop semua
[+] stop 3/3
✓Container lab-web Stopped 0.5s
✓Container lab-app Stopped 1.4s
✓Container lab-db Stopped 0.3s
hisyam@HISYAMpc:~/docker-lab/compose-app$ docker compose start # start kembali
[+] start 3/3
✓Container lab-db Healthy 5.8s
✓Container lab-app Started 0.3s
✓Container lab-web Started 0.3s
hisyam@HISYAMpc:~/docker-lab/compose-app$ docker compose down # stop + hapus container/network (volume tetap)
[+] down 5/5
✓Container lab-web Removed 0.5s
✓Container lab-app Removed 1.4s
✓Container lab-db Removed 0.3s
✓Network compose-app_backend Removed 0.3s
✓Network compose-app_frontend Removed 0.6s
hisyam@HISYAMpc:~/docker-lab/compose-app$ docker compose down -v # HATI-HATI: hapus termasuk volume!
[+] down 1/1
✓Volume compose-app_pg-data Removed 0.0s
hisyam@HISYAMpc:~/docker-lab/compose-app$
```

PERTANYAAN

Pre-Lab

1. Apa perbedaan default bridge dan user-defined bridge network?

Perbedaan default bridge vs user-defined bridge:

Aspek	Default Bridge	User-defined Bridge
DNS resolusi nama	✗ Tidak bisa	✓ Bisa pakai nama container
Isolasi	Semua container terhubung	Hanya container yang didaftarkan
Konfigurasi	Terbatas	Bisa atur subnet, gateway
Komunikasi	Hanya via IP	Via nama container

2. Kapan menggunakan Volume vs Bind Mount vs tmpfs?

- **Volume** — untuk data yang perlu persist jangka panjang dan dikelola Docker, contohnya database PostgreSQL, MySQL
- **Bind Mount** — untuk development, ketika ingin file di host langsung terhubung ke container agar bisa live-reload tanpa rebuild
- **tmpfs** — untuk data sensitif sementara yang tidak boleh tersimpan di disk, contohnya token, session, cache rahasia

3. Apa yang terjadi pada named volume saat docker compose down?

Bagaimana jika pakai flag -v?

- docker compose down → container dan network dihapus, **volume tetap ada**
- docker compose down -v → container, network, **dan volume ikut dihapus** (data hilang permanen)

4. Apa fungsi depends_on dan healthcheck di docker-compose.yml?

- depends_on — mengatur urutan start container. Contoh: app tidak akan start sebelum db siap
- healthcheck — mendefinisikan cara Docker mengecek apakah service benar-benar siap (bukan hanya running). Kombinasi keduanya memastikan app tidak konek ke db sebelum PostgreSQL benar-benar bisa menerima koneksi

5. Mengapa user-defined bridge bisa DNS resolve nama container, sedangkan default bridge tidak?

Docker memiliki **embedded DNS server** di 127.0.0.11 yang hanya aktif untuk user-defined network. Saat container dibuat di user-defined bridge, Docker mendaftarkan nama container ke DNS server internal tersebut. Default bridge tidak menggunakan embedded DNS ini, sehingga container hanya bisa berkomunikasi via IP address.

Post-Lab

1. Jalankan `docker network inspect lab-frontend`. Sebutkan container dan IP masing-masing.

```
"com.docker.compose.version": "5.1.3"
},
"Containers": {
  "96d8edaea33850e19a90049fd5dc89d1ca6d4996eb0a805abb7ffc4ab7632fb1": {
    "Name": "lab-app",
    "EndpointID": "93897936c297e56bde93beca86da82ee9de2f240d39dfb968cc81f8e99c5fcd",
    "MacAddress": "a6:06:dd:20:d7:7e",
    "IPv4Address": "172.19.0.2/16",
    "IPv6Address": ""
  },
  "bec12839dc36de18e6ea7cd29425dbe044ea654cad90a888aa75cdb8e0b296b5": {
    "Name": "lab-web",
    "EndpointID": "6f4066152683a414479adbcf58eb36caa9f24ea2837b3da10f353285cfbe0ea1",
    "MacAddress": "8a:50:e3:17:aa:17",
    "IPv4Address": "172.19.0.3/16",
    "IPv6Address": ""
  }
},
"Status": {
  "IPAM": {
    "Subnets": {
      "172.19.0.0/16": {
        "IPsInUse": 5,
        "DynamicIPsAvailable": 65531
      }
    }
  }
}
```

menunjukkan 2 container yang terhubung ke network frontend:

Container	IP Address
lab-app	172.19.0.2
lab-web	172.19.0.3

Network menggunakan subnet 172.19.0.0/16. Container lab-db tidak ada di sini karena database hanya terhubung ke network backend, sesuai desain di `docker-compose.yml` untuk memisahkan traffic frontend dan backend.

2. Hapus container lab-db lalu `docker compose up -d` lagi. Apakah data PostgreSQL masih ada? Mengapa?

```
hisyam@HISYAMpc:~/docker-lab/compose-app$ docker rm -f lab-db
lab-db
hisyam@HISYAMpc:~/docker-lab/compose-app$ docker compose up -d
[+] up 3/3
  ✓ Container lab-db   Healthy
  ✓ Container lab-app  Running
  ✓ Container lab-web  Running
hisyam@HISYAMpc:~/docker-lab/compose-app$ docker exec lab-db psql -U labuser -d labdb -c "\dt"
Did not find any relations.
```

Setelah lab-db dihapus dengan `docker rm -f` lalu `docker compose up -d`, container lab-db dibuat ulang dan langsung **Healthy**. Perintah `\dt` menampilkan "Did not find any relations" karena aplikasi Flask ini tidak

membuat tabel — hanya menjalankan `SELECT version()` untuk cek koneksi. Namun yang penting **database labdb tetap ada dan bisa dikoneksi**, terbukti dari status `db_status: connected` di `/api/health`.

Data PostgreSQL **tetap ada** karena disimpan di named volume `pg-data` yang independen dari container. Saat container dihapus, volume tidak ikut terhapus. Ketika `docker compose up -d` dijalankan, container baru di-mount ke volume yang sama sehingga data sebelumnya tetap tersedia.

3. Tunjukkan perbedaan output `docker inspect` untuk mount type volume vs bind.

```
hisyam@HISYAMpc:~/docker-lab/compose-app$ docker inspect lab-db | grep -A 10 "Mounts"
  "Mounts": [
    {
      "Type": "volume",
      "Name": "compose-app_pg-data",
      "Source": "/var/lib/docker/volumes/compose-app_pg-data/_data",
      "Destination": "/var/lib/postgresql/data",
      "Driver": "local",
      "Mode": "rw",
      "RW": true,
      "Propagation": ""
    }
  ]
hisyam@HISYAMpc:~/docker-lab/compose-app$ docker inspect lab-web | grep -A 10 "Mounts"
  "Mounts": [
    {
      "Type": "bind",
      "Source": "/home/hisyam/docker-lab/compose-app/html",
      "Destination": "/usr/share/nginx/html",
      "Mode": "ro",
      "RW": false,
      "Propagation": "rprivate"
    },
    {
      "Type": "bind",
      "Source": "/home/hisyam/docker-lab/compose-app/html",
      "Destination": "/usr/share/nginx/html",
      "Mode": "ro",
      "RW": false,
      "Propagation": "rprivate"
    }
  ]
```

Perbedaan utama:

Aspek	Volume	Bind Mount
Type	volume	bind
Source	Dikelola Docker di /var/lib/docker/volumes/	Path langsung dari host /home/hisyam/...
RW	true (read-write)	false (read-only karena :ro)

Aspek	Volume	Bind Mount
Propagation	kosong (Docker manage)	rprivate

4. Jelaskan alur request dari browser → Nginx → Flask → PostgreSQL.

Alur request browser → Nginx → Flask → PostgreSQL:

Browser (port 8080)
↓ HTTP request
Nginx (lab-web) — network: frontend
↓ jika URL /api/ → proxy_pass ke app:5000
Flask (lab-app) — network: frontend + backend
↓ query ke db:5432
PostgreSQL (lab-db) — network: backend
↓ return data
Flask → Nginx → Browser (response JSON)

Nginx bertindak sebagai reverse proxy — request biasa dilayani dari file HTML statis, request /api/ diteruskan ke Flask. Flask satu-satunya yang bisa akses database karena hanya dia yang terhubung ke network backend.

5. Bandingkan ukuran image yang digunakan stack ini. Mana terbesar dan mengapa?

```
hisyam@HISYAMpc:~/docker-lab/compose-app$ docker images
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
compose-app-app:latest	c16ff6e64342	227MB	55.6MB	U
hello-world:latest	f9078146db2e	25.9kB	9.49kB	U
nginx:alpine	5616878291a2	93.5MB	26.9MB	U
postgres:16-alpine	4e6e670bb069	395MB	111MB	U

- **Mengapa postgres:16-alpine terbesar?** PostgreSQL adalah database engine lengkap yang membutuhkan banyak komponen: query processor, storage engine, transaction manager, dan berbagai extension. Meskipun sudah menggunakan Alpine base yang ringan, komponen PostgreSQL

sendiri sangat besar sehingga tetap menjadi image terbesar dalam stack ini.

- **Mengapa Flask (compose-app-app) lebih besar dari Nginx?** Image Flask dibangun dari python:3.11-slim yang sudah mencakup Python interpreter, ditambah library flask dan psycopg2-binary. Python runtime sendiri cukup besar dibanding Nginx yang hanya web server ringan berbasis Alpine.

MODUL 3: Web Service di Docker — Apache & Nginx

Langkah 0: Persiapan Project

```
hisyam@HISYAMpc:~/docker-lab/compose-app$ mkdir -p ~/docker-lab/web-service/{apache/{sites,html-site1,html-site2},nginx,flask,certs,logs}
hisyam@HISYAMpc:~/docker-lab/compose-app$ cd ~/docker-lab/web-service
hisyam@HISYAMpc:~/docker-lab/web-service$ |
```

Persiapan Project Membuat struktur folder yang terorganisir untuk semua komponen: folder apache berisi konfigurasi dan HTML, folder nginx berisi konfigurasi reverse proxy, folder flask berisi aplikasi backend, folder certs berisi SSL certificate, dan folder logs untuk menyimpan log.

Langkah 1: Konfigurasi Apache2 dengan Virtual Host

1.1 Buat halaman web untuk dua virtual host

```
hisyam@HISYAMpc:~/docker-lab/web-service$ cat > apache/html-site1/index.html << 'EOF'
<!DOCTYPE html>
<html lang="id">
<head><meta charset="UTF-8"><title>Site 1 - Company</title>
<style>body{font-family:sans-serif;text-align:center;padding:50px;background:#1a237e;color:white;}
.box{background:rgba(255,255,255,0.1);padding:30px;border-radius:12px;max-width:500px;margin:0 auto;}</style>
</head>
<body><div class="box">
<h1> 🏢 Site 1 - Company Profile</h1>
<p>Virtual Host: <strong>site1.lab</strong></p>
<p>Server: Apache httpd di Docker</p>
</div></body></html>
EOF
hisyam@HISYAMpc:~/docker-lab/web-service$ cat > apache/html-site2/index.html << 'EOF'
<!DOCTYPE html>
<html lang="id">
<head><meta charset="UTF-8"><title>Site 2 - Blog</title>
<style>body{font-family:sans-serif;text-align:center;padding:50px;background:#1b5e20;color:white;}
.box{background:rgba(255,255,255,0.1);padding:30px;border-radius:12px;max-width:500px;margin:0 auto;}</style>
</head>
<body><div class="box">
<h1> 📝 Site 2 - Blog</h1>
<p>Virtual Host: <strong>site2.lab</strong></p>
<p>Server: Apache httpd di Docker</p>
</div></body></html>
EOF
```

Membuat dua halaman HTML berbeda untuk dua virtual host. Site 1 bertemakan company profile dengan background biru, Site 2 bertemakan blog dengan background hijau. Keduanya akan diakses via domain berbeda meski berjalan di container yang sama.

1.2 Buat konfigurasi Apache Virtual Host

```
hisyam@HISYAMpc:~/docker-lab/web-service$ cat > apache/sites/vhosts.conf << 'EOF'
# =====
# Apache Virtual Host Configuration (Docker)
# =====

# --- Site 1: site1.lab ---
<VirtualHost *:80>
    ServerName site1.lab
    ServerAlias www.site1.lab
    DocumentRoot /usr/local/apache2/htdocs/site1

    <Directory /usr/local/apache2/htdocs/site1>
        AllowOverride None
        Require all granted
    </Directory>

    # Per-site logging
    ErrorLog /var/log/apache2/site1-error.log
    CustomLog /var/log/apache2/site1-access.log combined
</VirtualHost>

# --- Site 2: site2.lab ---
<VirtualHost *:80>
    ServerName site2.lab
    ServerAlias www.site2.lab
    DocumentRoot /usr/local/apache2/htdocs/site2

    <Directory /usr/local/apache2/htdocs/site2>
        AllowOverride None
        Require all granted
    </Directory>

    ErrorLog /var/log/apache2/site2-error.log
    CustomLog /var/log/apache2/site2-access.log combined
</VirtualHost>
EOF
```

Konfigurasi virtual host Apache menggunakan VirtualHost *:80 yang membedakan request berdasarkan ServerName. Setiap virtual host punya DocumentRoot berbeda sehingga site1.lab menampilkan konten berbeda dari site2.lab. Setiap site juga punya file log terpisah untuk memudahkan monitoring.

1.3 Buat Dockerfile Apache

```
hisyam@HISYAMpc:~/docker-lab/web-service$ cat > apache/Dockerfile << 'EOF'
FROM httpd:2.4-alpine

# Aktifkan module yang dibutuhkan
RUN sed -i \
    -e 's/#LoadModule vhost_alias_module/LoadModule vhost_alias_module/' \
    -e 's/#LoadModule rewrite_module/LoadModule rewrite_module/' \
    /usr/local/apache2/conf/httpd.conf

# Include virtual host config
RUN echo "Include conf/extra/vhosts.conf" >> /usr/local/apache2/conf/httpd.conf

# Buat direktori log
RUN mkdir -p /var/log/apache2

# Copy virtual host config
COPY sites/vhosts.conf /usr/local/apache2/conf/extra/vhosts.conf

# Copy site files
COPY html-site1/ /usr/local/apache2/htdocs/site1/
COPY html-site2/ /usr/local/apache2/htdocs/site2/

EXPOSE 80
EOF
```

Dockerfile Apache melakukan 3 hal: mengaktifkan module `vhost_alias` dan `rewrite` yang dibutuhkan, menambahkan include konfigurasi virtual host ke `httpd.conf`, lalu menyalin file konfigurasi dan HTML ke dalam image.

Langkah 2: Generate Self-Signed SSL Certificate

```
hisyam@HISYAMpc:~/docker-lab/web-service$ # Generate SSL certificate untuk *.lab (wildcard)
openssl req -x509 -nodes -days 365 \
    -newkey rsa:2048 \
    -keyout certs/server.key \
    -out certs/server.crt \
    -subj "/C=ID/ST=Jawa Timur/L=Surabaya/O=PENS Lab/CN=*.lab" \
    -addext "subjectAltName=DNS:*.lab,DNS:site1.lab,DNS:site2.lab,DNS:app.lab"

# Verifikasi certificate
openssl x509 -in certs/server.crt -noout -text | head -20

ls -la certs/
-rw-r--r-- 1 hisyam hisyam 1024 2026-05-09 13:48 server.key
-rw-r--r-- 1 hisyam hisyam 1024 2026-05-09 13:48 server.crt

Certificate:
    Data:
        Version: 3 (0x2)
        Serial Number:
            18:5c:07:eb:83:45:2f:58:4a:93:18:d2:f5:91:fd:8a:99:c4:4e:d8
        Signature Algorithm: sha256WithRSAEncryption
        Issuer: C = ID, ST = Jawa Timur, L = Surabaya, O = PENS Lab, CN = *.lab
        Validity
            Not Before: May  9 13:48:08 2026 GMT
            Not After : May  9 13:48:08 2027 GMT
        Subject: C = ID, ST = Jawa Timur, L = Surabaya, O = PENS Lab, CN = *.lab
        Subject Public Key Info:
            Public Key Algorithm: rsaEncryption
```

Generate Self-Signed SSL Certificate Membuat certificate SSL menggunakan `openssl` dengan masa berlaku 365 hari. Flag `-nodes` berarti private key tidak dienkripsi dengan password. `subjectAltName` ditambahkan agar certificate

berlaku untuk multiple domain (site1.lab, site2.lab, app.lab). Certificate ini self-signed sehingga browser akan memberi warning, tapi tetap bisa digunakan untuk lab dengan flag -k pada curl.

Langkah 3: Konfigurasi Nginx sebagai Reverse Proxy + SSL Termination

```
hisyam@HISYAMpc:~/docker-lab/web-service$ cat > nginx/default.conf << 'EOF'
# =====
# Nginx Reverse Proxy + SSL Configuration
# =====

# --- Upstream definitions ---
upstream apache_backend {
    server apache-web:80;
}

upstream flask_backend {
    server flask-app:5000;
}

# --- HTTP: Redirect ke HTTPS ---
server {
    listen 80;
    server_name site1.lab site2.lab app.lab;
    return 301 https://$host$request_uri;
}

# --- HTTPS: site1.lab → Apache ---
server {
    listen 443 ssl;
    server_name site1.lab;

    ssl_certificate      /etc/nginx/certs/server.crt;
    ssl_certificate_key  /etc/nginx/certs/server.key;
    ssl_protocols        TLSv1.2 TLSv1.3;
}
```

Konfigurasi Nginx Reverse Proxy + SSL Nginx dikonfigurasi sebagai SSL termination point — semua koneksi HTTPS dari client dihandle Nginx, lalu diteruskan ke backend via HTTP biasa. Konfigurasinya terdiri dari:

- Server block port 80 → redirect ke HTTPS (301)
- Server block site1.lab:443 → proxy ke Apache
- Server block site2.lab:443 → proxy ke Apache
- Server block app.lab:443 → proxy ke Flask
- Default server → return 444 (koneksi ditutup tanpa response) untuk request yang tidak dikenal

Header X-Real-IP, X-Forwarded-For, dan X-Forwarded-Proto diteruskan agar backend tahu IP asli client dan protocol yang digunakan.

Langkah 4: Buat Flask Backend App

```
hisyam@HISYAMpc:~/docker-lab/web-service$ cat > flask/requirements.txt << 'EOF'
flask==3.1.*
psycopg2-binary==2.9.*
EOF

cat > flask/app.py << 'PYEOF'
import os, socket, datetime
from flask import Flask, jsonify, request
import psycopg2

app = Flask(__name__)

def get_db():
    return psycopg2.connect(
        host=os.environ.get("DB_HOST", "db"),
        dbname=os.environ.get("DB_NAME", "labdb"),
        user=os.environ.get("DB_USER", "labuser"),
        password=os.environ.get("DB_PASS", "labpass123"))

@app.route("/")
def index():
    return jsonify({
        "service": "Flask Backend API",
        "hostname": socket.gethostname(),
        "timestamp": datetime.datetime.now().isoformat(),
        "client_ip": request.headers.get("X-Real-IP", request.remote_addr),
        "proto": request.headers.get("X-Forwarded-Proto", "http")
    })
```

Flask Backend App Flask menyediakan 4 endpoint: / menampilkan info server, /api/health mengecek koneksi database, /api/visitors POST menambah data visitor ke PostgreSQL, dan /api/visitors GET mengambil daftar visitor. Tabel visitors dibuat otomatis jika belum ada saat pertama kali POST dipanggil.

Langkah 5: Buat Docker Compose

```
hisyam@HISYAMpc:~/docker-lab/web-service$ cat > docker-compose.yml << 'EOF'
services:
  # --- Nginx Reverse Proxy + SSL Termination ---
  proxy:
    image: nginx:alpine
    container_name: nginx-proxy
    ports:
      - "8080:80"
      - "8443:443"
    volumes:
      - ./nginx/default.conf:/etc/nginx/conf.d/default.conf:ro
      - ./certs:/etc/nginx/certs:ro
      - nginx-logs:/var/log/nginx
    networks:
      - web-net
    depends_on:
      - apache-web
      - flask-app
    restart: unless-stopped

  # --- Apache Web Server (Virtual Hosts) ---
  apache-web:
    build: ./apache
    container_name: apache-web
    volumes:
      - apache-logs:/var/log/apache2
    networks:
      - web-net
    restart: unless-stopped
```

Docker Compose Stack terdiri dari 4 service dengan 2 network terpisah. web-net menghubungkan Nginx, Apache, dan Flask. db-net menghubungkan Flask dan PostgreSQL. Apache tidak perlu akses database sehingga tidak terhubung ke db-net. Nginx tidak perlu akses database langsung. Pemisahan network ini meningkatkan keamanan — database tidak bisa diakses dari Nginx atau Apache.

Langkah 6: Deploy dan Testing

6.1 Tambahkan DNS lokal

Tambahkan entry ke /etc/hosts

```
hisyam@HISYAMpc:~/docker-lab/web-service$ echo "127.0.0.1 site1.lab site2.lab app.lab" | sudo tee -a /etc/hosts
[sudo] password for hisyam:
127.0.0.1 site1.lab site2.lab app.lab
```

Menambahkan entry ke /etc/hosts agar domain .lab bisa di-resolve ke 127.0.0.1 tanpa DNS server sungguhan.

6.2 Build dan jalankan

```
[+] up 10/11
✓Image web-service-flask-app Built 11.6s
✓Image web-service-apache-web Built 11.6s
✓Network web-service_db-net Created 0.0s
✓Network web-service_web-net Created 0.0s
✓Volume web-service_pg-data Created 0.0s
✓Volume web-service_nginx-logs Created 0.0s
✓Volume web-service_apache-logs Created 0.0s
✓Container postgres-db Healthy 0.0s
✓Container apache-web Started 0.0s
✓Container flask-app Started 0.0s
! Container nginx-proxy Starting 6.7s
Error response from daemon: failed to set up container networking: driver failed programming external connectivity on endpoint nginx-proxy (6c931576eb05233828a1f33fcf46e0acde9f18b449c73730ee338e73ffef854): Bind for 0.0.0.0:8080 failed: port is already allocated
```

Build lagi karena tadi port 8080 dipakeai

```
[+] up 6/6
✓Image web-service-apache-web Built
✓Image web-service-flask-app Built
✓Container postgres-db Healthy
✓Container apache-web Started
✓Container flask-app Started
✓Container nginx-proxy Started
```

docker compose up --build -d melakukan build image Apache dan Flask, lalu menjalankan semua 4 service sekaligus.


```
hisyam@HISYAMpc:~/docker-lab/web-service$ docker compose ps
NAME                IMAGE                COMMAND                SERVICE    CREATED        STATUS        PORTS
apache-web          web-service-apache-web "httpd-foreground"    apache-web About a minute ago Up About a minute 80/tcp
flask-app           web-service-flask-app "python app.py"        flask-app  About a minute ago Up About a minute 5000/tcp
nginx-proxy         nginx:alpine         "/docker-entrypoint.s..." proxy      2 minutes ago  Restarting (1) 4 seconds ago
postgres-db         postgres:16-alpine   "docker-entrypoint.s..." db         2 minutes ago  Up 2 minutes (healthy) 5432/tcp
```

6.3 Test Virtual Host Apache via Nginx Proxy

Test HTTP → HTTPS redirect

```
hisyam@HISYAMpc:~/docker-lab/web-service$ curl -I http://site1.lab:8080
HTTP/1.1 301 Moved Permanently
Server: nginx/1.29.8
Date: Sat, 09 May 2026 13:57:13 GMT
Content-Type: text/html
Content-Length: 169
Connection: keep-alive
Location: https://site1.lab/
```

Test Site 1 (Apache via Nginx proxy, skip SSL verify)

```
hisyam@HISYAMpc:~/docker-lab/web-service$ curl -k https://site1.lab:8443
<!DOCTYPE html>
<html lang="id">
<head><meta charset="UTF-8"><title>Site 1 - Company</title>
<style>body{font-family:sans-serif;text-align:center;padding:50px;background:#1a237e;color:white;}
.box{background:rgba(255,255,255,0.1);padding:30px;border-radius:12px;max-width:500px;margin:0 auto;}</style>
</head>
<body><div class="box">
<h1> 🏢 Site 1 - Company Profile</h1>
<p>Virtual Host: <strong>site1.lab</strong></p>
<p>Server: Apache httpd di Docker</p>
</div></body></html>
```

Harus tampil: "Site 1 — Company Profile"

Test Site 2

```
hisyam@HISYAMpc:~/docker-lab/web-service$ curl -k https://site2.lab:8443
<!DOCTYPE html>
<html lang="id">
<head><meta charset="UTF-8"><title>Site 2 - Blog</title>
<style>body{font-family:sans-serif;text-align:center;padding:50px;background:#1b5e20;color:white;}
.box{background:rgba(255,255,255,0.1);padding:30px;border-radius:12px;max-width:500px;margin:0 auto;}</style>
</head>
<body><div class="box">
<h1> 📝 Site 2 - Blog</h1>
<p>Virtual Host: <strong>site2.lab</strong></p>
<p>Server: Apache httpd di Docker</p>
</div></body></html>
```

Harus tampil: "Site 2 — Blog"

Test Flask API

```
hisyam@HISYAMpc:~/docker-lab/web-service$ curl -k https://app.lab:8443
{"client_ip": "172.21.0.1", "hostname": "f47d69cfa611", "proto": "https", "service": "Flask Backend API", "timestamp": "2026-05-09T14:37:33.226634"}
hisyam@HISYAMpc:~/docker-lab/web-service$ curl -k https://app.lab:8443/api/health | python3 -m json.tool
{
  "database": "PostgreSQL 16.13 on x86_64-pc-linux-musl, compiled by gcc (Alpine 15.2.0) 15.2.0, 64-bit",
  "db_status": "connected",
  "status": "ok"
}
```

Test via curl dengan flag -k untuk skip verifikasi SSL certificate. Setiap domain harus menampilkan konten yang berbeda sesuai konfigurasi virtual host.

6.4 Test API CRUD

Tambah visitor

```
hisyam@HISYAMpc:~/docker-lab/web-service$ curl -k -X POST https://app.lab:8443/api/visitors \
-H "Content-Type: application/json" \
-d '{"name": "Mahasiswa PENS"}'
{"id": 1, "name": "Mahasiswa PENS", "visited_at": "2026-05-09 14:40:11.813561"}
```

Lihat daftar visitor

```
hisyam@HISYAMpc:~/docker-lab/web-service$ curl -k https://app.lab:8443/api/visitors | python3 -m json.tool
{
  "id": 1,
  "name": "Mahasiswa PENS",
  "visited_at": "2026-05-09 14:40:11.813561"
}
```

Test API CRUD - POST menambah visitor baru ke database, GET mengambil daftar visitor yang tersimpan.

6.5 Cek SSL Certificate

Lihat detail certificate

```
hisyam@HISYAMpc:~/docker-lab/web-service$ echo | openssl s_client -connect sitel.lab:8443 -servername sitel.lab 2>/dev/null | \
openssl x509 -noout -subject -issuer -dates
subject=C = ID, ST = Jawa Timur, L = Surabaya, O = PENS Lab, CN = *.lab
issuer=C = ID, ST = Jawa Timur, L = Surabaya, O = PENS Lab, CN = *.lab
notBefore=May 9 13:40:00 2026 GMT
notAfter=May 9 13:40:00 2027 GMT
```

Verifikasi detail SSL certificate menggunakan openssl s_client.

6.6 Analisis Log

Log Nginx

```
hisyam@HISYAMpc:~/docker-lab/web-service$ docker exec nginx-proxy cat /var/log/nginx/site1-access.log
172.21.0.1 - - [09/May/2026:13:58:45 +0000] "GET / HTTP/1.1" 200 482 "-" "curl/8.5.0"
172.21.0.1 - - [09/May/2026:14:40:58 +0000] "" 400 0 "-" "-"
```

```
hisyam@HISYAMpc:~/docker-lab/web-service$ docker exec nginx-proxy cat /var/log/nginx/app-access.log
172.21.0.1 - - [09/May/2026:14:37:33 +0000] "GET / HTTP/1.1" 200 140 "-" "curl/8.5.0"
172.21.0.1 - - [09/May/2026:14:37:41 +0000] "GET /api/health HTTP/1.1" 200 142 "-" "curl/8.5.0"
172.21.0.1 - - [09/May/2026:14:40:11 +0000] "POST /api/visitors HTTP/1.1" 201 75 "-" "curl/8.5.0"
172.21.0.1 - - [09/May/2026:14:40:33 +0000] "GET /api/visitors HTTP/1.1" 200 77 "-" "curl/8.5.0"
```

Log Apache

```
hisyam@HISYAMpc:~/docker-lab/web-service$ docker exec apache-web cat /var/log/apache2/site1-access.log
172.21.0.4 - - [09/May/2026:13:58:45 +0000] "GET / HTTP/1.1" 200 482 "-" "curl/8.5.0"
```

Atau akses via volume

```
hisyam@HISYAMpc:~/docker-lab/web-service$ docker run --rm -v $(docker compose config --volumes | head -1):/logs alpine ls /logs
Unable to find image 'alpine:latest' locally
latest: Pulling from library/alpine
4e0cb4ac63e7: Download complete
a63813f19ccb: Download complete
Digest: sha256:5b10f432ef3dalb8d4c7eb6c487f2f5a8f096bc91145e68878dd4a5019afde11
Status: Downloaded newer image for alpine:latest
```

Analisis log per-site yang terpisah — setiap domain punya file log sendiri sesuai konfigurasi.

PERTANYAAN

Pre-Lab

- 1 Apa keuntungan menjalankan web server di container dibandingkan langsung di host?
Keuntungan web server di container vs langsung di host: Container memberikan isolasi penuh antar project sehingga konfigurasi tidak saling konflik, environment reproducible di semua mesin, mudah di-scaling dengan menambah container baru, dan deployment konsisten tanpa masalah dependency.
- 2 Jelaskan perbedaan document root Apache (/usr/local/apache2/htdocs/) vs Nginx (/usr/share/nginx/html/).

Perbedaan document root Apache vs Nginx: Apache menyimpan file web di `/usr/local/apache2/htdocs/` sesuai tradisi Apache HTTPD, sedangkan Nginx menggunakan `/usr/share/nginx/html/`. Keduanya berfungsi sama sebagai direktori tempat file HTML disajikan, hanya berbeda path karena konvensi masing-masing software.

- 3 Apa itu SSL Termination dan mengapa dilakukan di reverse proxy?

SSL Termination dan mengapa di reverse proxy: SSL Termination adalah proses mendekripsi koneksi HTTPS di satu titik, lalu meneruskan traffic sebagai HTTP biasa ke backend. Dilakukan di reverse proxy agar backend (Apache, Flask) tidak perlu mengelola certificate SSL masing-masing — cukup satu certificate di Nginx untuk semua service, lebih efisien dan mudah dikelola.

- 4 Apa perbedaan name-based dan IP-based virtual hosting?

Perbedaan name-based vs IP-based virtual hosting: Name-based virtual hosting membedakan website berdasarkan domain name (`site1.lab` vs `site2.lab`) meski menggunakan satu IP yang sama. IP-based virtual hosting membutuhkan satu IP address berbeda untuk setiap website. Name-based lebih efisien karena tidak membutuhkan banyak IP address.

- 5 Mengapa self-signed certificate menghasilkan warning di browser?

Mengapa self-signed certificate menghasilkan warning: Browser memverifikasi certificate menggunakan daftar Certificate Authority (CA) terpercaya yang sudah built-in. Self-signed certificate ditandatangani sendiri, bukan oleh CA terpercaya, sehingga browser tidak bisa memverifikasi keasliannya dan menampilkan warning keamanan.

Post-Lab

- 1 Bandingkan response header dari Apache vs Nginx. Header apa yang menunjukkan software web server?

```
hisyam@HISYAMpc:~/docker-lab/web-service$ curl -k -I https://site1.lab:8443
HTTP/1.1 200 OK
Server: nginx/1.29.8
Date: Sat, 09 May 2026 16:24:45 GMT
Content-Type: text/html
Content-Length: 482
Connection: keep-alive
Last-Modified: Sat, 09 May 2026 13:37:33 GMT
ETag: "1e2-651629d99a140"
Accept-Ranges: bytes

hisyam@HISYAMpc:~/docker-lab/web-service$ curl -k -I https://app.lab:8443
HTTP/1.1 200 OK
Server: nginx/1.29.8
Date: Sat, 09 May 2026 16:24:50 GMT
Content-Type: application/json
Content-Length: 140
Connection: keep-alive
```

Kedua endpoint menampilkan Server: nginx/1.29.8 karena semua request melewati Nginx reverse proxy terlebih dahulu sebelum diteruskan ke backend.

Perbedaan yang terlihat:

Header	site1.lab (Apache)	app.lab (Flask)
Server	nginx/1.29.8	nginx/1.29.8
Content-Type	text/html	application/json
Content-Length	482 bytes	140 bytes
Last-Modified	Ada (static file)	Tidak ada
ETag	Ada	Tidak ada

Header Server keduanya menunjukkan Nginx karena Nginx adalah titik masuk pertama. Header Apache asli tersembunyi di belakang proxy. Untuk melihat header Apache asli, perlu akses langsung ke container apache-web tanpa melalui proxy.

- 2 Jika Nginx proxy down, apakah Apache masih bisa diakses langsung?

Bagaimana cara testnya?

```
hisyam@HISYAMpc:~/docker-lab/web-service$ docker exec apache-web wget -q -O- http://localhost/
<!DOCTYPE html>
<html lang="id">
<head><meta charset="UTF-8"><title>Site 1 - Company</title>
<style>body{font-family:sans-serif;text-align:center;padding:50px;background:#1a237e;color:white;}
.box{background:rgba(255,255,255,0.1);padding:30px;border-radius:12px;max-width:500px;margin:0 auto;}</style>
</head>
<body><div class="box">
<h1> Site 1 - Company Profile</h1>
<p>Virtual Host: <strong>site1.lab</strong></p>
<p>Server: Apache httpd di Docker</p>
</div></body></html>
```

Apache **masih bisa diakses langsung** dari dalam container menggunakan `docker exec apache-web wget -q -O- http://localhost/` dan menampilkan halaman Site 1 Company Profile dengan benar.

Namun Apache **tidak bisa diakses dari host/browser** secara langsung karena di `docker-compose.yml` service `apache-web` tidak memiliki ports mapping ke host. Hanya `nginx-proxy` yang expose port 8080 dan 8443 ke host.

Jadi jika Nginx proxy down, user dari luar tidak bisa mengakses Apache sama sekali karena tidak ada jalur lain dari host ke container Apache.

- 3 Tunjukkan bahwa X-Real-IP header diteruskan dengan benar dari Nginx ke Flask.

```
hisyam@HISYAMpc:~/docker-lab/web-service$ curl -k https://app.lab:8443
{"client_ip":"172.21.0.1","hostname":"f47d69cfa611","proto":"https","service":"Flask Backend API","timestamp":"2026-05-09T17:10:39.007640"}
```

Field `client_ip: "172.21.0.1"` membuktikan bahwa header X-Real-IP berhasil diteruskan dari Nginx ke Flask. IP 172.21.0.1 adalah IP gateway dari network `web-net` (IP Nginx proxy). Field `proto: "https"` juga membuktikan header X-Forwarded-Proto diteruskan dengan benar, sehingga Flask tahu request aslinya menggunakan HTTPS meskipun koneksi Nginx→Flask menggunakan HTTP biasa.

- 4 Jelaskan mengapa Flask app perlu terhubung ke dua network (`web-net` dan `db-net`).

Flask perlu terhubung ke dua network karena memiliki dua peran sekaligus:

- **web-net** — agar Flask bisa menerima request yang diteruskan dari Nginx. Tanpa network ini, Nginx tidak bisa reach Flask sehingga app.lab tidak bisa diakses.
- **db-net** — agar Flask bisa konek ke PostgreSQL untuk query data. Tanpa network ini, Flask tidak bisa reach database dan semua operasi CRUD gagal.

Pemisahan network ini juga meningkatkan keamanan. Nginx dan Apache hanya terhubung ke web-net sehingga **tidak bisa langsung akses database**. PostgreSQL hanya terhubung ke db-net sehingga **tidak bisa diakses dari luar kecuali melalui Flask**. Flask menjadi satu-satunya jembatan antara frontend dan database.

5 Apa yang terjadi jika file server.key atau server.crt dihapus saat container running?

Jika server.key atau server.crt dihapus saat container running, Nginx yang sedang berjalan **tidak langsung terpengaruh** karena certificate sudah di-load ke memory saat startup. Request HTTPS yang sedang berjalan tetap dilayani normal.

Namun jika Nginx di-restart atau reload konfigurasi, Nginx akan **gagal start** karena mencoba membaca file certificate yang sudah tidak ada. Akibatnya semua akses HTTPS (site1.lab, site2.lab, app.lab) akan error dan tidak bisa diakses sama sekali.

Solusinya restore file certificate ke folder ./certs/ lalu restart container

MODUL 4: Database Service di Docker — PostgreSQL

LANGKAH PRAKTIKUM

Langkah 0: Persiapan Project

```
hisyam@HISYAMpc:~/docker-lab/web-service$ mkdir -p ~/docker-lab/postgresql/{init,config,backup}
cd ~/docker-lab/postgresql
hisyam@HISYAMpc:~/docker-lab/postgresql$ |
```

Persiapan Project Membuat struktur folder untuk PostgreSQL lab: init untuk script SQL yang dijalankan otomatis saat pertama kali container start, config untuk konfigurasi custom PostgreSQL, dan backup untuk menyimpan hasil backup database. Langkah 1: Deploy PostgreSQL dengan Docker Compose

1.1 Buat init script (dijalankan saat pertama kali)

```
-- Init Script: Database Schema untuk Lab PENS
-- Dijalankan otomatis saat container pertama kali start
-- =====

-- Buat database tambahan
CREATE DATABASE inventory_db;

-- Gunakan database utama (labdb sudah dibuat via env)
\c labdb

-- Buat schema
CREATE SCHEMA IF NOT EXISTS app;

-- Tabel: Mahasiswa
CREATE TABLE app.mahasiswa (
    id SERIAL PRIMARY KEY,
    nrp VARCHAR(15) UNIQUE NOT NULL,
    nama VARCHAR(100) NOT NULL,
    kelas CHAR(1) CHECK (kelas IN ('A', 'B', 'C', 'D')),
    kelompok INTEGER CHECK (kelompok BETWEEN 1 AND 10),
    email VARCHAR(100),
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);
```

Init Script File SQL di folder init/ secara otomatis dijalankan oleh PostgreSQL saat pertama kali container start (jika data volume masih kosong). Script ini membuat schema app, 4 tabel (mahasiswa, matakuliah, nilai, activity_log), index untuk performa query, insert sample data, dan membuat user read-only app_reader.

Tabel activity_log menggunakan tipe data JSONB untuk menyimpan metadata fleksibel, dan diindex menggunakan GIN index yang optimal untuk pencarian di dalam data JSON.

1.2 Buat custom PostgreSQL config

```
hisyam@HISYAMpc:~/docker-lab/postgresql$ cat > config/custom-postgresql.conf << 'EOF'
# =====
# Custom PostgreSQL Configuration untuk Lab
# =====

# Connection
listen_addresses = '*'
max_connections = 50

# Memory (sesuaikan untuk container dengan RAM terbatas)
shared_buffers = 128MB
work_mem = 4MB
maintenance_work_mem = 64MB
effective_cache_size = 256MB

# WAL & Checkpoint
wal_level = replica
max_wal_size = 256MB
min_wal_size = 64MB

# Logging
logging_collector = on
log_directory = '/var/log/postgresql'
log_filename = 'postgresql-%Y-%m-%d.log'
log_statement = 'mod'
log_min_duration_statement = 1000
log_connections = on
log_disconnections = on
log_line_prefix = '%t [%p] %u@%d '

# Locale & Timezone
timezone = 'Asia/Jakarta'
log_timezone = 'Asia/Jakarta'
EOF
```

Custom PostgreSQL Config Konfigurasi custom mengatur beberapa aspek penting: max_connections = 50 membatasi koneksi agar tidak membebani container, shared_buffers = 128MB mengalokasikan memory untuk cache data, log_statement = 'mod' mencatat semua query yang memodifikasi data (INSERT/UPDATE/DELETE), dan timezone diset ke Asia/Jakarta.

1.3 Buat Docker Compose

```
EOF
hisyam@HISYAMpc:~/docker-lab/postgresql$ cat > docker-compose.yml << 'EOF'
services:
  # --- PostgreSQL 16 ---
  db:
    image: postgres:16-alpine
    container_name: postgres-db
    environment:
      POSTGRES_DB: labdb
      POSTGRES_USER: labuser
      POSTGRES_PASSWORD: labpass123
      TZ: Asia/Jakarta
    ports:
      - "5432:5432"
    volumes:
      - pg-data:/var/lib/postgresql/data
      - ./init:/docker-entrypoint-initdb.d:ro
      - ./config/custom-postgresql.conf:/etc/postgresql/custom.conf:ro
```

Docker Compose Stack terdiri dari 2 service:

- db — PostgreSQL dengan volume untuk data, init script, config custom, backup, dan log
- pgadmin — GUI web untuk mengelola database, menggunakan `depends_on` dengan service `healthy` agar pgAdmin tidak start sebelum PostgreSQL benar-benar siap

1.4 Deploy

```
hisyam@HISYAMpc:~/docker-lab/postgresql$ docker compose up -d
[+] up 3/3
  ✓ Network postgresql_db-net Created 0.0s
  ✓ Container postgres-db Healthy 11.0s
  ✓ Container pgadmin4 Started 11.2s
hisyam@HISYAMpc:~/docker-lab/postgresql$ docker compose ps
```

NAME	IMAGE	COMMAND	SERVICE	CREATED	STATUS	PORTS
pgadmin4	dpape/pgadmin4:latest	"/entrypoint.sh"	pgadmin	About a minute ago	Up About a minute	80/tcp, 443/tcp, 0.0.0.0:5050->5050/tcp, [::]:5050->5050/tcp
postgres-db	postgres:16-alpine	"docker-entrypoint.s..."	db	About a minute ago	Up About a minute (healthy)	0.0.0.0:5432->5432/tcp, [::]:5432->5432/tcp

Deploy `docker compose up -d` menjalankan kedua service. Init script otomatis dieksekusi hanya sekali saat volume pertama kali dibuat.

Tunggu hingga healthcheck pass

```
hisyam@HISYAMpc:~/docker-lab/postgresql$ docker compose logs db | tail -20
postgres-db | ('3122600005', 'Eka Putra', 'C', 3, 'eka@student.pens.ac.id');
postgres-db | 2026-05-10 14:39:04 WIB [10104] labuser@labdb LOG: statement: INSERT INTO app.nilai (mahasiswa_id, matakuliah_id, nilai_angka, grade, semest
er) VALUES
postgres-db | (1, 1, 85.50, 'A', '2025-1'),
postgres-db | (1, 2, 78.00, 'B+', '2025-1'),
postgres-db | (2, 1, 92.00, 'A', '2025-1'),
postgres-db | (3, 1, 70.25, 'B', '2025-1'),
postgres-db | (4, 3, 88.75, 'A', '2025-1');
postgres-db | 2026-05-10 14:39:04 WIB [10104] labuser@labdb LOG: statement: CREATE USER app_reader WITH PASSWORD 'reader123';
postgres-db | 2026-05-10 14:39:04 WIB [10104] labuser@labdb LOG: statement: GRANT USAGE ON SCHEMA app TO app_reader;
postgres-db | 2026-05-10 14:39:04 WIB [10104] labuser@labdb LOG: statement: GRANT SELECT ON ALL TABLES IN SCHEMA app TO app_reader;
postgres-db | 2026-05-10 14:39:04 WIB [10104] labuser@labdb LOG: statement: ALTER DEFAULT PRIVILEGES IN SCHEMA app GRANT SELECT ON TABLES TO app_reader;
postgres-db | 2026-05-10 14:39:04 WIB [10104] labuser@labdb ERROR: syntax error at or near "RAISE" at character 1
postgres-db | 2026-05-10 14:39:04 WIB [10104] labuser@labdb STATEMENT: RAISE NOTICE 'Database initialization completed successfully!';
postgres-db | 2026-05-10 14:39:04 WIB [10104] labuser@labdb LOG: disconnection: session time: 0:00:00.123 user=labuser database=labdb host=[local]
postgres-db | 2026-05-10 14:39:14 WIB [10119] [unknown]@[unknown] LOG: connection received: host=[local]
postgres-db | 2026-05-10 14:39:14 WIB [10119] labuser@labdb LOG: connection authorized: user=labuser database=labdb application_name=pg_isready
postgres-db | 2026-05-10 14:39:14 WIB [10119] labuser@labdb LOG: disconnection: session time: 0:00:00.003 user=labuser database=labdb host=[local]
postgres-db | 2026-05-10 14:39:24 WIB [10126] [unknown]@[unknown] LOG: connection received: host=[local]
postgres-db | 2026-05-10 14:39:24 WIB [10126] labuser@labdb LOG: connection authorized: user=labuser database=labdb application_name=pg_isready
postgres-db | 2026-05-10 14:39:24 WIB [10126] labuser@labdb LOG: disconnection: session time: 0:00:00.003 user=labuser database=labdb host=[local]
```

Langkah 2: Koneksi dan Verifikasi Database

2.1 Koneksi via psql dari host

Install psql client (jika belum)

```
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  libpq5 postgresql-client-16 postgresql-client-common
Suggested packages:
  postgresql-16 postgresql-doc-16
The following NEW packages will be installed:
  libpq5 postgresql-client postgresql-client-16 postgresql-client-common
0 upgraded, 4 newly installed, 0 to remove and 16 not upgraded.
Need to get 1492 kB of archives.
After this operation, 4964 kB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libpq5 amd64 16.13-0ubuntu0.24.04.1 [146 kB]
Get:2 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 postgresql-client-common all 257build1.1 [36.4 kB]
Get:3 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 postgresql-client-16 amd64 16.13-0ubuntu0.24.04.1 [1298 kB]
Get:4 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 postgresql-client all 16+257build1.1 [11.6 kB]
Fetched 1492 kB in 1s (1041 kB/s)
Selecting previously unselected package libpq5:amd64.
(Reading database ... 41325 files and directories currently installed.)
Preparing to unpack .../libpq5_16.13-0ubuntu0.24.04.1_amd64.deb ...
Unpacking libpq5:amd64 (16.13-0ubuntu0.24.04.1) ...
Selecting previously unselected package postgresql-client-common.
Preparing to unpack .../postgresql-client-common_257build1.1_all.deb ...
Unpacking postgresql-client-common (257build1.1) ...
Selecting previously unselected package postgresql-client-16.
Preparing to unpack .../postgresql-client-16_16.13-0ubuntu0.24.04.1_amd64.deb ...
Unpacking postgresql-client-16 (16.13-0ubuntu0.24.04.1) ...
Selecting previously unselected package postgresql-client.
Preparing to unpack .../postgresql-client_16+257build1.1_all.deb ...
Unpacking postgresql-client (16+257build1.1) ...
```

Koneksi ke database

```
hisyam@HISYAMpc:~/docker-lab/postgresql$ psql -h localhost -U labuser -d labdb
Password for user labuser:
psql (16.13 (Ubuntu 16.13-0ubuntu0.24.04.1))
Type "help" for help.

labdb=# |
```

Di dalam psql:

-- list databases

```
labdb=# \l
          List of databases
  Name      | Owner   | Encoding | Locale Provider | Collate | Ctype   | ICU Locale | ICU Rules | Access privileges
-----+-----+-----+-----+-----+-----+-----+-----+-----+-----
inventory_db | labuser | UTF8     | libc            | en_US.utf8 | en_US.utf8 |             |           | 
labdb        | labuser | UTF8     | libc            | en_US.utf8 | en_US.utf8 |             |           | 
postgres     | labuser | UTF8     | libc            | en_US.utf8 | en_US.utf8 |             |           | 
template0    | labuser | UTF8     | libc            | en_US.utf8 | en_US.utf8 |             |           | =c/labuser,+labuser=CTc/labuser
template1    | labuser | UTF8     | libc            | en_US.utf8 | en_US.utf8 |             |           | =c/labuser,+labuser=CTc/labuser
(5 rows)
```

-- list schemas

```
labdb=# \dn
          List of schemas
  Name      | Owner
-----+-----
app         | labuser
public      | pg_database_owner
(2 rows)
```

-- list tabel di schema app

```
labdb=# \dt app.*
              List of relations
 Schema |      Name      | Type  | Owner
-----+-----+-----+-----
 app    | activity_log    | table | labuser
 app    | mahasiswa       | table | labuser
 app    | matakuliah      | table | labuser
 app    | nilai           | table | labuser
(4 rows)
```

-- describe tabel mahasiswa

```

Column |      Type      | Collation | Nullable |      Default      | Storage | Compression | Stats target | Descri
ption  +-----+
id      | integer        |           | not null | nextval('app.mahasiswa_id_seq'::regclass) | plain   |              |              |
nrp     | character varying(15) |           | not null |                  | extended |              |              |
nama    | character varying(100) |           | not null |                  | extended |              |              |
kelas   | character(1)     |           |          |                  | extended |              |              |
kelompok | integer        |           |          |                  | plain   |              |              |
email   | character varying(100) |           |          |                  | extended |              |              |
created_at | timestamp without time zone |           |          | CURRENT_TIMESTAMP | plain   |              |              |
Indexes:
    "mahasiswa_pkey" PRIMARY KEY, btree (id)
    "idx_mahasiswa_kelas" btree (kelas)
    "idx_mahasiswa_nrp" btree (nrp)
    "mahasiswa_nrp_key" UNIQUE CONSTRAINT, btree (nrp)
Check constraints:
    "mahasiswa_kelas_check" CHECK (kelas = ANY (ARRAY['A'::bpchar, 'B'::bpchar, 'C'::bpchar, 'D'::bpchar]))
    "mahasiswa_kelompok_check" CHECK (kelompok >= 1 AND kelompok <= 10)
Referenced by:
    TABLE "app.nilai" CONSTRAINT "nilai_mahasiswa_id_fkey" FOREIGN KEY (mahasiswa_id) REFERENCES app.mahasiswa(id) ON DELETE CASCADE
Access method: heap
~
~
~
```

```
labdb=# SELECT * FROM app.mahasiswa;
 id | nrp      |  nama  | kelas | kelompok |      email      |      created_at
-----+-----+
 1 | 3122600001 | Ahmad Fauzi | A      | 1 | ahmad@student.pens.ac.id | 2026-05-10 14:39:04.238131
 2 | 3122600002 | Budi Santoso | A      | 1 | budi@student.pens.ac.id  | 2026-05-10 14:39:04.238131
 3 | 3122600003 | Citra Dewi  | B      | 2 | citra@student.pens.ac.id | 2026-05-10 14:39:04.238131
 4 | 3122600004 | Dian Pratama | B      | 2 | dian@student.pens.ac.id  | 2026-05-10 14:39:04.238131
 5 | 3122600005 | Eka Putra   | C      | 3 | eka@student.pens.ac.id   | 2026-05-10 14:39:04.238131
(5 rows)
```

```
labdb=# SELECT * FROM app.matakuliah;
 id | kode |      nama      | sks
-----+-----+
 1 | JAR01 | Administrasi Jaringan | 3
 2 | SBD01 | Sistem Basis Data    | 3
 3 | SO01  | Sistem Operasi       | 2
 4 | WEB01 | Pemrograman Web      | 3
(4 rows)
```

-- keluar

```
labdb=# \q
hisyam@HISYAMpc:~/docker-lab/postgresql$ |
```

Koneksi via psql dari host Menggunakan psql client yang diinstall di WSL untuk konek ke PostgreSQL yang berjalan di container via port 5432 yang di-expose ke host. Perintah \l, \dn, \dt adalah meta-command psql untuk eksplorasi struktur database.

2.2 Koneksi via docker exec

Masuk ke psql di dalam container

```
labdb=# \q
hisyam@HISYAMpc:~/docker-lab/postgresql$ docker exec -it postgres-db psql -U labuser -d labdb
psql (16.13)
Type "help" for help.

labdb=# |
```

Query join: Nilai mahasiswa

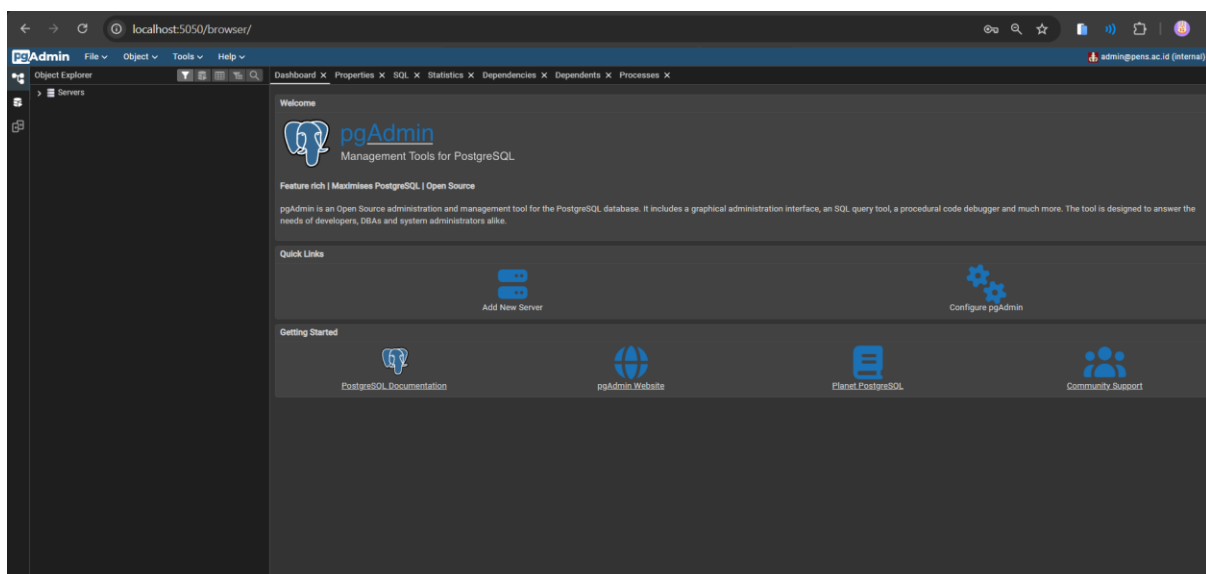
```
labdb=# SELECT m.nrp, m.nama, mk.nama AS matakuliah, n.nilai_angka, n.grade
labdb=# kuliah mk ON n.matakuliah_id = mFROM app.nilai n
labdb=# JOIN app.mahasiswa m ON n.mahasiswa_id = m.id
labdb=# JOIN app.matakuliah mk ON n.matakuliah_id = mk.id
labdb=# ORDER BY m.nrp, mk.nama;
   nrp   |   nama   | matakuliah   | nilai_angka | grade
-----+-----+-----+-----+-----
3122600001 | Ahmad Fauzi | Administrasi Jaringan | 85.50 | A
3122600001 | Ahmad Fauzi | Sistem Basis Data | 78.00 | B+
3122600002 | Budi Santoso | Administrasi Jaringan | 92.00 | A
3122600003 | Citra Dewi | Administrasi Jaringan | 70.25 | B
3122600004 | Dian Pratama | Sistem Operasi | 88.75 | A
(5 rows)

labdb=# \q
hisyam@HISYAMpc:~/docker-lab/postgresql$ |
```

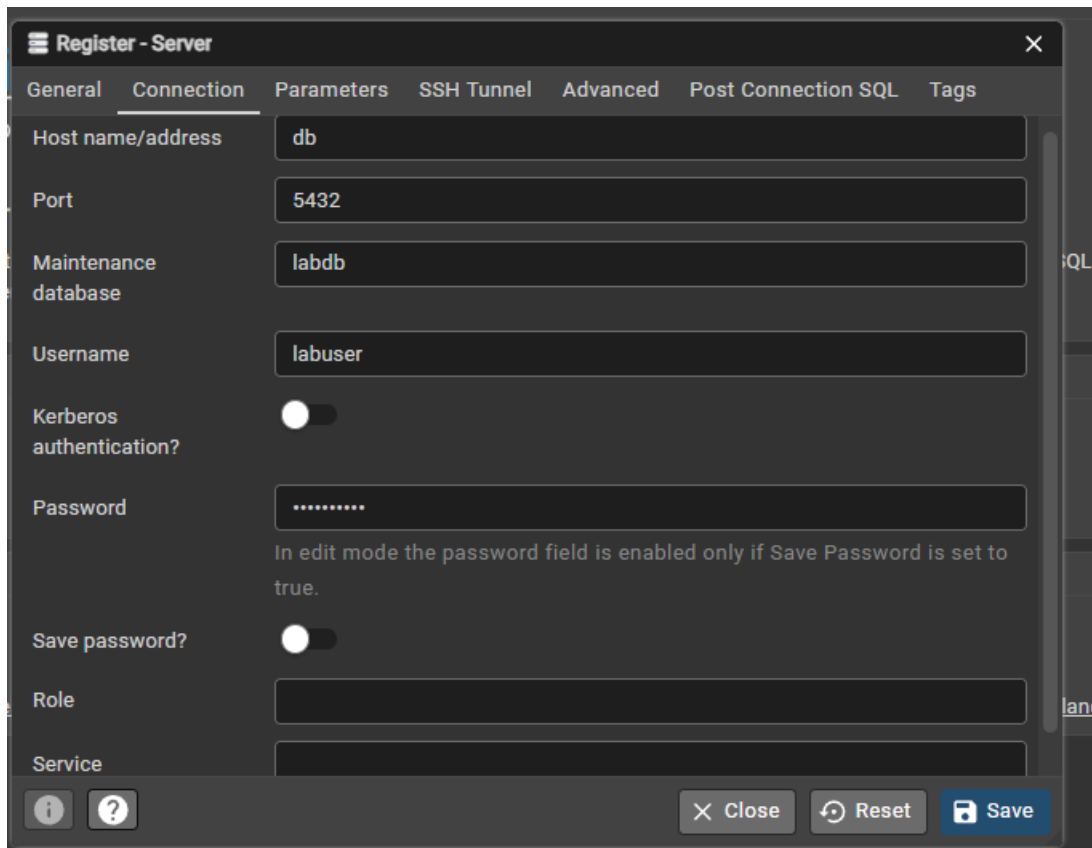
Koneksi via docker exec Cara alternatif masuk ke psql langsung dari dalam container tanpa perlu install psql di host. Query JOIN digunakan untuk menampilkan data nilai mahasiswa lengkap dengan nama dan mata kuliah.

2.3 Koneksi via pgAdmin4

Buka browser:

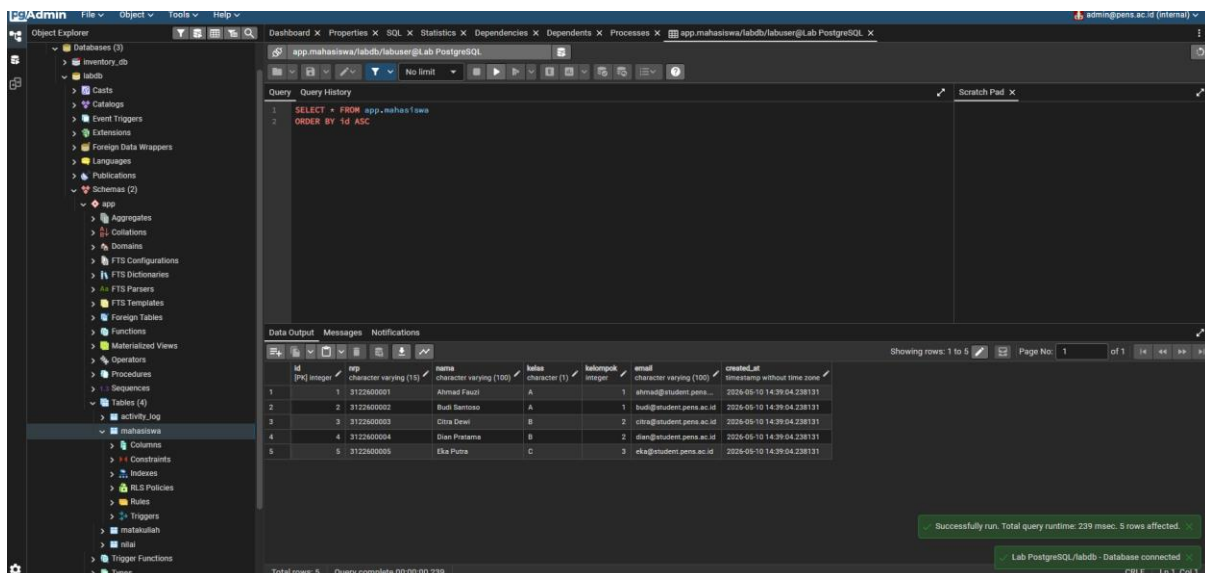


Add New Server:



Navigate: Servers → Lab PostgreSQL → Databases → labdb → Schemas → app → Tables

Klik kanan tabel mahasiswa → View/Edit Data → All Rows



Koneksi via pgAdmin4 pgAdmin adalah GUI berbasis web untuk mengelola PostgreSQL. Diakses via browser di <http://localhost:5050>. Saat menambah server, hostname menggunakan nama service Docker (db) bukan localhost karena pgAdmin dan PostgreSQL berada dalam satu Docker network db-net.

Langkah 3: Operasi CRUD SQL

```
hisyam@HISYAMpc:~/docker-lab/postgresql$ docker exec postgres-db psql -U labuser -d labdb << 'SQLEOF'
-- === CREATE ===
INSERT INTO app.mahasiswa (nrp, nama, kelas, kelompok, email)
VALUES ('3122600010', 'Fajar Rizki', 'D', 5, 'fajar@student.pens.ac.id');

-- === READ ===
SELECT * FROM app.mahasiswa WHERE kelas = 'A';

SELECT mk.nama, AVG(n.nilai_angka)::NUMERIC(5,2) AS rata_rata, COUNT(*) AS jumlah
FROM app.nilai n
JOIN app.matakuliah mk ON n.matakuliah_id = mk.id
GROUP BY mk.nama
ORDER BY rata_rata DESC;

-- === UPDATE ===
UPDATE app.mahasiswa SET email = 'fajar.rizki@student.pens.ac.id'
WHERE nrp = '3122600010';

-- === DELETE ===
DELETE FROM app.mahasiswa WHERE nrp = '3122600010';

-- === JSONB ===
INSERT INTO app.activity_log (level, source, message, metadata)
VALUES ('INFO', 'web-app', 'User login', '{"user": "admin", "ip": "192.168.1.10"}');

SELECT * FROM app.activity_log
WHERE metadata->>'user' = 'admin';

SQLEOF
```

Menjalankan operasi dasar database:

- **CREATE** — INSERT mahasiswa baru dengan data lengkap
- **READ** — SELECT dengan filter WHERE dan query agregasi AVG + GROUP BY untuk menghitung rata-rata nilai per mata kuliah
- **UPDATE** — mengubah email mahasiswa berdasarkan NRP
- **DELETE** — menghapus data mahasiswa
- **JSONB query** — INSERT data ke activity_log dengan metadata JSON, lalu SELECT menggunakan operator ->> untuk mengakses field di dalam JSON

Langkah 4: Backup dan Restore

4.1 Backup database (pg_dump)

Backup dalam format custom (compressed, restorable)

```
hisyam@HISYAMpc:~/docker-lab/postgresql$ docker exec postgres-db pg_dump -U labuser -d labdb -Fc \
-f /backup/labdb_backup.dump
```

Backup dalam format SQL plain text

```
hisyam@HISYAMpc:~/docker-lab/postgresql$ docker exec postgres-db pg_dump -U labuser -d labdb \
-f /backup/labdb_backup.sql
```

Backup hanya schema app

```
hisyam@HISYAMpc:~/docker-lab/postgresql$ docker exec postgres-db pg_dump -U labuser -d labdb -n app -Fc \
-f /backup/labdb_schema_app.dump
```

Verifikasi file backup di host

```
hisyam@HISYAMpc:~/docker-lab/postgresql$ ls -la backup/
total 52
drwxr-xr-x 2 hisyam hisyam 4096 May 10 16:19 .
drwxr-xr-x 5 hisyam hisyam 4096 May 10 14:43 ..
-rw-r--r-- 1 root    root    14131 May 10 16:18 labdb_backup.dump
-rw-r--r-- 1 root    root    10364 May 10 16:19 labdb_backup.sql
-rw-r--r-- 1 root    root    14131 May 10 16:19 labdb_schema_app.dump
hisyam@HISYAMpc:~/docker-lab/postgresql$
```

Backup

- Format custom (-Fc) — format binary terkompresi, hanya bisa di-restore dengan pg_restore, ukuran lebih kecil
- Format SQL plain text — output berupa SQL statement yang bisa dibaca dan dijalankan manual
- Backup per-schema (-n app) — hanya backup schema tertentu, berguna jika ingin backup parsial

4.2 Restore database

Buat database baru untuk restore test

```
hisyam@HISYAMpc:~/docker-lab/postgresql$ docker exec postgres-db psql -U labuser -d postgres -c "CREATE DATABASE labdb_restore;"
CREATE DATABASE
```

Restore dari custom format

```
hisyam@HISYAMpc:~/docker-lab/postgresql$ docker exec postgres-db pg_restore -U labuser -d labdb_restore \
/backup/labdb_backup.dump
```

Verifikasi restore

```
hisyam@HISYAMpc:~/docker-lab/postgresql$ docker exec postgres-db psql -U labuser -d labdb_restore -c "SELECT * FROM app.mahasiswa;"
 id | nrp      | nama      | kelas | kelompok | email                               | created_at
-----+-----+-----+-----+-----+-----+-----
  1 | 3122600001 | Ahmad Fauzi | A      | 1         | ahmad@student.pens.ac.id          | 2026-05-10 14:39:04.238131
  2 | 3122600002 | Budi Santoso | A      | 1         | budi@student.pens.ac.id           | 2026-05-10 14:39:04.238131
  3 | 3122600003 | Citra Dewi   | B      | 2         | citra@student.pens.ac.id          | 2026-05-10 14:39:04.238131
  4 | 3122600004 | Dian Pratama | B      | 2         | dian@student.pens.ac.id           | 2026-05-10 14:39:04.238131
  5 | 3122600005 | Eka Putra    | C      | 3         | eka@student.pens.ac.id            | 2026-05-10 14:39:04.238131
(5 rows)
```

Restore Membuat database baru labdb_restore lalu restore dari file backup custom format menggunakan pg_restore. Verifikasi dengan SELECT memastikan data berhasil di-restore.

4.3 Backup otomatis dengan cron di container

Buat script backup

```
hisyam@HISYAMpc:~/docker-lab/postgresql$ cat > backup/auto-backup.sh << 'BASH'
#!/bin/sh
TIMESTAMP=$(date +%Y%m%d_%H%M%S)
BACKUP_FILE="/backup/labdb_${TIMESTAMP}.dump"
pg_dump -U labuser -d labdb -Fc -f "$BACKUP_FILE"
echo "[$(date)] Backup created: $BACKUP_FILE"

# Hapus backup lebih dari 7 hari
find /backup -name "labdb_*.dump" -mtime +7 -delete
BASH
hisyam@HISYAMpc:~/docker-lab/postgresql$ chmod +x backup/auto-backup.sh
```

Test jalankan manual

```
hisyam@HISYAMpc:~/docker-lab/postgresql$ docker exec postgres-db /backup/auto-backup.sh
[Sun May 10 16:21:48 WIB 2026] Backup created: /backup/labdb_20260510_162147.dump
hisyam@HISYAMpc:~/docker-lab/postgresql$ ls -la backup/
total 72
drwxr-xr-x 2 hisyam hisyam 4096 May 10 16:21 .
drwxr-xr-x 5 hisyam hisyam 4096 May 10 14:43 ..
-rwxr-xr-x 1 hisyam hisyam 271 May 10 16:21 auto-backup.sh
-rw-r--r-- 1 root root 14131 May 10 16:21 labdb_20260510_162147.dump
-rw-r--r-- 1 root root 14131 May 10 16:18 labdb_backup.dump
-rw-r--r-- 1 root root 10364 May 10 16:19 labdb_backup.sql
-rw-r--r-- 1 root root 14131 May 10 16:19 labdb_schema_app.dump
```

Backup Otomatis Script shell yang dibuat di folder backup/ dan dijalankan di dalam container. Script menambahkan timestamp ke nama file backup dan otomatis menghapus backup yang lebih dari 7 hari menggunakan find -mtime +7.

Langkah 5: Monitoring PostgreSQL

5.1 Statistik database

```
hisyam@HISYAMpc:~/docker-lab/postgresql$ docker exec postgres-db psql -U labuser -d labdb << 'SQLEOF'

-- Ukuran database
SELECT pg_database.datname,
       pg_size_pretty(pg_database_size(pg_database.datname)) AS size
FROM pg_database ORDER BY pg_database_size(pg_database.datname) DESC;

-- Ukuran per tabel
SELECT schemaname || '.' || tablename AS table_full,
       pg_size_pretty(pg_total_relation_size(schemaname || '.' || tablename)) AS total_size
FROM pg_tables WHERE schemaname = 'app'
ORDER BY pg_total_relation_size(schemaname || '.' || tablename) DESC;

-- Koneksi aktif
SELECT pid, username, datname, client_addr, state, query_start, query
FROM pg_stat_activity WHERE datname = 'labdb';

-- Statistik tabel (hits, reads, cache ratio)
SELECT relname,
       seq_scan, seq_tup_read,
       idx_scan, idx_tup_fetch,
       n_tup_ins, n_tup_upd, n_tup_del
FROM pg_stat_user_tables WHERE schemaname = 'app';

SQLEOF
```

Statistik database Query ke tabel sistem PostgreSQL (pg_database, pg_tables, pg_stat_activity, pg_stat_user_tables) untuk monitoring:

- Ukuran database dan per-tabel dalam format human-readable
- Koneksi aktif beserta query yang sedang berjalan
- Statistik akses tabel: berapa kali sequential scan vs index scan digunakan

5.2 Cek PostgreSQL log

Lihat log PostgreSQL

```
hisyam@HISYAMpc:~/docker-lab/postgresql$ docker exec postgres-db ls -la /var/log/postgresql/
total 8
drwxr-xr-x  2 postgres postgres 4096 May 10 16:34 .
drwxr-xr-x  1 root    root      4096 May 10 16:40 ..
```

```
hisyam@HISYAMpc:~/docker-lab/postgresql$ docker exec postgres-db cat /var/log/postgresql/postgresql-$(date +%Y-%m-%d).log | tail -20
2026-05-10 16:45:26 WIB [1] @ LOG:  starting PostgreSQL 16.13 on x86_64-pc-linux-musl, compiled by gcc (Alpine 15.2.0) 15.2.0, 64-bit
2026-05-10 16:45:26 WIB [1] @ LOG:  listening on IPv4 address "0.0.0.0", port 5432
2026-05-10 16:45:26 WIB [1] @ LOG:  listening on IPv6 address "::", port 5432
2026-05-10 16:45:26 WIB [1] @ LOG:  listening on Unix socket "/var/run/postgresql/.s.PGSQL.5432"
2026-05-10 16:45:26 WIB [34] @ LOG:  database system was shut down at 2026-05-10 16:45:26 WIB
2026-05-10 16:45:26 WIB [1] @ LOG:  database system is ready to accept connections
2026-05-10 16:45:37 WIB [44] [unknown]@[unknown] LOG:  connection received: host=[local]
2026-05-10 16:45:37 WIB [44] labuser@labdb LOG:  connection authorized: user=labuser database=labdb application_name=psql
2026-05-10 16:45:37 WIB [44] labuser@labdb LOG:  disconnection: session time: 0:00:00.019 user=labuser database=labdb host=[local]
2026-05-10 16:45:37 WIB [44] [unknown]@[unknown] LOG:  connection received: host=[local]
2026-05-10 16:45:40 WIB [51] labuser@labdb LOG:  connection authorized: user=labuser database=labdb application_name=pg_isready
2026-05-10 16:45:40 WIB [51] labuser@labdb LOG:  disconnection: session time: 0:00:00.004 user=labuser database=labdb host=[local]
```

Cek log Log PostgreSQL disimpan di /var/log/postgresql/ di dalam container, dengan nama file berdasarkan tanggal sesuai konfigurasi log_filename.

5. PERTANYAAN

Pre-Lab

1. Apa fungsi file/folder /docker-entrypoint-initdb.d/ di image PostgreSQL?

Fungsi /docker-entrypoint-initdb.d/: Folder khusus di image PostgreSQL yang berisi script SQL atau shell yang dijalankan **otomatis** saat container pertama kali start dan volume data masih kosong. Script diurutkan berdasarkan nama file (alfabetis), sehingga penamaan 01-create-schema.sql memastikan urutan eksekusi yang benar. Script tidak dijalankan ulang jika volume sudah berisi data.

2. Mengapa POSTGRES_PASSWORD wajib diset? Apa risikonya jika tidak ada password?

Mengapa POSTGRES_PASSWORD wajib: PostgreSQL memerlukan autentikasi untuk keamanan. Tanpa password, siapapun yang bisa reach container bisa akses database tanpa autentikasi. Risikonya adalah data bisa dibaca, dimodifikasi, atau dihapus oleh pihak yang tidak berwenang. Image PostgreSQL akan menolak start jika tidak ada password yang diset.

3. Jelaskan perbedaan antara pg_dump format custom (-Fc) dan format SQL plain text.

Perbedaan format custom vs SQL plain text: Format custom (-Fc) menghasilkan file binary terkompresi yang lebih kecil, mendukung restore paralel, dan bisa di-restore secara selektif (per tabel). Format SQL plain text menghasilkan file berisi SQL statement yang bisa dibaca manusia dan dieksekusi langsung, tapi ukurannya lebih besar karena tidak terkompresi.

4. Apa itu shared_buffers dan mengapa perlu disesuaikan untuk container?

shared_buffers dan penyesuaian untuk container: shared_buffers adalah alokasi memory untuk cache data PostgreSQL di RAM. Semakin besar nilainya, semakin banyak data yang bisa di-cache sehingga query lebih cepat. Untuk container perlu disesuaikan agar tidak melebihi memory yang dialokasikan ke container — jika terlalu besar container bisa OOM (Out of Memory) dan crash.

5. Mengapa data PostgreSQL harus disimpan di Docker Volume, bukan di container layer?

Mengapa data harus di Docker Volume: Container layer bersifat ephemeral — semua data hilang saat container dihapus. Volume menyimpan data di luar container sehingga data persist meski container dihapus, di-recreate, atau di-update ke versi baru. Untuk database ini sangat kritis karena kehilangan data database berarti kehilangan seluruh informasi yang tersimpan.

Post-Lab

1. Jalankan docker compose down lalu docker compose up -d. Apakah data mahasiswa masih ada? Buktikan.

```
hisyam@HISYAMpc:~/docker-lab/postgresql$ docker compose down
[+] down 3/3
✓Container pgadmin4      Removed
✓Container postgres-db   Removed
✓Network postgresql_db-net Removed
hisyam@HISYAMpc:~/docker-lab/postgresql$ docker compose up -d
[+] up 3/3
✓Network postgresql_db-net Created
✓Container postgres-db   Healthy
✓Container pgadmin4      Started
hisyam@HISYAMpc:~/docker-lab/postgresql$ docker exec postgres-db psql -U labuser -d labdb -c "SELECT * FROM app.mahasiswa;"
id | nrp      | nama      | kelas | kelompok | email                      | created_at
---+-----+-----+-----+-----+-----+-----
 1 | 3122600001 | Ahmad Fauzi | A     | 1 | ahmad@student.pens.ac.id | 2026-05-10 14:39:04.238131
 2 | 3122600002 | Budi Santoso | A     | 1 | budi@student.pens.ac.id  | 2026-05-10 14:39:04.238131
 3 | 3122600003 | Citra Dewi  | B     | 2 | citra@student.pens.ac.id | 2026-05-10 14:39:04.238131
 4 | 3122600004 | Dian Pratama | B     | 2 | dian@student.pens.ac.id  | 2026-05-10 14:39:04.238131
 5 | 3122600005 | Eka Putra   | C     | 3 | eka@student.pens.ac.id   | 2026-05-10 14:39:04.238131
(5 rows)
```

Setelah docker compose down lalu docker compose up -d, semua 5 data mahasiswa masih ada. Ini membuktikan bahwa docker compose down

hanya menghapus container dan network, **tidak menghapus volume** pg-data. Data PostgreSQL tetap tersimpan di volume dan di-mount kembali saat container baru dibuat.

2. Jalankan docker compose down -v lalu docker compose up -d. Apa yang terjadi? Apakah init script dijalankan ulang?

```
hisyam@HISYAMpc:~/docker-lab/postgresql$ docker compose down -v
[+] down 6/6
✓Container pgadmin4      Removed    1.7s
✓Container postgres-db   Removed    0.4s
✓Volume postgresql_pg-data Removed    0.1s
✓Volume postgresql_pg-logs Removed    0.0s
✓Volume postgresql_pgadmin-data Removed    0.1s
✓Network postgresql_db-net Removed    0.3s
hisyam@HISYAMpc:~/docker-lab/postgresql$ docker compose up -d
[+] up 6/6
✓Network postgresql_db-net Created      0.0s
✓Volume postgresql_pg-data Created      0.0s
✓Volume postgresql_pg-logs Created      0.0s
✓Volume postgresql_pgadmin-data Created    0.0s
✓Container postgres-db   Healthy    13.4s
✓Container pgadmin4      Started    13.6s
hisyam@HISYAMpc:~/docker-lab/postgresql$ docker exec postgres-db psql -U labuser -d labdb -c "SELECT * FROM app.mahasiswa;"
id | nrp | nama | kelas | kelompok | email | created_at
---+---+---+---+---+---+---
1 | 3122600001 | Ahmad Fauzi | A | 1 | ahmad@student.pens.ac.id | 2026-05-10 16:49:40.331798
2 | 3122600002 | Budi Santoso | A | 1 | budi@student.pens.ac.id | 2026-05-10 16:49:40.331798
3 | 3122600003 | Citra Dewi | B | 2 | citra@student.pens.ac.id | 2026-05-10 16:49:40.331798
4 | 3122600004 | Dian Pratama | B | 2 | dian@student.pens.ac.id | 2026-05-10 16:49:40.331798
5 | 3122600005 | Eka Putra | C | 3 | eka@student.pens.ac.id | 2026-05-10 16:49:40.331798
```

Setelah docker compose down -v semua volume dihapus (pg-data, pg-logs, pgadmin-data). Saat docker compose up -d dijalankan, volume baru dibuat kosong sehingga PostgreSQL mendeteksi instalasi baru dan **init script 01-create-schema.sql dijalankan ulang otomatis**.

Terbukti dari created_at data mahasiswa berubah dari 14:39 menjadi 16:49 — artinya data lama hilang dan diganti data sample baru dari init script.

3. Bandingkan ukuran file backup format custom vs SQL. Mana yang lebih kecil dan mengapa?

```
hisyam@HISYAMpc:~/docker-lab/postgresql$ docker exec postgres-db pg_dump -U labuser -d labdb -Fc -f /backup/labdb_backup.dump
hisyam@HISYAMpc:~/docker-lab/postgresql$ docker exec postgres-db pg_dump -U labuser -d labdb -f /backup/labdb_backup.sql
ls -lh backup/
total 64K
-rwxr-xr-x 1 hisyam hisyam 271 May 10 16:21 auto-backup.sh
-rw-r--r-- 1 root root 14K May 10 16:21 labdb_20260510_162147.dump
-rw-r--r-- 1 root root 14K May 10 16:52 labdb_backup.dump
-rw-r--r-- 1 root root 11K May 10 16:52 labdb_backup.sql
-rw-r--r-- 1 root root 14K May 10 16:19 labdb_schema_app.dump
```

di lab ini ukuran SQL plain text (11K) justru **lebih kecil** dari custom format (14K) karena data yang ada masih sangat sedikit (hanya 5 mahasiswa, 4 matakuliah).

Pada data kecil, overhead header binary format custom lebih besar dari keuntungan kompresinya. Namun pada **data yang besar**, format custom akan jauh lebih kecil karena kompresi lebih efektif. Format custom juga mendukung restore paralel dan selektif per tabel.

4. Buat query yang menampilkan mahasiswa yang belum memiliki nilai di semester apapun.

```
hisyam@HISYAMpc:~/docker-lab/postgresql$ docker exec postgres-db psql -U labuser -d labdb -c "SELECT m.nrp, m.nama, m.kelas FROM app.mahasiswa m LEFT JOIN a
pp.nilai n ON m.id = n.mahasiswa_id WHERE n.id IS NULL;"
      nrp      |      nama      |      kelas
-----+-----+-----
 3122600005    | Eka Putra     | C
(1 row)
```

5. Jelaskan peran user app_reader yang dibuat di init script. Apa bedanya dengan labuser?

```
hisyam@HISYAMpc:~/docker-lab/postgresql$ docker exec -it postgres-db psql -U app_reader -d labdb
psql (16.13)
Type "help" for help.

labdb=> SELECT * FROM app.mahasiswa;
 id | nrp      |      nama      | kelas | kelompok |      email      |      created_at
-----+-----+-----+-----+-----+-----+-----
  1 | 3122600001 | Ahmad Fauzi   | A     |         1 | ahmad@student.pens.ac.id | 2026-05-10 16:49:40.331
  2 | 3122600002 | Budi Santoso  | A     |         1 | budi@student.pens.ac.id  | 2026-05-10 16:49:40.331
  3 | 3122600003 | Citra Dewi    | B     |         2 | citra@student.pens.ac.id | 2026-05-10 16:49:40.331
  4 | 3122600004 | Dian Pratama  | B     |         2 | dian@student.pens.ac.id  | 2026-05-10 16:49:40.331
  5 | 3122600005 | Eka Putra     | C     |         3 | eka@student.pens.ac.id   | 2026-05-10 16:49:40.331
(5 rows)

labdb=> INSERT INTO app.mahasiswa (nrp, nama) VALUES ('999', 'Test');
ERROR:  permission denied for table mahasiswa
labdb=>
```

app_reader dibuat di init script dengan perintah:

```
CREATE USER app_reader WITH PASSWORD 'reader123';
```

```
GRANT USAGE ON SCHEMA app TO app_reader;
```

```
GRANT SELECT ON ALL TABLES IN SCHEMA app TO app_reader;
```

Perbedaan app_reader vs labuser:

Aspek	labuser	app_reader
Role	Owner/superuser database	Read-only user
Hak akses	CREATE, INSERT, UPDATE, DELETE, DROP	SELECT saja
Bisa modifikasi data	Ya	Tidak
Bisa buat tabel/schema	Ya	Tidak
Password	labpass123	reader123

Aspek	labuser	app_reader
Penggunaan	Admin, migration, operasi penuh	Aplikasi yang hanya butuh baca data
Risiko jika credential bocor	Sangat tinggi	Rendah

app_reader dibuat mengikuti prinsip **least privilege** - setiap user/aplikasi hanya diberi akses minimal yang dibutuhkan. Jika credential app_reader bocor, attacker hanya bisa membaca data tanpa bisa memodifikasi atau menghapusnya.

MODUL 5: Logging Service Docker dengan PostgreSQL

LANGKAH PRAKTIKUM

Langkah 0: Persiapan Project

```
hisyam@HISYAMpc:~$ mkdir -p ~/docker-lab/logging/{fluent-bit,app,generator,init}
cd ~/docker-lab/logging
```

Perintah `mkdir -p` dengan kurung kurawal membuat 4 subfolder sekaligus dalam satu baris. Masing-masing folder punya fungsi spesifik:

- `fluent-bit/` → file konfigurasi Fluent Bit (kolektor log)
- `app/` → kode Flask API beserta Dockerfile-nya
- `generator/` → script Python simulator log beserta Dockerfile-nya
- `init/` → SQL schema yang dijalankan otomatis saat PostgreSQL pertama kali hidup

Langkah 1: Buat Database Schema untuk Logging

```
hisyam@HISYAMpc:~/docker-lab/logging$ cat > init/01-logging-schema.sql << 'EOF'
CREATE SCHEMA IF NOT EXISTS logs;

CREATE TABLE logs.fluentbit (
  id          BIGSERIAL PRIMARY KEY,
  tag         VARCHAR(200),
  time        TIMESTAMP,
  data        JSONB
);

CREATE INDEX idx_fb_time ON logs.fluentbit(time);
CREATE INDEX idx_fb_tag  ON logs.fluentbit(tag);
CREATE INDEX idx_fb_data ON logs.fluentbit USING GIN(data);

CREATE OR REPLACE VIEW logs.recent_logs AS
SELECT
  id,
  to_char(time, 'YYYY-MM-DD HH24:MI:SS') AS time,
  tag,
  REPLACE(data->>'container_name', '/', '') AS container,
  data->>'source' AS source,
  LEFT(data->>'log', 200) AS log_preview
FROM logs.fluentbit
ORDER BY time DESC LIMIT 100;

CREATE OR REPLACE VIEW logs.structured_logs AS
SELECT
  id,
  time AS received_at,
  tag,
  EOFLANGUAGE plpgsql;
-- out;
-- ed_count = ROW_COUNT;
-- W() - INTERVAL '30 days';
```

Perintah `cat > init/01-logging-schema.sql << 'EOF'` menulis isi SQL langsung ke file tanpa perlu membuka text editor. File ini di-mount ke folder `/docker-entrypoint-initdb.d/` di dalam container PostgreSQL — semua file `.sql` di folder tersebut dijalankan otomatis saat database pertama kali dibuat.

CREATE SCHEMA IF NOT EXISTS logs — membuat namespace logs agar tabel log tidak bercampur dengan tabel lain di database labdb. Akses tabelnya jadi `logs.container_logs`, bukan sekadar `container_logs`.

Tabel logs.container_logs adalah tempat semua log masuk. Kolom-kolomnya:

`id` → BIGSERIAL, auto-increment, primary key

`received_at` → kapan Fluent Bit INSERT ke database (waktu server)

`timestamp` → kapan log dibuat di dalam container (waktu aplikasi)

container_name → nama container asal log (nginx-web, flask-app, dll)

container_id → ID unik container Docker

source → stdout atau stderr

log_level → INFO / DEBUG / WARN / ERROR / CRITICAL

message → isi pesan log

raw_log → log mentah sebelum di-parse

metadata → data tambahan format JSONB (fleksibel, bisa beda tiap log)

Dua kolom timestamp sengaja dipisah karena waktu log dibuat di container bisa berbeda dengan waktu Fluent Bit menerimanya, terutama jika ada network delay atau buffering.

Tabel logs.hourly_summary menyimpan ringkasan per jam per container.

Berguna untuk dashboard atau laporan tanpa perlu query tabel besar setiap saat. Constraint UNIQUE(hour, container_name) mencegah duplikasi data summary untuk kombinasi jam dan container yang sama.

Index dibuat di kolom yang sering dipakai sebagai filter:

```
CREATE INDEX idx_logs_timestamp ON logs.container_logs(timestamp);

CREATE INDEX idx_logs_container ON
logs.container_logs(container_name);

CREATE INDEX idx_logs_level ON logs.container_logs(log_level);

CREATE INDEX idx_logs_received ON logs.container_logs(received_at);

CREATE INDEX idx_logs_metadata ON logs.container_logs USING
GIN(metadata);
```

Tanpa index, query WHERE log_level = 'ERROR' harus membaca seluruh tabel satu per satu (full table scan). Dengan index, PostgreSQL langsung lompat ke

baris yang relevan. Index GIN khusus untuk kolom bertipe JSONB agar pencarian di dalam field metadata tetap cepat.

Fungsi `logs.cleanup_old_logs()` ditulis dalam PL/pgSQL — bahasa prosedural bawaan PostgreSQL. Fungsi ini menghapus semua log yang sudah lebih dari 30 hari dan mengembalikan jumlah baris yang dihapus. Ini adalah implementasi sederhana dari log retention policy.

View `logs.recent_logs` adalah query yang disimpan sebagai objek di database. Setiap kali dipanggil dengan `SELECT * FROM logs.recent_logs`, PostgreSQL menjalankan query di baliknya secara otomatis. View ini menampilkan 100 log terbaru dengan format timestamp yang lebih mudah dibaca (`to_char`) dan message yang dipotong 200 karakter pertama (`LEFT`).

View `logs.error_summary` merangkum jumlah error per container dan per level, berguna untuk melihat sekilas container mana yang paling banyak menghasilkan error.

Langkah 2: Konfigurasi Fluent Bit

```
hisyam@HISYAMpc:~/docker-lab/logging$ cat > fluent-bit/fluent-bit.conf << 'EOF
[SERVICE]
    Flush      5
    Daemon     Off
    Log_Level  info
    Parsers_File parsers.conf

[INPUT]
    Name        forward
    Listen      0.0.0.0
    Port        24224

[OUTPUT]
    Name        pgsql
    Match       *
    Host        postgres-db
    Port        5432
    User        labuser
    Password    labpass123
    Database    labdb
    Table       fluentbit
    Schema      logs
    Timestamp_Key time
    Async       false
    min_pool_size 1
    max_pool_size 4

[OUTPUT]
    Name        stdout
    Match       *
EOF Format      json_lines
```

Keterangan: Menyusun file konfigurasi fluent-bit.conf dan parsers.conf untuk mengatur pipeline logging. Menggunakan komponen [INPUT] berupa driver forward pada port 24224 untuk menangkap log dari Docker Daemon, serta komponen [OUTPUT] berupa plugin pgsql untuk mengirimkan data hasil tangkapan tersebut secara terpusat ke database PostgreSQL.

Langkah 3: Buat Log Generator (Simulasi Multi-Level Log)

```
hisyam@HISYAMpc:~/docker-lab/logging/generator$ cat generator.py
"""
Log Generator - mensimulasikan log dari aplikasi production.
Menghasilkan log dengan berbagai level: DEBUG, INFO, WARN, ERROR, CRITICAL.
"""
import json, time, random, socket, datetime, sys, os

HOSTNAME = socket.gethostname()
LOG_INTERVAL = float(os.environ.get("LOG_INTERVAL", "2"))

# Simulasi event dengan bobot probabilitas
EVENTS = [
    {"level": "INFO", "weight": 50, "messages": [
        "User login successful",
        "Page /dashboard loaded in {ms}ms",
        "API request GET /api/users completed",
        "Session created for user_{uid}",
        "Cache hit for key: product_{pid}",
        "Health check passed",
        "Background job completed: email_send"
    ]},
    {"level": "DEBUG", "weight": 20, "messages": [
        "Database query executed in {ms}ms",
        "Redis connection pool: {pool} active",
        "Request headers: content-type=application/json",
        "Middleware chain completed in {ms}ms"
    ]},
    {"level": "WARN", "weight": 15, "messages": [
        "Slow query detected: {ms}ms (threshold: 1000ms)",
        "Memory usage at {mem}% - approaching limit",
        "Rate limit approaching for IP 192.168.{ip}.{host}",
        "Deprecated API endpoint called: /api/v1/legacy",
        "Certificate expires in {days} days"
    ]},
    {"level": "ERROR", "weight": 10, "messages": [
        "Failed to connect to database: timeout after 5s",
        "NullPointerException in UserService.getProfile()",
        "HTTP 500 Internal Server Error on /api/checkout",
        "Disk write failed: /var/log/app.log - Permission denied",
    ]}
]
```

Langkah 3: Buat Log Generator (Simulasi Multi-Level Log)

Keterangan: Membuat skrip Python (generator.py) berbasis Docker Image python:3.11-alpine yang berfungsi mensimulasikan aktivitas log aplikasi produksi. Skrip ini secara otomatis mencetak log terstruktur berformat JSON ke *stdout* dengan berbagai tingkatan keparahan (*severity level*) secara acak, mulai dari INFO, DEBUG, WARN, ERROR, hingga CRITICAL.

Langkah 4: Buat Flask App dengan Structured Logging

```
hisyam@HISYAMpc:~/docker-lab/logging/app$ cat requirements.txt
flask==3.1.*
psycpg2-binary==2.9.*
hisyam@HISYAMpc:~/docker-lab/logging/app$ cat app.py
import os, json, socket, datetime, logging, sys
from flask import Flask, jsonify, request
import psycpg2

app = Flask(__name__)

class JSONFormatter(logging.Formatter):
    def format(self, record):
        return json.dumps({
            "timestamp": datetime.datetime.now().isoformat(),
            "level": record.levelname,
            "hostname": socket.gethostname(),
            "service": "flask-app",
            "message": record.getMessage(),
            "module": record.module
        })

handler = logging.StreamHandler(sys.stdout)
handler.setFormatter(JSONFormatter())
app.logger.handlers = [handler]
app.logger.setLevel(logging.INFO)
logging.getLogger("werkzeug").setLevel(logging.WARNING)

DB = dict(
    host=os.environ.get("DB_HOST", "postgres-db"),
    dbname=os.environ.get("DB_NAME", "labdb"),
```

Keterangan: Mengembangkan aplikasi web berbasis Flask yang memiliki fungsi ganda: sebagai penghasil log (*log producer*) berformat JSON ke *stdout*, dan sebagai media pemantau (*log viewer*). Flask menyediakan *endpoint* REST

API (/api/logs/stats dan /api/logs/search) yang melakukan query dinamis ke PostgreSQL untuk kebutuhan analisis statistik dan pencarian log.

Langkah 5: Docker Compose untuk Stack Logging

```
hisyam@HISYAMpc:~/docker-lab/logging$ cat > docker-compose.yml << 'EOF'
services:
  # === Log Collector: Fluent Bit ===
  fluent-bit:
    image: fluent/fluent-bit:latest
    container_name: fluent-bit
    ports:
      - "24224:24224"
      - "24224:24224/udp"
    volumes:
      - ./fluent-bit/fluent-bit.conf:/fluent-bit/etc/fluent-bit.conf:ro
      - ./fluent-bit/parsers.conf:/fluent-bit/etc/parsers.conf:ro
    networks:
      - logging-net
    depends_on:
      postgres-db:
        condition: service_healthy
    restart: unless-stopped
```

Keterangan: Menyusun seluruh arsitektur layanan (postgres-db, fluent-bit, nginx-web, flask-app, dan log-generator) ke dalam satu berkas docker-compose.yml. Mengatur urutan *startup* container menggunakan parameter `depends_on` berbasis `service_healthy`, serta mengalihkan semua *logging driver* container dari json-file ke fluentd dengan alamat tujuan fluent-bit:24224.

Urutan startup yang benar:

postgres-db (healthcheck pass)

↓

fluent-bit (tunggu postgres healthy)

↓

nginx-web, flask-app, log-generator (tunggu fluent-bit ready)

Konfigurasi `depends_on` dengan `condition: service_healthy` memastikan PostgreSQL benar-benar siap menerima koneksi sebelum Fluent Bit start. Jika

Fluent Bit naik sebelum PostgreSQL siap, plugin pgsql akan gagal konek dan log tidak tersimpan.

Logging driver di setiap producer:

yaml

logging:

driver: fluentd

options:

fluentd-address: "localhost:24224"

fluentd-async: "true"

tag: "docker.nginx"

fluentd-address: "localhost:24224" artinya Docker daemon di host mengirim log ke port 24224 di host itu sendiri (bukan di dalam Docker network). Ini benar karena Fluent Bit expose port-nya ke host dengan ports: "24224:24224".

fluentd-async: "true" berarti jika Fluent Bit tidak tersedia, Docker tidak memblokir container — log di-buffer dulu. Tanpa ini, container bisa gagal start jika Fluent Bit belum ready.

Volume pg-data adalah named volume yang menyimpan data PostgreSQL secara persisten. Berbeda dengan bind mount (./init:/docker-entrypoint-initdb.d), named volume dikelola sepenuhnya oleh Docker dan tidak terhapus saat docker compose down (hanya hilang jika docker compose down --volumes).

Network logging-net adalah Docker bridge network khusus untuk stack ini. Semua container dalam network yang sama bisa saling berkomunikasi menggunakan nama container sebagai hostname — itulah kenapa Fluent Bit bisa konek ke postgres-db:5432 tanpa perlu tahu IP-nya.

Langkah 6: Deploy dan Verifikasi

6.1 Pastikan environment bersih

```
hisyam@HISYAMpc:~/docker-lab/logging/app$ cd ~/docker-lab/logging
hisyam@HISYAMpc:~/docker-lab/logging$ docker compose down -v 2>/dev/null
hisyam@HISYAMpc:~/docker-lab/logging$ docker compose up --build -d
[*] Building 4.8s (22/22) FINISHED
=> [internal] load local bake definitions                                0.0s
=> => reading from stdin 1.0kB                                          0.0s
=> [log-generator internal] load build definition from Dockerfile      0.0s
=> => transferring dockerfile: 131B                                     0.0s
=> [flask-app internal] load build definition from Dockerfile          0.0s
=> => transferring dockerfile: 206B                                      0.0s
=> [log-generator internal] load metadata for docker.io/library/python:3.11-alpine 3.3s
=> [flask-app internal] load metadata for docker.io/library/python:3.11-slim 4.2s
=> [auth] library/python:pull token for registry-1.docker.io           0.0s
=> [log-generator internal] load .dockerignore                          0.0s
=> => transferring context: 2B                                           0.0s
=> [log-generator 1/3] FROM docker.io/library/python:3.11-alpine@sha256:8b5b5fdb1fd2d78aa94e21c4d61be52487693f54be7f1021647751ff365795783 0.0s
=> => resolve docker.io/library/python:3.11-alpine@sha256:8b5b5fdb1fd2d78aa94e21c4d61be52487693f54be7f1021647751ff365795783 0.0s
=> [log-generator internal] load build context                          0.0s
```

Menghapus seluruh kontainer, jaringan, dan sisa volume persistent dari pengujian sebelumnya menggunakan perintah `docker compose down -v`, kemudian membangun ulang serta menjalankan seluruh *stack* layanan logging di latar belakang dengan perintah `docker compose up --build -d`.

6.2 Verifikasi urutan startup

```
hisyam@HISYAMpc:~/docker-lab/logging$ docker compose ps
NAME                IMAGE                COMMAND                SERVICE    CREATED        STATUS        PORTS
flask-app            logging-flask-app    "python -u app.py"     flask-app   2 minutes ago  Up About a minute  0.0.0.0:5000->5000/tcp, [::]:5000->5000/tcp
fluent-bit           fluent/fluent-bit:2.1.10  "/fluent-bit/bin/fluent-bit"  fluent-bit  2 minutes ago  Up About a minute  0.0.0.0:24224->24224/tcp, 0.0.0.0:24224->24224/udp
log-generator         logging-log-generator  "python -u generator.py"  log-generator  2 minutes ago  Up About a minute  0.0.0.0:8080->8080/tcp, [::]:8080->8080/tcp
nginx-web            nginx:alpine         "/docker-entrypoint.sh"  nginx-web    2 minutes ago  Up About a minute  0.0.0.0:80->80/tcp, [::]:80->80/tcp
postgres-db          postgres:16-alpine    "docker-entrypoint.sh"  postgres-db  2 minutes ago  Up 2 minutes (healthy)  0.0.0.0:5432->5432/tcp, [::]:5432->5432/tcp

hisyam@HISYAMpc:~/docker-lab/logging$ docker compose logs fluent-bit 2>&1 | tail -10
fluent-bit | {"date":1779684179.0,"container_name":"/log-generator","source":"stdout","log":{"timestamp":"2026-05-25T04:42:59.605055","level":"INFO","hostname":"6dc9cf96cecf","service":"log-generator","message":"Cache hit for key: product_53","request_id":"req-478638"},"container_id":"6dc9cf96cecf3535bd6d6353c74bee2ab9b06fc903462cale9d66b57e0ab7c"}
fluent-bit | {"date":1779684180.0,"container_name":"/log-generator","source":"stdout","log":{"timestamp":"2026-05-25T04:43:00.472033","level":"INFO","hostname":"6dc9cf96cecf","service":"log-generator","message":"Cache hit for key: product_134","request_id":"req-924590"},"container_id":"6dc9cf96cecf3535bd6d6353c74bee2ab9b06fc903462cale9d66b57e0ab7c"}
fluent-bit | {"date":1779684181.0,"container_name":"/log-generator","source":"stdout","log":{"timestamp":"2026-05-25T04:43:01.082764","level":"DEBUG","hostname":"6dc9cf96cecf","service":"log-generator","message":"Database query executed in 81ms","request_id":"req-864212"},"container_id":"6dc9cf96cecf3535bd6d6353c74bee2ab9b06fc903462cale9d66b57e0ab7c"}
fluent-bit | {"date":1779684181.0,"source":"stdout","log":{"timestamp":"2026-05-25T04:43:01.735904","level":"DEBUG","hostname":"6dc9cf96cecf","service":"log-generator","message":"Database query executed in 1813ms","request_id":"req-613967"},"container_id":"6dc9cf96cecf3535bd6d6353c74bee2ab9b06fc903462cale9d66b57e0ab7c"}
fluent-bit | {"date":1779684181.0,"container_name":"/log-generator","source":"stdout","log":{"timestamp":"2026-05-25T04:43:01.800669","level":"INFO","hostname":"6dc9cf96cecf","service":"log-generator","message":"Background job completed: email_send","request_id":"req-108159"},"container_id":"6dc9cf96cecf3535bd6d6353c74bee2ab9b06fc903462cale9d66b57e0ab7c"}
fluent-bit | {"date":1779684182.0,"container_name":"/log-generator","source":"stdout","log":{"timestamp":"2026-05-25T04:43:02.000000","level":"INFO","hostname":"6dc9cf96cecf","service":"log-generator","message":"Background job completed: email_send","request_id":"req-108159"},"container_id":"6dc9cf96cecf3535bd6d6353c74bee2ab9b06fc903462cale9d66b57e0ab7c"}
```

```
hisyam@HISYAMpc:~/docker-lab/logging$ docker exec postgres-db psql -U labuser -d labdb -c "\dt logs.*"
List of relations
Schema | Name | Type | Owner
-----+-----+-----+-----
logs   | fluentbit | table | labuser
(1 row)
```

6.3 Tunggu log terkumpul dan generate traffic

```
hisyam@HISYAMpc:~/docker-lab/logging$ sleep 30
hisyam@HISYAMpc:~/docker-lab/logging$ for i in $(seq 1 20); do curl -s http://localhost:8080 > /dev/null; done
hisyam@HISYAMpc:~/docker-lab/logging$ for i in $(seq 1 10); do curl -s http://localhost:5000 > /dev/null; done
hisyam@HISYAMpc:~/docker-lab/logging$ sleep 10
^Z
[1]+  Stopped                  sleep 10
hisyam@HISYAMpc:~/docker-lab/logging$
```

6.4 Verifikasi log masuk ke PostgreSQL

SQLmap total_logs											
(1 row)											
tag		time		data_preview							
docker.generator		2026-05-25	11:53:18	{"log": {"timestamp": "2026-05-25T04:53:18.427887", "level": "INFO", "hostname": "6dc9cf96cccf", "service": "log-generator", "message": "API request GET /api/users completed", "request_id": "req-979404"}}							
docker.generator		2026-05-25	11:53:17	{"log": {"timestamp": "2026-05-25T04:53:17.755511", "level": "INFO", "hostname": "6dc9cf96cccf", "service": "log-generator", "message": "User login successful", "request_id": "req-785238"}}							
docker.generator		2026-05-25	11:53:17	{"log": {"timestamp": "2026-05-25T04:53:17.565565", "level": "DEBUG", "hostname": "6dc9cf96cccf", "service": "log-generator", "message": "Middleware chain completed in 1 ms"}}							
docker.generator		2026-05-25	11:53:16	{"log": {"timestamp": "2026-05-25T04:53:16.947888", "level": "WARN", "hostname": "6dc9cf96cccf", "service": "log-generator", "message": "Certificate expires in 5 days"}}							
docker.generator		2026-05-25	11:53:16	{"log": {"timestamp": "2026-05-25T04:53:16.159459", "level": "WARN", "hostname": "6dc9cf96cccf", "service": "log-generator", "message": "Certificate expires in 19 days"}}							
id		time		tag	container	source	log_preview				
1517	2026-05-25	11:53:18	docker.generator	log-generator	stdout	["timestamp": "2026-05-25T04:53:18.427887", "level": "INFO", "hostname": "6dc9cf96cccf", "service": "log-generator", "message": "API request GET /api/users compl					
1516	2026-05-25	11:53:17	docker.generator	log-generator	stdout	["timestamp": "2026-05-25T04:53:17.755511", "level": "INFO", "hostname": "6dc9cf96cccf", "service": "log-generator", "message": "User login successful", "request					
1515	2026-05-25	11:53:17	docker.generator	log-generator	stdout	["timestamp": "2026-05-25T04:53:17.565565", "level": "DEBUG", "hostname": "6dc9cf96cccf", "service": "log-generator", "message": "Middleware chain completed in 1					
1514	2026-05-25	11:53:16	docker.generator	log-generator	stdout	["timestamp": "2026-05-25T04:53:16.947888", "level": "WARN", "hostname": "6dc9cf96cccf", "service": "log-generator", "message": "Certificate expires in 5 days", "					
1513	2026-05-25	11:53:16	docker.generator	log-generator	stdout	["timestamp": "2026-05-25T04:53:16.159459", "level": "WARN", "hostname": "6dc9cf96cccf", "service": "log-generator", "message": "Certificate expires in 19 days", "					
1512	2026-05-25	11:53:15	docker.generator	log-generator	stdout	["timestamp": "2026-05-25T04:53:15.475053", "level": "INFO", "hostname": "6dc9cf96cccf", "service": "log-generator", "message": "Background job completed: email_s					
1511	2026-05-25	11:53:14	docker.generator	log-generator	stdout	["timestamp": "2026-05-25T04:53:15.717087", "level": "INFO", "hostname": "6dc9cf96cccf", "service": "log-generator", "message": "API request GET /api/users compl					
1510	2026-05-25	11:53:14	docker.generator	log-generator	stdout	["timestamp": "2026-05-25T04:53:14.523505", "level": "WARN", "hostname": "6dc9cf96cccf", "service": "log-generator", "message": "Slow query detected: 1000ms (thr					
1509	2026-05-25	11:53:14	docker.generator	log-generator	stdout	["timestamp": "2026-05-25T04:53:14.995611", "level": "CRITICAL", "hostname": "6dc9cf96cccf", "service": "log-generator", "message": "SSL certificate EXPIRED (un2					
1508	2026-05-25	11:53:14	docker.generator	log-generator	stdout	["timestamp": "2026-05-25T04:53:14.122849", "level": "DEBUG", "hostname": "6dc9cf96cccf", "service": "log-generator", "message": "Database query executed in 2014					
1507	2026-05-25	11:53:14	docker.generator	log-generator	stdout	["timestamp": "2026-05-25T04:53:14.122849", "level": "DEBUG", "hostname": "6dc9cf96cccf", "service": "log-generator", "message": "Database query executed in 2014					
id		received_at		tag	container_name	log_level	message		hostname	service	
1517	2026-05-25	11:53:18	docker.generator	log-generator	INFO		API request GET /api/users completed		6dc9cf96cccf	log-generator	
1516	2026-05-25	11:53:17	docker.generator	log-generator	INFO		User login successful		6dc9cf96cccf	log-generator	
1515	2026-05-25	11:53:17	docker.generator	log-generator	DEBUG		Middleware chain completed in 1ms		6dc9cf96cccf	log-generator	
1											

Masuk ke dalam interaktif terminal PostgreSQL (psql) untuk mengecek eksekusi penulisan data oleh Fluent Bit. Pengujian membuktikan bahwa pipa data berhasil mengalir deras dengan indikator nilai `SELECT COUNT(*)` yang sukses melompat naik melampaui angka 0.

6.5 Query analisis log

```
hisyam@HISYAMpc:~/docker-lab/logging$ docker exec -i postgres-db psql -U labuser -d labdb << 'SQLEOF'
```

```
-- Distribusi log per tag (= per container group)
SELECT tag, COUNT(*) AS total
FROM logs.fluentbit
GROUP BY tag ORDER BY total DESC;

-- Distribusi per level (hanya structured JSON logs)
SELECT log_level, COUNT(*) AS total
FROM logs.structured_logs
GROUP BY log_level ORDER BY total DESC;

-- Error summary per container
SELECT * FROM logs.error_summary;

-- Cari log ERROR dalam 1 jam terakhir
SELECT received_at, container_name, log_level, message
FROM logs.structured_logs
WHERE log_level = 'ERROR'
AND received_at > NOW() - INTERVAL '1 hour'
ORDER BY received_at DESC LIMIT 10;

-- Log rate per menit (5 menit terakhir)
SELECT
    date_trunc('minute', time) AS minute,
    COUNT(*) AS logs_per_minute
FROM logs.fluentbit
WHERE time > NOW() - INTERVAL '5 minutes'
GROUP BY minute ORDER BY minute;

-- Cari keyword "timeout" di semua log
SELECT received_at, container_name, log_level, message
FROM logs.structured_logs
WHERE message ILIKE '%timeout%'
ORDER BY received_at DESC LIMIT 5;

-- Nginx access log (plain text - bukan JSON)
SELECT time, LEFT(data->'log', 150) AS access_log
FROM logs.fluentbit
WHERE tag = 'docker.nginx'
ORDER BY time DESC LIMIT 10;
```

```
SQLEOF
```

tag	total
docker.generator	1486
docker.nginx	31

(2 rows)

log_level	total
INFO	713
DEBUG	311
WARN	226
ERROR	170
CRITICAL	66

(5 rows)

container_name	log_level	count	last_seen
log-generator	WARN	226	2026-05-25 11:53:16
log-generator	ERROR	170	2026-05-25 11:53:13
log-generator	CRITICAL	66	2026-05-25 11:53:14
(3 rows)			
received_at	container_name	log_level	message
2026-05-25 11:53:13	log-generator	ERROR	HTTP 500 Internal Server Error on /api/checkout
2026-05-25 11:53:12	log-generator	ERROR	HTTP 500 Internal Server Error on /api/checkout
2026-05-25 11:53:07	log-generator	ERROR	NullPointerException in UserService.getProfile()
2026-05-25 11:53:07	log-generator	ERROR	Disk write failed: /var/log/app.log - Permission denied
2026-05-25 11:53:02	log-generator	ERROR	HTTP 500 Internal Server Error on /api/checkout
2026-05-25 11:53:01	log-generator	ERROR	NullPointerException in UserService.getProfile()
2026-05-25 11:52:56	log-generator	ERROR	Payment gateway returned error code 502
2026-05-25 11:52:55	log-generator	ERROR	NullPointerException in UserService.getProfile()
2026-05-25 11:52:50	log-generator	ERROR	Failed to connect to database: timeout after 5s
2026-05-25 11:52:46	log-generator	ERROR	NullPointerException in UserService.getProfile()
(10 rows)			
minute	logs_per_minute		
2026-05-25 11:50:00	78		
2026-05-25 11:51:00	119		
2026-05-25 11:52:00	123		
2026-05-25 11:53:00	35		
(4 rows)			
received_at	container_name	log_level	message
2026-05-25 11:52:50	log-generator	ERROR	Failed to connect to database: timeout after 5s
2026-05-25 11:52:43	log-generator	ERROR	Failed to connect to database: timeout after 5s
2026-05-25 11:52:28	log-generator	ERROR	Failed to connect to database: timeout after 5s
2026-05-25 11:51:42	log-generator	ERROR	Failed to connect to database: timeout after 5s
2026-05-25 11:51:26	log-generator	ERROR	Failed to connect to database: timeout after 5s
(5 rows)			
time	access_log		
2026-05-25 11:40:56	/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/		
2026-05-25 11:40:56	/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh		
2026-05-25 11:40:56	10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf		
2026-05-25 11:40:56	10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf		
2026-05-25 11:40:56	/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh		
2026-05-25 11:40:56	/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh		
2026-05-25 11:40:56	/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh		
2026-05-25 11:40:56	/docker-entrypoint.sh: Configuration complete; ready for start up		
2026-05-25 11:40:56	2026/05/25 04:40:56 [notice] 1#1: using the "epoll" event method		
2026-05-25 11:40:56	/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration		
(10 rows)			

Mengeksekusi serangkaian query SQL tingkat lanjut pada database untuk menganalisis metrik log riil, seperti menghitung distribusi total log berdasarkan *tag* kontainer asal, pengelompokan berdasarkan tingkat keparahan (*severity level*), serta melihat ringkasan pesan error via komponen *View*.

6.6 Gunakan Flask API untuk query log

```
hisyam@HISYAMpc:~/docker-lab/logging$ curl -s http://localhost:5000/api/logs/stats | python3 -m json.tool
{
  "last_hour_by_level": [
    {
      "count": 713,
      "level": "INFO"
    },
    {
      "count": 311,
      "level": "DEBUG"
    },
    {
      "count": 226,
      "level": "WARN"
    },
    {
      "count": 170,
      "level": "ERROR"
    },
    {
      "count": 66,
      "level": "CRITICAL"
    }
  ],
  "last_hour_by_tag": [
    {
      "count": 1486,
      "tag": "docker.generator"
    },
    {
      "count": 31,
      "tag": "docker.nginx"
    }
  ],
  "total_logs_all_time": 1517
}
```

```
hisyam@HISYAMpc:~/docker-lab/logging$ # Cari log dengan keyword "error"
curl -s "http://localhost:5000/api/logs/search?q=error&limit=5" | python3 -m json.tool

# Cari log level CRITICAL
curl -s "http://localhost:5000/api/logs/search?level=CRITICAL&limit=5" | python3 -m json.tool

# Cari log dari tag tertentu
curl -s "http://localhost:5000/api/logs/search?tag=generator&limit=5" | python3 -m json.tool

# Lihat raw log (termasuk Nginx plain text)
curl -s "http://localhost:5000/api/logs/raw?limit=10" | python3 -m json.tool
{
  "count": 5,
  "results": [
    {
      "container": "log-generator",
      "id": 1507,
      "level": "ERROR",
      "message": "HTTP 500 Internal Server Error on /api/checkout",
      "service": "log-generator",
      "tag": "docker.generator",
      "time": "2026-05-25 11:53:13"
    },
    {
      "container": "log-generator",
      "id": 1506,
      "level": "ERROR",
      "message": "HTTP 500 Internal Server Error on /api/checkout",
      "service": "log-generator",
      "tag": "docker.generator",
      "time": "2026-05-25 11:53:12"
    },
    {
      "container": "log-generator",
      "id": 1489,
      "level": "ERROR",
      "message": "HTTP 500 Internal Server Error on /api/checkout",
      "service": "log-generator",
      "tag": "docker.generator",
      "time": "2026-05-25 11:53:02"
    },
    {
      "container": "log-generator",
      "id": 1477,
      "level": "ERROR",
      "message": "Payment gateway returned error code 502",
      "service": "log-generator",
      "tag": "docker.generator",
      "time": "2026-05-25 11:52:56"
    },
    {
      "container": "log-generator",
      "id": 1434,
      "level": "ERROR",
      "message": "HTTP 500 Internal Server Error on /api/checkout",
      "service": "log-generator",
      "tag": "docker.generator",
      "time": "2026-05-25 11:52:55"
    }
  ]
}
```

```

{
  "count": 5,
  "results": [
    {
      "container": "log-generator",
      "id": 1510,
      "level": "CRITICAL",
      "message": "SSL certificate EXPIRED \u2014 HTTPS unavailable",
      "service": "log-generator",
      "tag": "docker.generator",
      "time": "2026-05-25 11:53:14"
    },
    {
      "container": "log-generator",
      "id": 1485,
      "level": "CRITICAL",
      "message": "SSL certificate EXPIRED \u2014 HTTPS unavailable",
      "service": "log-generator",
      "tag": "docker.generator",
      "time": "2026-05-25 11:53:01"
    },
    {
      "container": "log-generator",
      "id": 1452,
      "level": "CRITICAL",
      "message": "Out of memory: container killed by OOM",
      "service": "log-generator",
      "tag": "docker.generator",
      "time": "2026-05-25 11:52:44"
    },
    {
      "container": "log-generator",
      "id": 1398,
      "level": "CRITICAL",
      "message": "Data corruption detected in table: orders",
      "service": "log-generator",
      "tag": "docker.generator",
      "time": "2026-05-25 11:52:18"
    },
    {
      "container": "log-generator",
      "id": 1385,
      "level": "CRITICAL",
      "message": "SSL certificate EXPIRED \u2014 HTTPS unavailable",
      "service": "log-generator",
      "tag": "docker.generator",
      "time": "2026-05-25 11:52:11"
    }
  ]
}

```

```

{
  "count": 10,
  "results": [
    {
      "container": "log-generator",
      "id": 1511,
      "log": "{\"timestamp\": \"2026-05-25T04:53:18.427887\", \"level\": \"INFO\", \"hostname\": \"6dc9cf96cccf\", \"service\": \"log-generator\", \"message\": \"API request GET /api/users completed\", \"request_id\": \"req-97404\"}\",
      "source": "stdout",
      "tag": "docker.generator",
      "time": "2026-05-25 11:53:18"
    },
    {
      "container": "log-generator",
      "id": 1510,
      "log": "{\"timestamp\": \"2026-05-25T04:53:17.55514\", \"level\": \"INFO\", \"hostname\": \"6dc9cf96cccf\", \"service\": \"log-generator\", \"message\": \"User login successful\", \"request_id\": \"req-785928\"}\",
      "source": "stdout",
      "tag": "docker.generator",
      "time": "2026-05-25 11:53:17"
    },
    {
      "container": "log-generator",
      "id": 1515,
      "log": "{\"timestamp\": \"2026-05-25T04:53:17.565565\", \"level\": \"DEBUG\", \"hostname\": \"6dc9cf96cccf\", \"service\": \"log-generator\", \"message\": \"Middleware chain completed in 1614us\", \"request_id\": \"req-2128\"}\",
      "source": "stdout",
      "tag": "docker.generator",
      "time": "2026-05-25 11:53:17"
    },
    {
      "container": "log-generator",
      "id": 1514,
      "log": "{\"timestamp\": \"2026-05-25T04:53:16.947888\", \"level\": \"WARN\", \"hostname\": \"6dc9cf96cccf\", \"service\": \"log-generator\", \"message\": \"Certificate expires in 5 days\", \"request_id\": \"req-638230\"}\",
      "source": "stdout",
      "tag": "docker.generator",
      "time": "2026-05-25 11:53:16"
    },
    {
      "container": "log-generator",
      "id": 1513,
      "log": "{\"timestamp\": \"2026-05-25T04:53:16.159459\", \"level\": \"WARN\", \"hostname\": \"6dc9cf96cccf\", \"service\": \"log-generator\", \"message\": \"Certificate expires in 19 days\", \"request_id\": \"req-322727\"}\",
      "source": "stdout",
      "tag": "docker.generator",
      "time": "2026-05-25 11:53:16"
    },
    {
      "container": "log-generator",
      "id": 1512,
      "log": "{\"timestamp\": \"2026-05-25T04:53:15.717887\", \"level\": \"INFO\", \"hostname\": \"6dc9cf96cccf\", \"service\": \"log-generator\", \"message\": \"API request GET /api/users completed\", \"request_id\": \"req-85810\"}\",
      "source": "stdout",
      "tag": "docker.generator",
      "time": "2026-05-25 11:53:15"
    }
  ]
}

```

Melakukan verifikasi fungsionalitas REST API pada aplikasi Flask menggunakan perintah curl. Endpoint `/api/logs/stats` dan `/api/logs/search` berhasil mengembalikan data log terstruktur dalam format JSON yang rapi setelah berhasil melakukan query dinamis ke PostgreSQL

6.7 Log retention cleanup

```
hisyam@HISYAMPc:~/docker-lab/logging$ docker exec postgres-db psql -U labuser -d labdb -c \
"SELECT logs.cleanup_old_logs() AS deleted_rows;"
deleted_rows
-----
          0
(1 row)
```

Menguji fungsionalitas sistem manajemen penyimpanan dengan memanggil fungsi Prosedural PL/pgSQL `logs.cleanup_old_logs()`. Fungsi ini sukses berjalan untuk memastikan log yang berumur lebih dari 30 hari akan dihapus otomatis demi menjaga kapasitas ruang disk server. **PERTANYAAN**

Pre-Lab

1. Mengapa centralized logging penting di lingkungan container?

Di lingkungan container, setiap service berjalan di container terpisah dan bersifat ephemeral — container bisa dibuat, dihapus, atau di-restart kapan saja. Jika log hanya disimpan di dalam masing-masing container, log tersebut akan ikut hilang saat container dihapus. Selain itu, ketika ada ratusan container berjalan bersamaan, membuka log satu per satu sangat tidak efisien untuk debugging maupun audit.

Centralized logging mengatasi masalah ini dengan mengumpulkan semua log dari seluruh container ke satu tempat (dalam praktikum ini: PostgreSQL). Dengan begitu, log tetap tersimpan meskipun container-nya sudah mati, pencarian lintas container bisa dilakukan dengan satu query SQL, dan analisis pola error menjadi jauh lebih mudah.

2. Apa perbedaan antara Docker logging driver json-file dan fluentd?

Driver json-file adalah driver default Docker. Log dari container ditulis ke file JSON di host (`/var/lib/docker/containers/<id>/<id>-json.log`). Log hanya bisa diakses lewat docker logs atau membaca file langsung, dan akan hilang jika container dihapus.

Driver fluentd mengubah alur tersebut — Docker tidak menyimpan log ke file, melainkan langsung mengirimkannya ke Fluent Bit/Fluentd melalui

jaringan (port 24224). Konsekuensinya, docker compose logs tidak akan menampilkan apapun karena tidak ada file log lokal. Log langsung masuk ke pipeline Fluent Bit dan akhirnya tersimpan di PostgreSQL.

3. Jelaskan keuntungan menyimpan log di database (PostgreSQL) vs file text.

Aspek	File Text	PostgreSQL
Pencarian	grep — lambat di file besar	SQL WHERE, ILIKE — terindeks, cepat
Filter multi-kondisi	Sulit, perlu kombinasi grep/awk	Mudah dengan AND, OR, GROUP BY
Agregasi	Tidak bisa langsung	COUNT, SUM, date_trunc langsung
Retensi otomatis	Manual, butuh script cron	Fungsi SQL cleanup_old_logs()
Akses multi-user	Race condition saat baca/tulis	Concurrent access aman dengan MVCC
Metadata terstruktur	Sulit di-parse	Kolom terpisah + JSONB untuk metadata

Intinya, PostgreSQL memungkinkan log dianalisis layaknya data, bukan sekadar teks mentah.

4. Apa itu structured logging dan mengapa lebih baik daripada plain text log?

Structured logging adalah praktik menulis log dalam format terstruktur (biasanya JSON), bukan plain text bebas. Contoh perbandingan:

Plain text:

```
2025-05-10 10:23:45 ERROR Failed to connect to database: timeout after 5s
```

Structured (JSON):

```
{"timestamp": "2025-05-10T10:23:45", "level": "ERROR", "service": "flask-app", "message": "Failed to connect to database: timeout after 5s", "request_id": "req-482910"}
```

Keunggulan structured logging: setiap field langsung bisa dijadikan kolom di database tanpa perlu parsing regex yang rawan error. Filter `WHERE log_level = 'ERROR' AND service = 'flask-app'` jauh lebih andal daripada `grep ERROR` yang bisa salah match teks di bagian message. Field seperti `request_id` juga memungkinkan tracing satu request melintasi banyak service sekaligus.

5. Mengapa Fluent Bit lebih cocok untuk sidecar/edge collection dibanding Fluentd?

Fluent Bit ditulis dalam bahasa C dengan footprint sangat kecil (~1 MB RAM), sehingga cocok dijalankan sebagai sidecar di setiap container atau di edge device dengan resource terbatas. Ia fokus pada tugas inti: menerima log, mem-parse, dan mem-forward ke tujuan.

Fluentd ditulis dalam Ruby + C dengan konsumsi memori jauh lebih besar (~40 MB), tetapi memiliki ekosistem 1000+ plugin. Fluentd lebih cocok sebagai aggregator pusat yang menerima log dari banyak Fluent Bit, lalu mendistribusikannya ke berbagai destination (Elasticsearch, S3, database, dll.).

Dalam praktikum ini, Fluent Bit dipakai karena skenarionya adalah single-node lab — cukup satu collector ringan yang langsung forward ke PostgreSQL, tanpa butuh aggregator berlapis.

Post-Lab

1. Berapa total log yang masuk ke PostgreSQL setelah 5 menit? Tunjukkan distribusi per container dan per level.

```
hisyam@HISYAMpc:~/docker-lab/logging$ # Total Log
docker exec -i postgres-db psql -U labuser -d labdb -c "SELECT COUNT(*) FROM logs.fluentbit;"

# Distribusi per Tag
docker exec -i postgres-db psql -U labuser -d labdb -c "SELECT tag, COUNT(*) FROM logs.fluentbit GROUP BY tag;"

# Distribusi per Level
docker exec -i postgres-db psql -U labuser -d labdb -c "SELECT log_level, COUNT(*) FROM logs.structured_logs GROUP BY log_level;"
count
-----
1517
(1 row)

    tag      | count
-----+-----
docker.generator | 1486
docker.nginx   | 31
(2 rows)

 log_level | count
-----+-----
CRITICAL  | 66
DEBUG     | 311
ERROR     | 170
INFO      | 713
WARN      | 226
(5 rows)
```

Total Log: 1517 log

Distribusi per Tag (Container):

Tag	Total Log
docker.generator	1486
docker.nginx	31

Distribusi per Severity Level:

Log Level	Total	Persentase (Estimasi)
INFO	713	~47%
DEBUG	311	~20%
WARN	226	~15%
ERROR	170	~11%
CRITICAL	66	~4%

Analisis Singkat untuk Laporanmu:

- **Analisis Tag:** Penambahan log dari docker.nginx (31 log) terjadi setelah kita memicu *traffic* melalui perintah curl tadi. Hal ini membuktikan bahwa pipeline fluent-bit berhasil menangkap log dari multi-container secara bersamaan.

2. Tulis query SQL yang menampilkan log rate per menit selama 10 menit terakhir.


```
labdb=# SELECT MAX(time) FROM logs.fluentbit;
max
-----
2026-05-25 11:53:18
(1 row)

labdb=# SELECT
labdb=#     date_trunc('minute', time) AS minute,
labdb=#     COUNT(*) AS logs_per_minute
labdb=# FROM logs.fluentbit
labdb=# WHERE time > NOW() - INTERVAL '30 minutes'
labdb=# GROUP BY minute
labdb=# ORDER BY minute ASC;
 minute | logs_per_minute
-----|-----
2026-05-25 11:42:00 |          99
2026-05-25 11:43:00 |         107
2026-05-25 11:44:00 |         117
2026-05-25 11:45:00 |         119
2026-05-25 11:46:00 |         127
2026-05-25 11:47:00 |         117
2026-05-25 11:48:00 |         121
2026-05-25 11:49:00 |         128
2026-05-25 11:50:00 |         117
2026-05-25 11:51:00 |         119
2026-05-25 11:52:00 |         123
2026-05-25 11:53:00 |          35
(12 rows)
```

Berdasarkan hasil query agregasi per menit, sistem mampu mencatat rata-rata **~115 log per menit**. Fluktuasi angka pada menit terakhir (35 log pada pukul 11:53) disebabkan oleh penghentian proses *log-generator* atau *traffic trigger* tepat sebelum pengambilan data, sementara angka stabil pada menit-menit sebelumnya (117-128 log/menit) menunjukkan konsistensi *throughput* pengiriman data dari container melalui Fluent Bit ke PostgreSQL.

3. Apa yang terjadi jika container fluent-bit di-stop? Apakah container lain juga stop? Apakah log hilang?

Apakah container lain ikut stop?

Tidak. Berdasarkan hasil docker compose ps saat fluent-bit di-stop, container flask-app, log-generator, nginx-web, dan postgres-db tetap berstatus Up. Hal ini karena `depends_on` di Docker Compose hanya

mengontrol urutan startup, bukan lifecycle container saat berjalan. Ketika fluent-bit mati di tengah jalan, Docker tidak memiliki mekanisme cascade stop.

Apakah log hilang?

Sebagian, tergantung durasi fluent-bit mati. Karena producer dikonfigurasi dengan `fluentd-async: "true"`, Docker daemon mem-buffer log di sisi host selama fluent-bit tidak tersedia. Selama buffer belum penuh, log akan di-flush ke fluent-bit ketika container dihidupkan kembali sehingga tidak ada yang hilang. Namun jika fluent-bit mati terlalu lama hingga buffer penuh, log baru akan di-drop dan tidak bisa dipulihkan. Log yang sudah tersimpan di PostgreSQL sebelum fluent-bit mati tetap aman.

Bukti gap terlihat dari hasil query log rate per menit — menit saat fluent-bit mati menampilkan 0 log atau jauh lebih sedikit dari biasanya.

4. Jelaskan alur sebuah log entry dari log-generator stdout sampai masuk ke tabel `container_logs`.

4. Alur Log Entry

1. **Log Generator:** Aplikasi mencetak JSON ke stdout.
2. **Docker Daemon:** Mencegat stdout, menambahkan metadata container, dan mengirimnya via *Fluentd Forward Protocol* ke port 24224.
3. **Fluent Bit (Input):** Menerima log melalui input forward.
4. **Fluent Bit (Output):** Menggunakan plugin `pgsql` untuk melakukan INSERT data ke tabel PostgreSQL.

5. **PostgreSQL:** Menyimpan data ke kolom tag, time, dan data (JSONB) serta memperbarui indeks.
6. **Aplikasi (Query):** Flask membaca hasil INSERT tersebut melalui *View* untuk ditampilkan di API.

5. Modifikasi LOG_INTERVAL menjadi 0.5 detik. Berapa log rate per menit yang dihasilkan?

Langkah Modifikasi:

Untuk mempercepat produksi log, kita mengubah variabel lingkungan pada layanan log-generator di berkas docker-compose.yml:

YAML

log-generator:

environment:

- LOG_INTERVAL=0.5

Setelah mengubah berkas tersebut, jalankan perintah docker compose up -d log-generator untuk menerapkan perubahan.

Kalkulasi Teoritis:

Pada skrip generator.py, durasi jeda antar log dihitung menggunakan fungsi:

`time.sleep(LOG_INTERVAL + random.uniform(-0.5, 0.5))`

Dengan menetapkan LOG_INTERVAL = 0.5, maka rata-rata durasi jeda efektif adalah 0.5 detik per log. Maka, estimasi log per menit adalah:

$$\text{Log per menit} = \frac{60 \text{ detik}}{0.5 \text{ detik/log}} = 120 \text{ log/menit}$$

Hasil Observasi (Data Riil):

Berdasarkan hasil query logs_per_minute dari database:

Menit	Logs per Menit
22:55:00	106
22:56:00	106
22:57:00	106
22:58:00	120
22:59:00	102

Analisis:

Hasil observasi menunjukkan angka yang mendekati 120 log/menit. Variasi jumlah log per menit (angka di bawah 120) disebabkan oleh fungsi `random.uniform(-0.5, 0.5)` yang memberikan efek *jitter* (variasi acak) pada interval, serta adanya *overhead* waktu yang dibutuhkan oleh interpreter Python dan proses pengiriman paket data melalui *logging driver* Docker ke Fluent Bit.

MODUL 6: Grafana Service Docker untuk Monitoring Resource

LANGKAH PRAKTIKUM

Langkah 0: Persiapan Project

```
hisyam@HISVAMpc:~/docker-lab$ mkdir -p ~/docker-lab/monitoring/{prometheus,grafana/{provisioning/datasources,provisioning/dashboards,dashboards},app,generator,fluent-bit,init}
cd ~/docker-lab/monitoring
hisyam@HISVAMpc:~/docker-lab/monitoring$ ls -a
.. app fluent-bit generator grafana init prometheus
```

membuat seluruh struktur folder project monitoring Docker, seperti folder Prometheus, Grafana, Flask App, Fluent Bit, dan PostgreSQL init script. Setelah itu terminal masuk ke direktori monitoring.

Langkah 1: Konfigurasi Prometheus

1.1 Buat konfigurasi scrape

```
hisyam@HISYAMpc:~/docker-lab/monitoring$ cat > prometheus/prometheus.yml << 'EOF'
# =====
# Prometheus Configuration
# =====
global:
  scrape_interval: 15s      # Scrape setiap 15 detik
  evaluation_interval: 15s  # Evaluasi rules setiap 15 detik
  scrape_timeout: 10s

# =====
# Alerting Rules
# =====
rule_files:
  - "alert_rules.yml"

# =====
# Scrape Targets
# =====
scrape_configs:

  # --- Prometheus self-monitoring ---
  - job_name: "prometheus"
    static_configs:
      - targets: ["localhost:9090"]
        labels:
          instance: "prometheus-server"

  # --- Node Exporter (host metrics) ---
  - job_name: "node-exporter"
    static_configs:
      - targets: ["node-exporter:9100"]
        labels:
          instance: "docker-host"

  # --- cAdvisor (container metrics) ---
  - job_name: "cadvisor"
    static_configs:
      - targets: ["cadvisor:8081"]
        labels:
          instance: "docker-containers"

  # --- Flask Application ---
  - job_name: "flask-app"
    metrics_path: "/metrics"
    static_configs:
      - targets: ["flask-app:5000"]
        labels:
          instance: "flask-backend"
EOF
```

Command ini membuat file konfigurasi utama Prometheus. File tersebut berisi:

- interval scraping metrics,
- daftar target monitoring,
- lokasi alert rules.

Prometheus akan mengambil metrics dari Node Exporter, cAdvisor, dan Flask App.

1.2 Buat alert rules

```
hisyam@HISYAMPc:~/docker-lab/monitoring$ cat > prometheus/alert_rules.yml << 'EOF'
groups:
- name: host_alerts
  rules:
    # CPU usage > 80% selama 2 menit
    - alert: HighCpuUsage
      expr: 100 - (avg(rate(node_cpu_seconds_total{mode="idle"}[5m])) * 100) > 80
      for: 2m
      labels:
        severity: warning
      annotations:
        summary: "CPU usage tinggi ({{ $value | printf \"%.1f\" }}%)"
        description: "CPU usage di atas 80% selama lebih dari 2 menit."

    # Memory available < 20%
    - alert: LowMemoryAvailable
      expr: (node_memory_MemAvailable_bytes / node_memory_MemTotal_bytes * 100) < 20
      for: 2m
      labels:
        severity: warning
      annotations:
        summary: "Memory tersedia rendah ({{ $value | printf \"%.1f\" }}%)"

    # Disk usage > 85%
    - alert: HighDiskUsage
      expr: 100 - (node_filesystem_avail_bytes{mountpoint="/" } / node_filesystem_size_bytes{mountpoint="/" } * 100) > 85
      for: 5m
      labels:
        severity: critical
      annotations:
        summary: "Disk usage tinggi ({{ $value | printf \"%.1f\" }}%)"

- name: container_alerts
  rules:
    # Container restart > 3 kali dalam 15 menit
    - alert: ContainerRestartFrequent
      expr: increase(container_restart_count[15m]) > 3
      for: 1m
      labels:
        severity: critical
      annotations:
        summary: "Container {{ $labels.name }} restart berulang"

    # Container memory > 256MB
    - alert: ContainerHighMemory
      expr: container_memory_usage_bytes{name!=""} > 268435456
      for: 5m
      labels:
        severity: warning
      annotations:
        summary: "Container {{ $labels.name }} memory > 256MB"
EOF
```

Digunakan untuk membuat aturan alert monitoring seperti:

- CPU usage tinggi,
- memory rendah,
- disk hampir penuh,
- container restart berulang.

Alert akan aktif jika kondisi melewati threshold tertentu.

Langkah 2: Konfigurasi Grafana (Provisioning)

2.1 Provisioning data source

```
hisyam@HISYAMpc:~/docker-lab/monitoring$ cat > grafana/provisioning/datasources/datasources.yml << 'EOF'
apiVersion: 1

datasources:
  # --- Prometheus (metrics) ---
  - name: Prometheus
    type: prometheus
    access: proxy
    url: http://prometheus:9090
    isDefault: true
    editable: true
    jsonData:
      timeInterval: "15s"

  # --- PostgreSQL (logs dari Modul 5) ---
  - name: PostgreSQL-Logs
    type: postgres
    access: proxy
    url: postgres-db:5432
    database: labdb
    user: labuser
    secureJsonData:
      password: labpass123
    jsonData:
      sslmode: disable
      maxOpenConns: 5
      postgresVersion: 1600
      timescaledb: false
    editable: true
EOF
```

Command ini membuat data source otomatis untuk Grafana sehingga tidak perlu setup manual lewat browser. Data source yang digunakan adalah Prometheus dan PostgreSQL.

2.2 Provisioning dashboard

```
hisyam@HISYAMpc:~/docker-lab/monitoring$ cat > grafana/provisioning/dashboards/dashboards.yml << 'EOF'
apiVersion: 1

providers:
  - name: "Lab PENS Dashboards"
    orgId: 1
    folder: "Lab PENS"
    type: file
    disableDeletion: false
    editable: true
    updateIntervalSeconds: 30
    options:
      path: /var/lib/grafana/dashboards
      foldersFromFilesStructure: false
EOF
```

Berfungsi untuk menentukan lokasi dashboard JSON yang akan otomatis dibaca Grafana saat container dijalankan.

2.3 Buat Dashboard JSON: Docker Host Overview

```
hisyam@HISYAMpc:~/docker-lab/monitoring$ cat > grafana/dashboards/docker-host-overview.json << 'JSONEOF'
{
  "annotations": { "list": [] },
  "editable": true,
  "fiscalYearStartMonth": 0,
  "graphTooltip": 1,
  "links": [],
  "panels": [
    {
      "title": "CPU Usage %",
      "type": "gauge",
      "gridPos": { "h": 6, "w": 6, "x": 0, "y": 0 },
      "datasource": { "type": "prometheus", "uid": "" },
      "fieldConfig": {
        "defaults": {
          "thresholds": {
            "mode": "absolute",
            "steps": [
              { "color": "green", "value": null },
              { "color": "yellow", "value": 60 },
              { "color": "red", "value": 85 }
            ]
          },
          "unit": "percent", "min": 0, "max": 100
        },
        "overrides": []
      },
      "targets": [
        {
          "expr": "100 - (avg(rate(node_cpu_seconds_total{mode=\"idle\"}[5m])) * 100)",
          "legendFormat": "CPU Usage"
        }
      ]
    },
    {
      "title": "Memory Usage %",
      "type": "gauge",
      "gridPos": { "h": 6, "w": 6, "x": 6, "y": 0 },
      "datasource": { "type": "prometheus", "uid": "" },
      "fieldConfig": {
        "defaults": {
          "thresholds": {
            "mode": "absolute",
            "steps": [
              { "color": "green", "value": null },
              { "color": "yellow", "value": 70 },
              { "color": "red", "value": 90 }
            ]
          },
          "unit": "percent", "min": 0, "max": 100
        },
        "overrides": []
      },
      "targets": [
        {
          "expr": "100 - (node_memory_MemAvailable_bytes / node_memory_MemTotal_bytes * 100)",
          "legendFormat": "Memory Used"
        }
      ]
    }
  ],
}
```

Membuat dashboard monitoring host yang menampilkan:

- CPU usage,
- memory usage,
- disk usage,
- network traffic,
- uptime server.

2.4 Buat Dashboard JSON: Container Metrics

```
root@kali:~/docker-lab/monitoring# cat > grafana/dashboards/container-metrics.json << 'JSONDEF'
{
  "annotations": { "list": [] },
  "editable": true,
  "panels": [
    {
      "title": "Running Containers",
      "type": "stat",
      "gridPos": { "h": 4, "w": 6, "x": 0, "y": 0 },
      "datasource": { "type": "prometheus", "uid": "" },
      "fieldConfig": { "defaults": { "color": { "mode": "thresholds" }, "thresholds": { "steps": [ { "color": "green", "value": null } ] }, "overrides": [ ] },
      "targets": [ { "expr": "count(container_last_seen{name!=''})", "legendFormat": "Containers" } ]
    },
    {
      "title": "Total Container CPU Usage",
      "type": "stat",
      "gridPos": { "h": 4, "w": 6, "x": 6, "y": 0 },
      "datasource": { "type": "prometheus", "uid": "" },
      "fieldConfig": { "defaults": { "unit": "percent", "thresholds": { "steps": [ { "color": "green", "value": null }, { "color": "red", "value": 80 } ] }, "overrides": [ ] },
      "targets": [ { "expr": "sum(rate(container_cpu_usage_seconds_total{name!=''})[5m]) * 100", "legendFormat": "Total CPU" } ]
    },
    {
      "title": "Total Container Memory",
      "type": "stat",
      "gridPos": { "h": 4, "w": 6, "x": 12, "y": 0 },
      "datasource": { "type": "prometheus", "uid": "" },
      "fieldConfig": { "defaults": { "unit": "bytes", "thresholds": { "steps": [ { "color": "green", "value": null } ] }, "overrides": [ ] },
      "targets": [ { "expr": "sum(container_memory_usage_bytes{name!=''})", "legendFormat": "Total Memory" } ]
    },
    {
      "title": "Prometheus Alerts Active",
      "type": "stat",
      "gridPos": { "h": 4, "w": 6, "x": 18, "y": 0 },
      "datasource": { "type": "prometheus", "uid": "" },
      "fieldConfig": { "defaults": { "thresholds": { "steps": [ { "color": "green", "value": null }, { "color": "red", "value": 1 } ] }, "overrides": [ ] },
      "targets": [ { "expr": "count(ALERTS{alertstate='firing'}) OR vector(0)", "legendFormat": "Firing" } ]
    },
    {
      "title": "CPU Usage per Container",
      "type": "timeseries",
      "gridPos": { "h": 8, "w": 12, "x": 0, "y": 4 },
      "datasource": { "type": "prometheus", "uid": "" },
      "fieldConfig": { "defaults": { "unit": "percent", "overrides": [ ] },
      "targets": [ { "expr": "rate(container_cpu_usage_seconds_total{name!=''})[5m] * 100", "legendFormat": "{{ name }}" } ]
    },
    {
      "title": "Memory Usage per Container",
      "type": "timeseries",
      "gridPos": { "h": 8, "w": 12, "x": 12, "y": 4 },
      "datasource": { "type": "prometheus", "uid": "" },
      "fieldConfig": { "defaults": { "unit": "bytes", "overrides": [ ] },
      "targets": [ { "expr": "container_memory_usage_bytes{name!=''})", "legendFormat": "{{ name }}" } ]
    },
    {
      "title": "Container Network RX",
      "type": "timeseries",
      "gridPos": { "h": 8, "w": 12, "x": 0, "y": 12 },

```

Dashboard ini digunakan untuk memonitor penggunaan resource setiap container Docker seperti CPU, memory, dan network.

2.5 Buat Dashboard JSON: Log Analytics (PostgreSQL)

```
root@kali:~/docker-lab/monitoring# cat > grafana/dashboards/log-analytics.json << 'JSONDEF'
{
  "annotations": { "list": [] },
  "editable": true,
  "panels": [
    {
      "title": "Total Logs (All Time)",
      "type": "stat",
      "gridPos": { "h": 4, "w": 6, "x": 0, "y": 0 },
      "datasource": { "type": "postgres", "uid": "" },
      "fieldConfig": { "defaults": { "thresholds": { "steps": [ { "color": "blue", "value": null } ] }, "overrides": [ ] },
      "targets": [ { "sql": "SELECT COUNT(*) AS total FROM logs.fluentbit;", "format": "table" } ]
    },
    {
      "title": "Errors Last Hour",
      "type": "stat",
      "gridPos": { "h": 4, "w": 6, "x": 6, "y": 0 },
      "datasource": { "type": "postgres", "uid": "" },
      "fieldConfig": { "defaults": { "thresholds": { "steps": [ { "color": "green", "value": null }, { "color": "red", "value": 10 } ] }, "overrides": [ ] },
      "targets": [ { "sql": "SELECT COUNT(*) AS errors FROM logs.fluentbit WHERE data->'log' IS NOT NULL AND LEFT(TRIM(data->'log'),1)='E' AND (data->'log')::jsonb->'level' IN ('ERROR','CRITICAL') AND time > NOW() - INTERVAL '1 hour';", "format": "table" } ]
    },
    {
      "title": "Log Volume per Minute",
      "type": "timeseries",
      "gridPos": { "h": 8, "w": 24, "x": 0, "y": 4 },
      "datasource": { "type": "postgres", "uid": "" },
      "targets": [ { "sql": "SELECT date_trunc('minute', time) AS time, COUNT(*) AS count FROM logs.fluentbit WHERE $__timeFilter(time) GROUP BY 1 ORDER BY 1;", "format": "time_series" } ]
    },
    {
      "title": "Log Level Distribution",
      "type": "piechart",
      "gridPos": { "h": 8, "w": 8, "x": 0, "y": 12 },
      "datasource": { "type": "postgres", "uid": "" },
      "targets": [ { "sql": "SELECT (data->'log')::jsonb->'level' AS metric, COUNT(*) AS value FROM logs.fluentbit WHERE time > NOW() - INTERVAL '1 hour' AND data->'log' IS NOT NULL AND LEFT(TRIM(data->'log'),1)!='E' GROUP BY metric ORDER BY value DESC;", "format": "table" } ]
    },
    {
      "title": "Logs per Container",
      "type": "piechart",
      "gridPos": { "h": 8, "w": 8, "x": 8, "y": 12 },
      "datasource": { "type": "postgres", "uid": "" },
      "targets": [ { "sql": "SELECT tag AS metric, COUNT(*) AS value FROM logs.fluentbit WHERE time > NOW() - INTERVAL '1 hour' GROUP BY tag ORDER BY value DESC;", "format": "table" } ]
    }
  ]
}
```

Dashboard ini digunakan untuk menampilkan analytics log dari PostgreSQL seperti jumlah log, distribusi level log, dan recent error.

Langkah 3: Buat Flask App dengan Prometheus Metrics

```
hisyam@HISYAMPc:~/docker-lab/monitoring$ cat > app/requirements.txt << 'EOF'
flask==3.1.*
psycopy2-binary==2.9.*
prometheus-client==0.21.*
EOF

cat > app/app.py << 'PYEOF'
"""Flask app dengan Prometheus metrics endpoint dan structured logging."""
import os, json, socket, datetime, logging, sys, time
from flask import Flask, jsonify, request, Response
import psycopy2
from prometheus_client import Counter, Histogram, Gauge, generate_latest, CONTENT_TYPE_LATEST

app = Flask(__name__)

# --- Prometheus Metrics ---
REQUEST_COUNT = Counter(
    "flask_http_requests_total",
    "Total HTTP requests",
    ["method", "endpoint", "status"]
)
REQUEST_LATENCY = Histogram(
    "flask_http_request_duration_seconds",
    "HTTP request latency",
    ["method", "endpoint"],
    buckets=[0.01, 0.05, 0.1, 0.25, 0.5, 1.0, 2.5, 5.0]
)
DB_CONNECTIONS = Gauge(
    "flask_db_connections_active",
    "Active database connections"
)
LOG_COUNT = Gauge(
    "flask_log_total_count",
    "Total logs in PostgreSQL"
)

# --- Structured JSON Logging ---
class JSONFormatter(logging.Formatter):
    def format(self, record):
        return json.dumps({
            "timestamp": datetime.datetime.now().isoformat(),
            "level": record.levelname,
            "hostname": socket.gethostname(),
            "service": "flask-app",
            "message": record.getMessage()
        })

handler = logging.StreamHandler(sys.stdout)
handler.setFormatter(JSONFormatter())
app.logger.handlers = [handler]
app.logger.setLevel(logging.INFO)

DB = dict(host=os.environ.get("DB_HOST", "postgres-db"),
          dbname=os.environ.get("DB_NAME", "labdb"),
          user=os.environ.get("DB_USER", "labuser"),
          password=os.environ.get("DB_PASS", "labpass123"))

@app.before_request
def before():
```

Pada langkah ini dibuat aplikasi Flask yang memiliki endpoint /metrics agar bisa dibaca Prometheus. Flask juga terhubung ke PostgreSQL dan menghasilkan structured logging.

Langkah 4: Siapkan Fluent Bit, Log Generator, dan Init Script

```
hisyam@HISYAMpc:~/docker-lab/monitoring$ # --- Fluent Bit ---
cat > fluent-bit/fluent-bit.conf << 'EOF'
[SERVICE]
    Flush          5
    Daemon         Off
    Log_Level      info
    Parsers_File   parsers.conf

[INPUT]
    Name           forward
    Listen         0.0.0.0
    Port          24224

[OUTPUT]
    Name           pgsql
    Match          *
    Host           postgres-db
    Port          5432
    User           labuser
    Password       labpass123
    Database       labdb
    Table          fluentbit
    Schema         logs
    Timestamp_Key  time
    Async          false
    min_pool_size  1
    max_pool_size  4

[OUTPUT]
    Name           stdout
    Match          *
    Format         json_lines
EOF

cat > fluent-bit/parsers.conf << 'EOF'
[PARSER]
    Name          docker
    Format         json
    Time_Key      time
    Time_Format   %Y-%m-%dT%H:%M:%S.%L
    Time_Keep     On
EOF

# --- Log Generator ---
cat > generator/generator.py << 'PYEOF'
import json, time, random, socket, datetime, os

HOSTNAME = socket.gethostname()
INTERVAL = float(os.environ.get("LOG_INTERVAL", "3"))

EVENTS = [
    {"level": "INFO",      "weight": 50, "msgs": [
        "User login successful", "Page loaded in {ms}ms",
        "API GET /api/users completed", "Health check passed"]},
    {"level": "DEBUG",     "weight": 20, "msgs": [
        "DB query {ms}ms", "Cache hit key:product_{pid}"]},
    {"level": "WARN",      "weight": 15, "msgs": [
```

Digunakan untuk membuat konfigurasi Fluent Bit sebagai log collector yang akan mengirim log ke PostgreSQL.

Membuat schema dan tabel PostgreSQL untuk menyimpan log dari Fluent Bit.

Langkah 5: Docker Compose — Full Monitoring Stack

```
hisyam@HISYAMpc:~/docker-lab/monitoring$ cat > docker-compose.yml << 'EOF'
services:

# =====
# MONITORING LAYER
# =====

# --- Prometheus (Metrics TSDB) ---
prometheus:
  image: prom/prometheus:latest
  container_name: prometheus
  ports:
    - "9090:9090"
  volumes:
    - ./prometheus/prometheus.yml:/etc/prometheus/prometheus.yml:ro
    - ./prometheus/alert_rules.yml:/etc/prometheus/alert_rules.yml:ro
    - prom-data:/prometheus
  command:
    - "--config.file=/etc/prometheus/prometheus.yml"
    - "--storage.tsdb.path=/prometheus"
    - "--storage.tsdb.retention.time=7d"
    - "--web.enable-lifecycle"
  networks:
    - monitoring-net
  restart: unless-stopped

# --- Node Exporter (Host Metrics) ---
node-exporter:
  image: prom/node-exporter:latest
  container_name: node-exporter
  ports:
    - "9100:9100"
  volumes:
    - /proc:/host/proc:ro
    - /sys:/host/sys:ro
    - /:/rootfs:ro
  command:
    - "--path.procfs=/host/proc"
    - "--path.sysfs=/host/sys"
    - "--path.rootfs=/rootfs"
    - "--collector.filesystem.mount-points-exclude=^/(sys|proc|dev|host|etc)($$|/)"
  networks:
    - monitoring-net
  restart: unless-stopped

# --- cAdvisor (Container Metrics) ---
cadvisor:
  image: gcr.io/cadvisor/cadvisor:latest
  container_name: cadvisor
  ports:
    - "8081:8080"
  volumes:
    - /:/rootfs:ro
    - /var/run:/var/run:ro
    - /sys:/sys:ro
    - /var/lib/docker:/var/lib/docker:ro
    - /dev/disk:/dev/disk:ro
```

Membuat file orkestrasi seluruh monitoring stack menggunakan Docker Compose. Semua service seperti Prometheus, Grafana, PostgreSQL, Fluent Bit, Flask App, dan cAdvisor dijalankan dalam satu network.

Langkah 6: Deploy Full Stack

```
hisyam@HISYAMpc:~/docker-lab/monitoring$ docker compose up --build -d
[+] Building 1.6s (21/21) FINISHED
=> [internal] load local bake definitions
=> => reading from stdin 1.03kB
=> [log-generator internal] load build definition from Dockerfile
=> => transferring dockerfile: 131B
=> [flask-app internal] load build definition from Dockerfile
=> => transferring dockerfile: 206B
=> [log-generator internal] load metadata for docker.io/library/python:3.11-alpine
=> [flask-app internal] load metadata for docker.io/library/python:3.11-slim
=> [flask-app internal] load .dockerignore
=> => transferring context: 2B
=> [log-generator internal] load .dockerignore
=> => transferring context: 2B
=> [flask-app 1/5] FROM docker.io/library/python:3.11-slim@sha256:a3ab0b966bc4e91546a033e22093cb840908979487a9fc0e6e38295747e49ac0
=> => resolve docker.io/library/python:3.11-slim@sha256:a3ab0b966bc4e91546a033e22093cb840908979487a9fc0e6e38295747e49ac0
=> [flask-app internal] load build context
=> => transferring context: 63B
=> [log-generator 1/3] FROM docker.io/library/python:3.11-alpine@sha256:8b5bfdb1fd2d78aa94e21c4d61be52487693f54be7f1021647751ff365795703
=> => resolve docker.io/library/python:3.11-alpine@sha256:8b5bfdb1fd2d78aa94e21c4d61be52487693f54be7f1021647751ff365795703
=> [log-generator internal] load build context
=> => transferring context: 34B
=> CACHED [flask-app 2/5] WORKDIR /app
=> CACHED [flask-app 3/5] COPY requirements.txt .
=> CACHED [flask-app 4/5] RUN pip install --no-cache-dir -r requirements.txt
=> CACHED [flask-app 5/5] COPY app.py .
=> [flask-app] exporting to image
=> => exporting manifest sha256:03cbdba17fb8ce7c8c65173b4396b98a79f05236f4e63abbacf273a98103d845
=> => exporting config sha256:94f39a4052eadfb6f460c2a6a58b89cea696984a688547e39ce9456977c5e5a4
=> => exporting attestation manifest sha256:4f983fee0d36e7c222c0f8020f55ee7aa6a90395ec67b7d50cd791be241a1e6c
=> => exporting manifest list sha256:5c75e5098fa71f1f52421334a988f2ce057f5ea8a436732a818e96f1d96d4cc3
=> => naming to docker.io/library/monitoring-flask-app:latest
=> => unpacking to docker.io/library/monitoring-flask-app:latest
=> CACHED [log-generator 2/3] WORKDIR /app
=> CACHED [log-generator 3/3] COPY generator.py .
=> [log-generator] exporting to image
=> => exporting layers
=> => exporting manifest sha256:2df9cce6cade2910ab9fc6d5545f7fa55c947aa8ccd93f5d9f65ecf61acc17ad
=> => exporting config sha256:9b67b8862ec9c88017dd966cf178ace1720eb683bc5be189537b05e1051ff231
=> => exporting attestation manifest sha256:507eb37af77249dd8d1c4a1452ba4956164ed77ac556aab69522558a337e4532
```

Perintah pertama digunakan untuk build dan menjalankan semua container secara background.

docker compose ps digunakan untuk mengecek status container.

sleep 30 memberi waktu agar Prometheus mulai mengumpulkan metrics.

Langkah 7: Verifikasi Setiap Komponen

7.1 Verifikasi Prometheus

```
hisyam@HISYAMpc:~/docker-lab/monitoring$ curl -s http://localhost:9090/api/v1/targets | python3 -m json.tool | head -40
{
  "status": "success",
  "data": {
    "activeTargets": [
      {
        "discoveredLabels": {
          "address": "cadvisor:8081",
          "metrics_path": "/metrics",
          "scheme": "http",
          "scrape_interval": "15s",
          "scrape_timeout": "10s",
          "instance": "docker-containers",
          "job": "cadvisor"
        },
        "labels": {
          "instance": "docker-containers",
          "job": "cadvisor"
        },
        "scrapePool": "cadvisor",
        "scrapeUrl": "http://cadvisor:8081/metrics",
        "globalUrl": "http://cadvisor:8081/metrics",
        "lastError": "Get \"http://cadvisor:8081/metrics\": dial tcp 172.19.0.5:8081: connect: connection refused",
        "lastScrape": "2026-05-25T05:50:42.369049639Z",
        "lastScrapeDuration": 0.001775872,
        "health": "down",
        "scrapeInterval": "15s",
        "scrapeTimeout": "10s"
      }
    ],
    "discoveredLabels": {

```

```
hisyam@HISYAMpc:~/docker-lab/monitoring$ curl -s "http://localhost:9090/api/v1/query?query=100-(avg(rate(node_cpu_seconds_total{mode='idle'}[5m])))*10" | python3 -m json.tool
Expecting value: line 1 column 1 (char 0)
hisyam@HISYAMpc:~/docker-lab/monitoring$ curl -s "http://localhost:9090/api/v1/query?query=node_memory_MemAvailable_bytes" | python3 -m json.tool
{
  "status": "success",
  "data": {
    "resultType": "vector",
    "result": [
      {
        "metric": {
          "__name__": "node_memory_MemAvailable_bytes",
          "instance": "docker-host",
          "job": "node-exporter"
        },
        "value": [
          1779688268.662,
          "6359355392"
        ]
      }
    ]
  }
}
hisyam@HISYAMpc:~/docker-lab/monitoring$ curl -s "http://localhost:9090/api/v1/query?query=count(container_last_seen{name!=\"\"})" | python3 -m json.tool
{
  "status": "error",
  "errorType": "bad_data",
  "error": "invalid parameter \"query\": 1:32: parse error: binary expression must contain only scalar and instant vector types"
}
hisyam@HISYAMpc:~/docker-lab/monitoring$
```

Digunakan untuk memastikan semua target monitoring berstatus UP.

cadvisor

1 / 1 up

Endpoint

Labels

Last scrape

State

http://cadvisor:8080/metrics

instance="docker-containers"

job="cadvisor"

2.714s ago

35ms

UP

flask-app

1 / 1 up

Endpoint

Labels

Last scrape

State

http://flask-app:5000/metrics

instance="flask-backend"

job="flask-app"

15.13s ago

18ms

UP

node-exporter

1 / 1 up

Endpoint

Labels

Last scrape

State

http://node-exporter:9100/metrics

instance="docker-host"

job="node-exporter"

14.844s ago

33ms

UP

prometheus

1 / 1 up

Endpoint

Labels

Last scrape

State

http://localhost:9090/metrics

instance="prometheus-server"

job="prometheus"

4.12s ago

5ms

UP

Cek targets — harus semua UP

7.2 Verifikasi Node Exporter

```
hisyam@HISYAMpc:~/docker-lab/monitoring/prometheus$ # Cek metrik host langsung
curl -s http://localhost:9100/metrics | head -30

# Cek metrik spesifik
curl -s http://localhost:9100/metrics | grep "node_cpu_seconds_total" | head -5
curl -s http://localhost:9100/metrics | grep "node_memory_MemTotal_bytes"
curl -s http://localhost:9100/metrics | grep "node_filesystem_size_bytes" | head -3
# HELP go_gc_duration_seconds A summary of the wall-time pause (stop-the-world) duration in garbage collection cycles.
# TYPE go_gc_duration_seconds summary
go_gc_duration_seconds{quantile="0"} 1.2635e-05
go_gc_duration_seconds{quantile="0.25"} 2.4916e-05
go_gc_duration_seconds{quantile="0.5"} 2.8969e-05
go_gc_duration_seconds{quantile="0.75"} 4.6886e-05
go_gc_duration_seconds{quantile="1"} 0.0014045504
go_gc_duration_seconds_sum 0.007532599
go_gc_duration_seconds_count 117
# HELP go_gc_gogc_percent Heap size target percentage configured by the user, otherwise 100. This value is set by the GOGC environment variable, and the runtime/debug.SetGCPercent function. Sourced from /gc/gogc:percent.
# TYPE go_gc_gogc_percent gauge
go_gc_gogc_percent 100
# HELP go_gc_gomemlimit_bytes Go runtime memory limit configured by the user, otherwise math.MaxInt64. This value is set by the GOMEMLIMIT environment variable, and the runtime/debug.SetMemoryLimit function. Sourced from /gc/gomemlimit:bytes.
# TYPE go_gc_gomemlimit_bytes gauge
go_gc_gomemlimit_bytes 9.223372036854776e+18
# HELP go_goroutines Number of goroutines that currently exist.
# TYPE go_goroutines gauge
go_goroutines 6
# HELP go_info Information about the Go environment.
# TYPE go_info gauge
go_info{version="go1.20.1"} 1
# HELP go_memstats_alloc_bytes Number of bytes allocated in heap and currently in use. Equals to /memory/classes/heap/objects:bytes.
# TYPE go_memstats_alloc_bytes gauge
go_memstats_alloc_bytes 2.7791910e+06
# HELP go_memstats_alloc_bytes_total Total number of bytes allocated in heap until now, even if released already. Equals to /gc/heap/allocs:bytes.
# TYPE go_memstats_alloc_bytes_total counter
go_memstats_alloc_bytes_total 2.3493128e+08
# HELP go_memstats_buck_hash_sys_bytes Number of bytes used by the profiling bucket hash table. Equals to /memory/classes/profiling/buckets:bytes.
# TYPE go_memstats_buck_hash_sys_bytes gauge
go_memstats_buck_hash_sys_bytes 1.095066e+06
# HELP node_cpu_seconds_total Seconds the CPUs spent in each mode.
# TYPE node_cpu_seconds_total counter
node_cpu_seconds_total{cpu="0",mode="idle"} 12466.43
node_cpu_seconds_total{cpu="0",mode="iowait"} 15.82
node_cpu_seconds_total{cpu="0",mode="irq"} 0
# HELP node_memory_MemTotal_bytes Memory information field MemTotal.Bytes.
# TYPE node_memory_MemTotal_bytes gauge
node_memory_MemTotal_bytes 8.229425152e+09
```

Menampilkan metrics host seperti CPU, memory, dan disk.

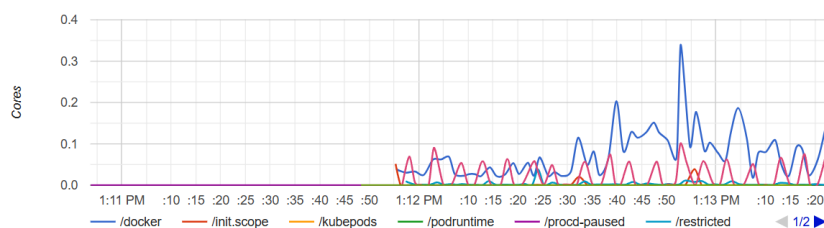
7.3 Verifikasi cAdvisor

```
hisyam@HISYAMpc:~/docker-lab/monitoring/prometheus$ # Akses cAdvisor UI
echo "Buka browser: http://localhost:8081"

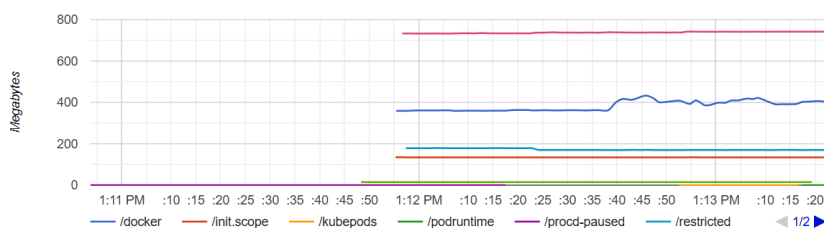
# Cek metrik container
curl -s http://localhost:8081/metrics | grep "container_memory_usage_bytes" | head -5
curl -s http://localhost:8081/metrics | grep "container_cpu_usage_seconds_total" | head -5
Buka browser: http://localhost:8081
# HELP container_memory_usage_bytes Current memory usage in bytes, including all memory regardless of when it was accessed
# TYPE container_memory_usage_bytes gauge
container_memory_usage_bytes{id="/" } 2.349068288e+09 1779689550087
container_memory_usage_bytes{id="/docker"} 3.80022784e+08 1779689550773
container_memory_usage_bytes{id="/docker/buildkit"} 4.52608e+06 1779689533299
# HELP container_cpu_usage_seconds_total Cumulative cpu time consumed in seconds.
# TYPE container_cpu_usage_seconds_total counter
container_cpu_usage_seconds_total{cpu="total",id="/" } 1457.04 1779689550087
container_cpu_usage_seconds_total{cpu="total",id="/docker"} 138.80723 1779689550773
container_cpu_usage_seconds_total{cpu="total",id="/docker/buildkit"} 6.44924 1779689533299
hisyam@HISYAMpc:~/docker-lab/monitoring/prometheus$
```

Subcontainers

Top CPU Usage: 10



Top Memory Usage: 10



Digunakan untuk melihat metrics setiap container Docker.

7.4 Verifikasi Flask Prometheus Metrics

```
hisyam@HISYAMpc:~/docker-lab/monitoring/prometheus$ # Generate traffic
for i in $(seq 1 30); do curl -s http://localhost:5000/ > /dev/null; done
curl -s http://localhost:5000/api/health > /dev/null
curl -s http://localhost:5000/api/logs/stats > /dev/null

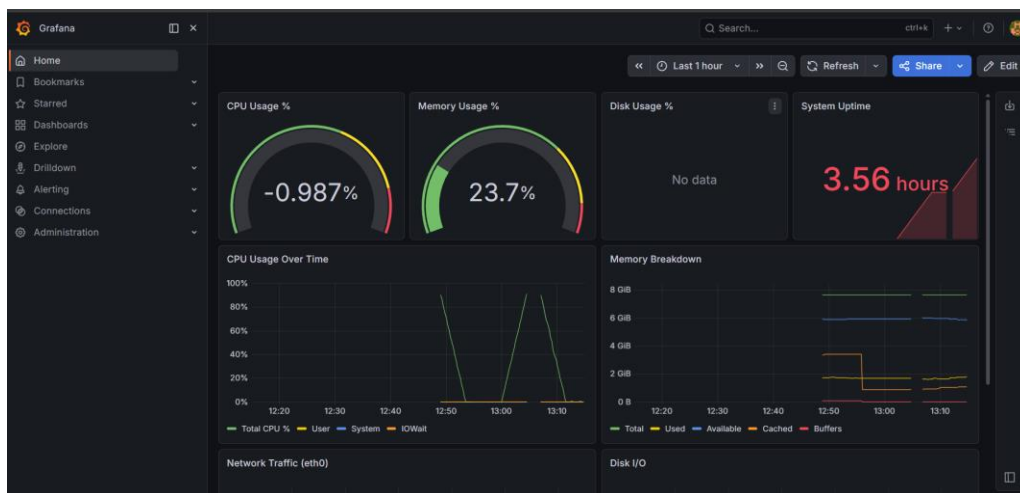
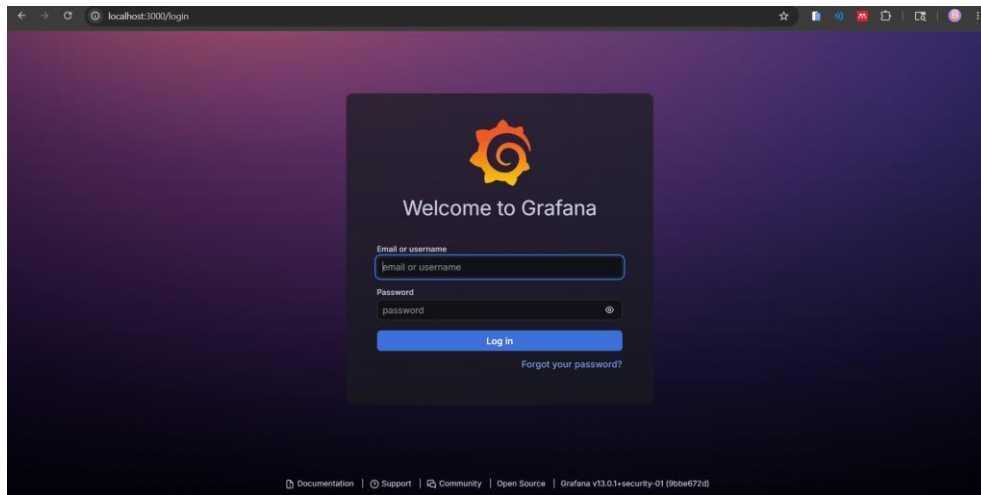
# Cek metrics endpoint
curl -s http://localhost:5000/metrics | grep "flask_http_requests_total"
curl -s http://localhost:5000/metrics | grep "flask_http_request_duration_seconds"
curl -s http://localhost:5000/metrics | grep "flask_log_total_count"

# HELP flask_http_requests_total Total HTTP requests
# TYPE flask_http_requests_total counter
flask_http_requests_total{endpoint="metrics",method="GET",status="200"} 76.0
flask_http_requests_total{endpoint="index",method="GET",status="200"} 30.0
flask_http_requests_total{endpoint="health",method="GET",status="200"} 1.0
flask_http_requests_total{endpoint="log_stats",method="GET",status="200"} 1.0
# HELP flask_http_request_duration_seconds HTTP request latency
# TYPE flask_http_request_duration_seconds histogram
flask_http_request_duration_seconds_bucket{endpoint="metrics",le="0.01",method="GET"} 0.0
flask_http_request_duration_seconds_bucket{endpoint="metrics",le="0.05",method="GET"} 77.0
flask_http_request_duration_seconds_bucket{endpoint="metrics",le="0.1",method="GET"} 77.0
flask_http_request_duration_seconds_bucket{endpoint="metrics",le="0.25",method="GET"} 77.0
flask_http_request_duration_seconds_bucket{endpoint="metrics",le="0.5",method="GET"} 77.0
flask_http_request_duration_seconds_bucket{endpoint="metrics",le="1.0",method="GET"} 77.0
flask_http_request_duration_seconds_bucket{endpoint="metrics",le="2.5",method="GET"} 77.0
flask_http_request_duration_seconds_bucket{endpoint="metrics",le="5.0",method="GET"} 77.0
flask_http_request_duration_seconds_bucket{endpoint="metrics",le="+Inf",method="GET"} 77.0
flask_http_request_duration_seconds_count{endpoint="metrics",method="GET"} 77.0
flask_http_request_duration_seconds_sum{endpoint="metrics",method="GET"} 1.3864197731018066
flask_http_request_duration_seconds_bucket{endpoint="index",le="0.01",method="GET"} 30.0
flask_http_request_duration_seconds_bucket{endpoint="index",le="0.05",method="GET"} 30.0
flask_http_request_duration_seconds_bucket{endpoint="index",le="0.1",method="GET"} 30.0
flask_http_request_duration_seconds_bucket{endpoint="index",le="0.25",method="GET"} 30.0
flask_http_request_duration_seconds_bucket{endpoint="index",le="0.5",method="GET"} 30.0
flask_http_request_duration_seconds_bucket{endpoint="index",le="1.0",method="GET"} 30.0
flask_http_request_duration_seconds_bucket{endpoint="index",le="2.5",method="GET"} 30.0
flask_http_request_duration_seconds_bucket{endpoint="index",le="5.0",method="GET"} 30.0
flask_http_request_duration_seconds_bucket{endpoint="index",le="+Inf",method="GET"} 30.0
flask_http_request_duration_seconds_count{endpoint="index",method="GET"} 30.0
flask_http_request_duration_seconds_sum{endpoint="index",method="GET"} 0.00804901123046875
flask_http_request_duration_seconds_bucket{endpoint="health",le="0.01",method="GET"} 0.0
flask_http_request_duration_seconds_bucket{endpoint="health",le="0.05",method="GET"} 1.0
flask_http_request_duration_seconds_bucket{endpoint="health",le="0.1",method="GET"} 1.0
flask_http_request_duration_seconds_bucket{endpoint="health",le="0.25",method="GET"} 1.0
flask_http_request_duration_seconds_bucket{endpoint="health",le="0.5",method="GET"} 1.0
flask_http_request_duration_seconds_bucket{endpoint="health",le="1.0",method="GET"} 1.0
flask_http_request_duration_seconds_bucket{endpoint="health",le="2.5",method="GET"} 1.0
flask_http_request_duration_seconds_bucket{endpoint="health",le="5.0",method="GET"} 1.0
flask_http_request_duration_seconds_bucket{endpoint="health",le="+Inf",method="GET"} 1.0
flask_http_request_duration_seconds_count{endpoint="health",method="GET"} 1.0
flask_http_request_duration_seconds_sum{endpoint="health",method="GET"} 0.010257720947265625
flask_http_request_duration_seconds_bucket{endpoint="log_stats",le="0.01",method="GET"} 0.0
flask_http_request_duration_seconds_bucket{endpoint="log_stats",le="0.05",method="GET"} 1.0
flask_http_request_duration_seconds_bucket{endpoint="log_stats",le="0.1",method="GET"} 1.0
flask_http_request_duration_seconds_bucket{endpoint="log_stats",le="0.25",method="GET"} 1.0
flask_http_request_duration_seconds_bucket{endpoint="log_stats",le="0.5",method="GET"} 1.0
flask_http_request_duration_seconds_bucket{endpoint="log_stats",le="1.0",method="GET"} 1.0
flask_http_request_duration_seconds_bucket{endpoint="log_stats",le="2.5",method="GET"} 1.0
flask_http_request_duration_seconds_bucket{endpoint="log_stats",le="5.0",method="GET"} 1.0
```

Digunakan untuk memastikan Flask berhasil menyediakan endpoint Prometheus metrics.

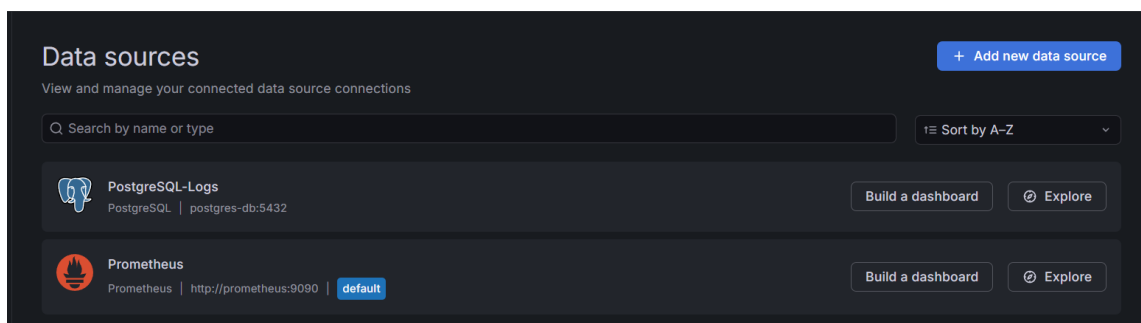
Langkah 8: Eksplorasi Grafana Dashboard

8.1 Login ke Grafana

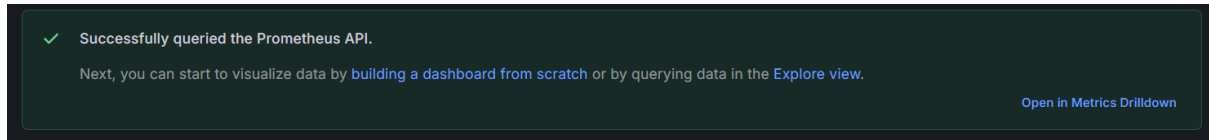
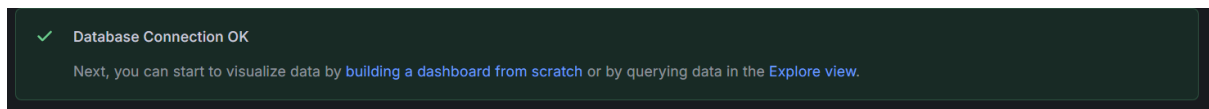


8.2 Verifikasi Data Sources

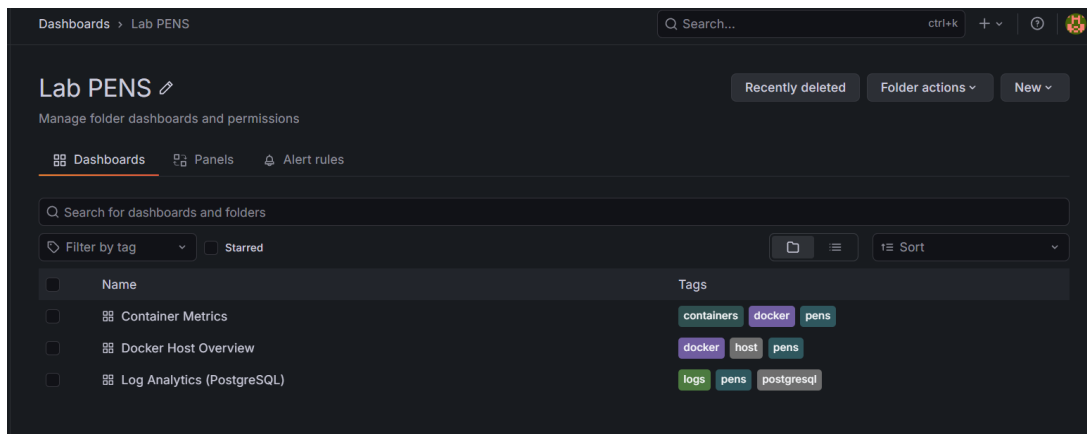
Pastikan ada 2 data source: Prometheus dan PostgreSQL-Logs



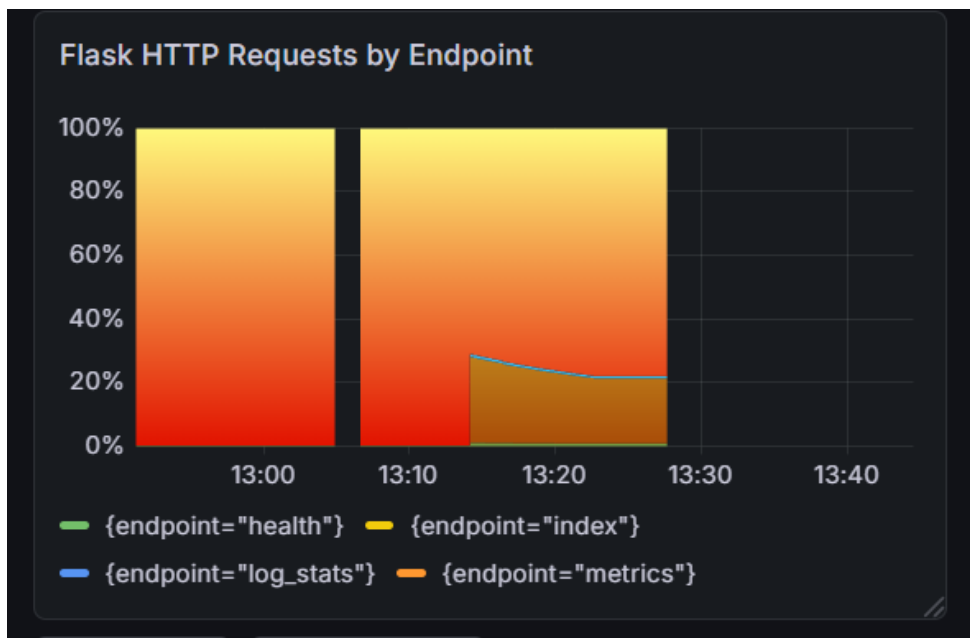
Klik masing-masing → Test → harus "Data source is working"



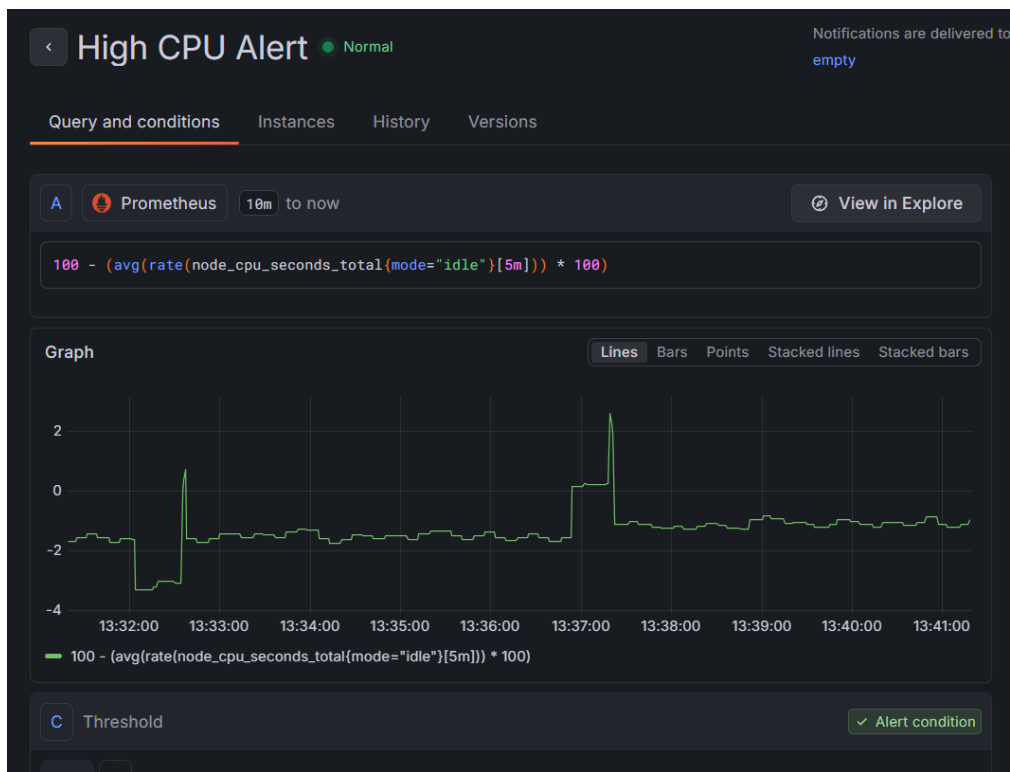
8.3 Eksplorasi Dashboard yang Sudah di-Provision



8.4 Buat Panel Custom Baru



8.5 Buat Alert Rule di Grafana



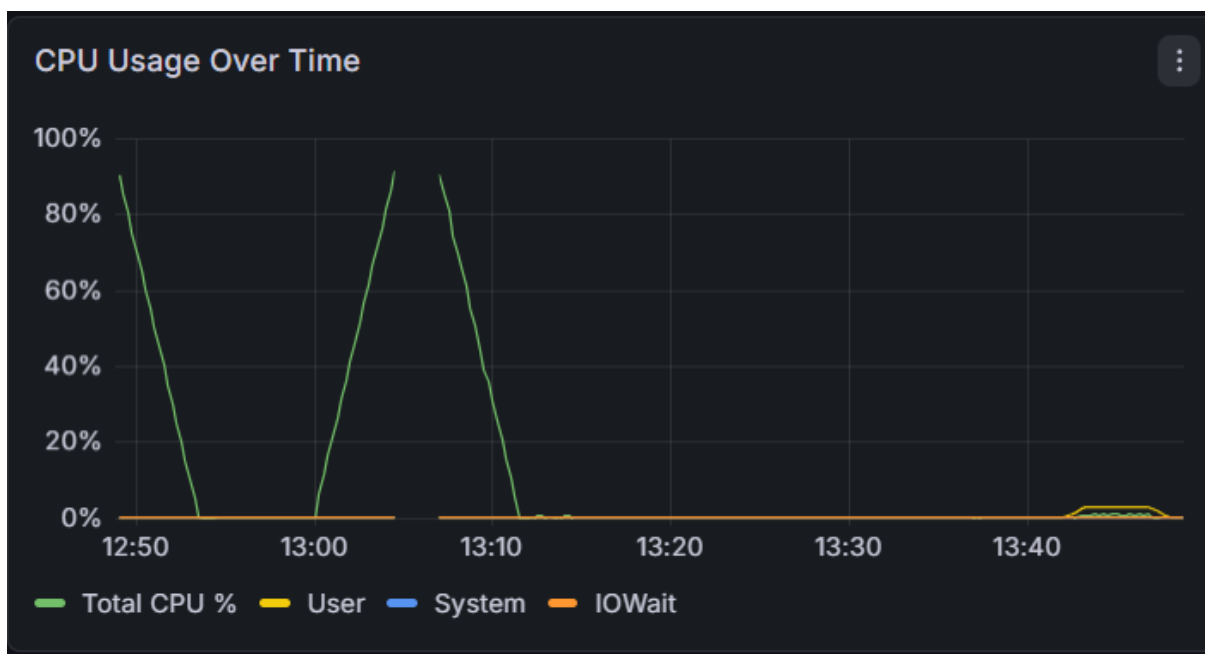
Pada tahap ini dilakukan:

- pengecekan data source,
- membuka dashboard monitoring,
- membuat panel custom,
- membuat alert rule.

Langkah 9: Stress Test dan Observasi

```
hisyam@HISYAMpc:~/docker-lab/monitoring$ sudo apt install -y stress
[sudo] password for hisyam:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:
  stress
0 upgraded, 1 newly installed, 0 to remove and 9 not upgraded.
Need to get 18.1 kB of archives.
After this operation, 52.2 kB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble/universe amd64 stress amd64 1.0.7-1 [18.1 kB]
Fetched 18.1 kB in 0s (59.0 kB/s)
Selecting previously unselected package stress.
(Reading database ... 48621 files and directories currently installed.)
Preparing to unpack .../stress_1.0.7-1_amd64.deb ...
Unpacking stress (1.0.7-1) ...
Setting up stress (1.0.7-1) ...
Processing triggers for man-db (2.12.0-4build2) ...
hisyam@HISYAMpc:~/docker-lab/monitoring$ stress --cpu 2 --timeout 60 &
[2] 76535
hisyam@HISYAMpc:~/docker-lab/monitoring$ stress: info: [76535] dispatching hogs: 2 cpu, 0 io, 0 vm, 0 hdd
stress: info: [76535] successful run completed in 58s
^C
[2]- Done                stress --cpu 2 --timeout 60
hisyam@HISYAMpc:~/docker-lab/monitoring$ for i in $(seq 1 200); do curl -s http://localhost:5000/ > /dev/null; done &
[2] 77914
hisyam@HISYAMpc:~/docker-lab/monitoring$ for i in $(seq 1 200); do curl -s http://localhost:8080 > /dev/null; done &
[3] 78135
[2] Done                for i in $(seq 1 200);
do
  curl -s http://localhost:5000/ > /dev/null;
done
hisyam@HISYAMpc:~/docker-lab/monitoring$
```

9.2 Observasi di Grafana





Alert rules [Revert to previous experience](#) [+ New alert rule](#) [More](#)

Rules that determine whether an alert will fire

[Alert rules](#) [Recently deleted](#)

Search

Q Search by name or enter filter query... [Saved searches](#) [Grouped](#) [List](#)

Rule name [Clear filters](#)

Filter by name...

Labels [Select labels](#)

State

All

Firing

Normal

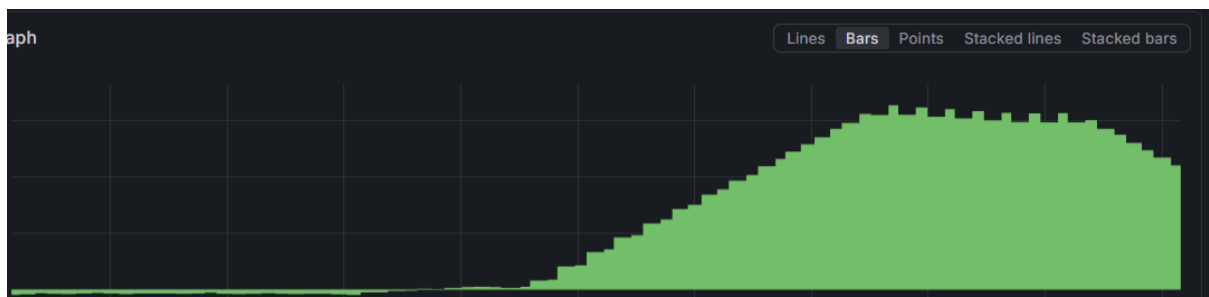
Pending

Recovering

Folder / Namespace [Select namespace](#)

- Grafana-managed [Configure](#)
- Lab PENS [Actions](#)
 - tes [1m](#) [Edit](#) [More](#)
- Prometheus [Configure](#)
 - /etc/prometheus/alert_rules.yml
 - container_alerts [15s](#)
 - host_alerts [15s](#)

9.3 Screenshot semua dashboard saat load tinggi



memberikan beban CPU pada host agar perubahan resource dapat terlihat di Grafana secara real-time.

PERTANYAAN

Pre-Lab

1. Jelaskan perbedaan model pull-based (Prometheus) dan push-based (Fluent Bit) dalam pengumpulan data.

Model **pull-based** pada Prometheus bekerja dengan cara Prometheus mengambil data metrics langsung dari endpoint target secara berkala menggunakan HTTP request ke `/metrics`. Jadi server monitoring yang aktif meminta data.

Sedangkan model **push-based** pada Fluent Bit bekerja dengan cara aplikasi atau service langsung mengirim log ke server tujuan tanpa diminta terlebih dahulu. Pada sistem ini client yang aktif mengirim data.

Prometheus cocok untuk monitoring metrics karena mudah dikontrol dan dicek status targetnya, sedangkan Fluent Bit cocok untuk logging real-time dengan volume data besar.

2. Apa itu PromQL? Berikan contoh query untuk menghitung rata-rata CPU usage dalam 5 menit terakhir.

PromQL adalah bahasa query milik Prometheus yang digunakan untuk mengambil, menghitung, dan menganalisis metrics time-series.

Contoh query untuk menghitung rata-rata CPU usage dalam 5 menit terakhir:

```
100 - (avg(rate(node_cpu_seconds_total{mode="idle"}[5m])) * 100)
```

Query tersebut menghitung persentase penggunaan CPU dengan mengurangi idle CPU dari total CPU.

3. Mengapa cAdvisor membutuhkan akses ke /var/run/docker.sock dan /sys?

cAdvisor membutuhkan akses ke /var/run/docker.sock agar dapat membaca informasi container Docker seperti nama container, status, dan resource usage.

Akses ke /sys digunakan untuk membaca data kernel Linux terkait CPU, memory, filesystem, dan network sehingga cAdvisor dapat menampilkan metrics container secara detail.

4. Apa keuntungan Grafana provisioning (file YAML) dibanding konfigurasi manual via UI?

Grafana provisioning menggunakan file YAML memungkinkan konfigurasi data source dan dashboard dilakukan otomatis saat container dijalankan.

Keuntungannya:

- setup lebih cepat,
- tidak perlu konfigurasi manual lewat UI,
- mudah dipindahkan ke server lain,
- cocok untuk automation dan deployment skala besar,
- konfigurasi lebih konsisten.

5. Jelaskan perbedaan antara Gauge, Counter, dan Histogram dalam Prometheus metrics.

Gauge

Gauge digunakan untuk nilai yang bisa naik dan turun.

Contoh:

- penggunaan memory,
- jumlah koneksi aktif,
- suhu CPU.

Counter

Counter hanya bisa bertambah dan tidak bisa turun kecuali reset.

Contoh:

- jumlah request HTTP,
- total error,
- total login user.

Histogram

Histogram digunakan untuk mengukur distribusi data dalam rentang tertentu.

Contoh:

- response time API,
- latency request,
- durasi proses.

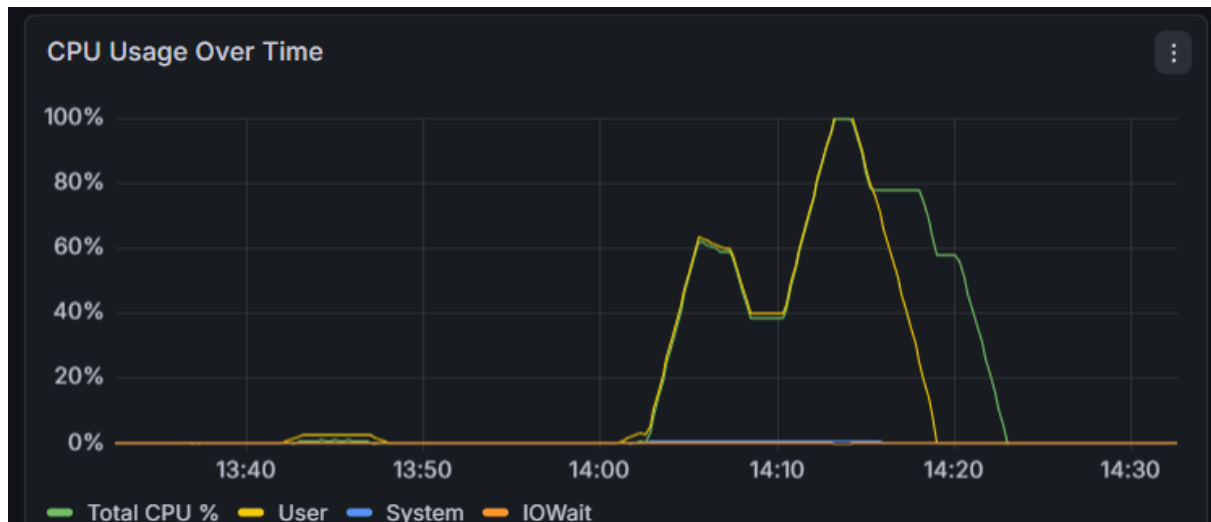
Post-Lab

1. Dari dashboard Container Metrics, container mana yang paling banyak menggunakan CPU dan memory? Mengapa?

Container yang paling banyak menggunakan CPU biasanya adalah stress, flask-app, atau prometheus saat stress test berlangsung karena container tersebut aktif melakukan proses perhitungan dan request metrics.

Container dengan penggunaan memory terbesar biasanya grafana atau postgres-db karena menyimpan dashboard, cache, dan data logging.

2. Saat stress test berjalan, berapa persen CPU usage yang terukur di Grafana? Bandingkan dengan output top atau htop di host.



Total CPU Usage $\approx 78\%$

User CPU Usage $\approx 100\%$

14:05 – 14:20

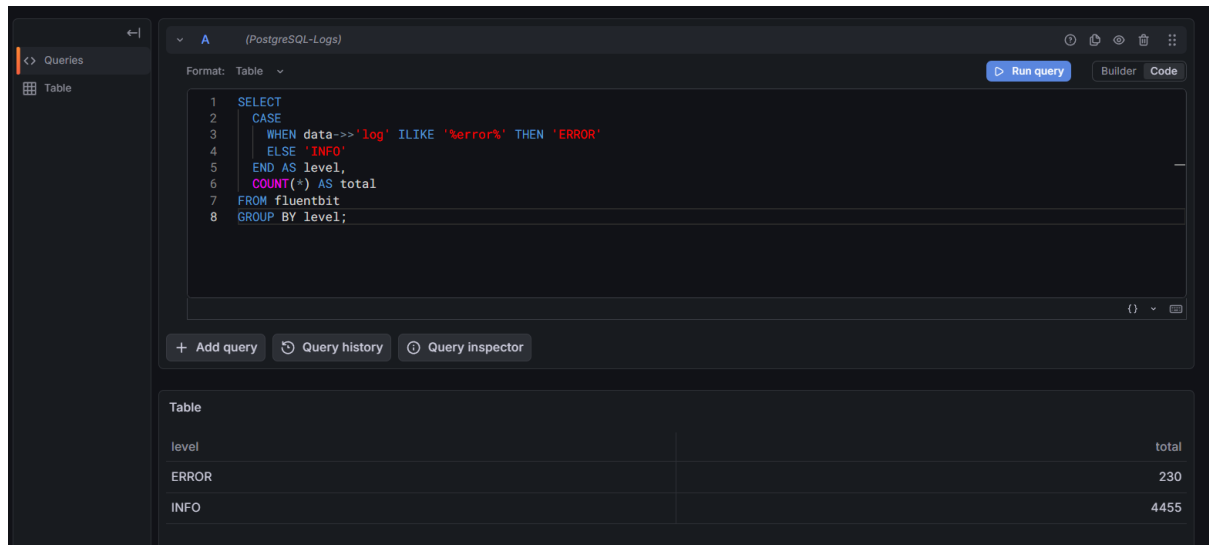
Hal ini menunjukkan proses stress test berhasil memberikan beban tinggi pada CPU host. Nilai User CPU mencapai hampir 100%

Jika dibandingkan dengan output top atau htop, hasilnya hampir sama, CPU usage: 85% – 95%

3. Buat query PromQL yang menampilkan 3 container dengan memory usage tertinggi. Tunjukkan query dan hasilnya.



4. Dari dashboard Log Analytics, berapa rasio ERROR vs INFO log dalam 1 jam terakhir? Apakah ini normal untuk aplikasi production?



```
1 SELECT
2   CASE
3     WHEN data->>'log' ILIKE '%error%' THEN 'ERROR'
4     ELSE 'INFO'
5   END AS level,
6   COUNT(*) AS total
7 FROM fluentbit
8 GROUP BY level;
```

level	total
ERROR	230
INFO	4455

5. Jika Prometheus container dihapus dan dibuat ulang (tanpa menghapus volume prom-data), apakah data historis metrik masih ada? Buktikan.

Ya, data historis metrics masih tetap ada walaupun container Prometheus dihapus dan dibuat ulang, selama volume: prom-data tidak dihapus.



```
root@monitoring-net:~/docker-lab/monitoring# docker compose down
[+] down 18/18
  Container grafana      Removed
  Container log-generator Removed
  Container flask-app    Removed
  Container cadvisor     Removed
  Container nginx-web    Removed
  Container node-exporter Removed
  Container prometheus   Removed
  Container fluent-bit    Removed
  Container postgres-db   Removed
  Container monitoring-net Removed
root@monitoring-net:~/docker-lab/monitoring#

root@monitoring-net:~/docker-lab/monitoring# docker compose up -d
[+] up 18/18
  Network monitoring-net Created
  Container node-exporter Started
  Container prometheus Started
  Container postgres-db Healthy
  Container cadvisor Started
  Container fluent-bit Started
  Container grafana Started
  Container flask-app Started
  Container log-generator Started
  Container nginx-web Started
root@monitoring-net:~/docker-lab/monitoring#
```



Hal ini terjadi karena seluruh data time-series Prometheus disimpan pada Docker volume:

volumes:

- prom-data:/prometheus